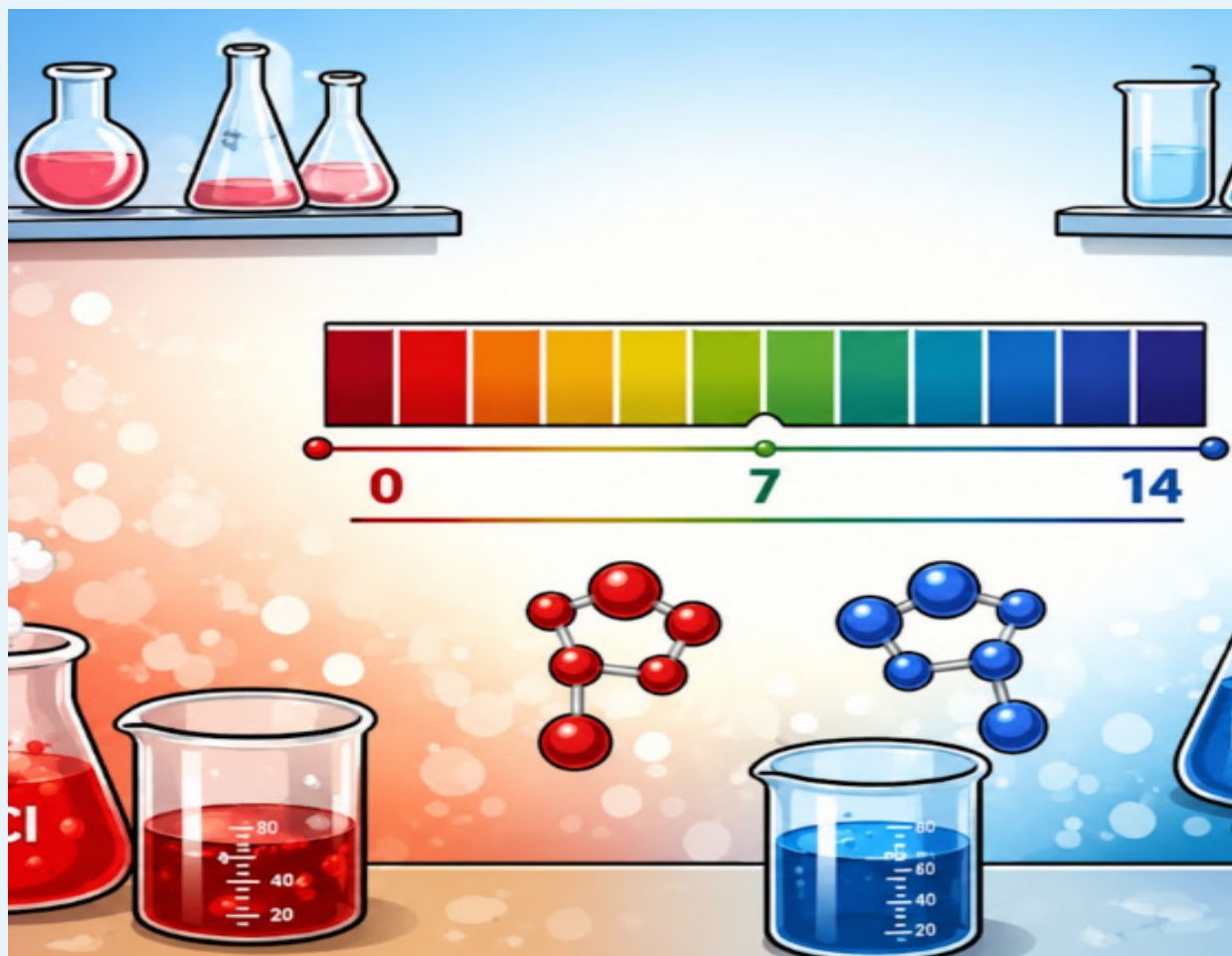




REB RWANDA BASIC
EDUCATION BOARD

CHEMISTRY



Student's Book

Senior 2

Experimental version

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FOREWORD

Dear Student,

The Rwanda Basic Education Board (REB) is honored to present the Chemistry student book for senior two. This Textbook is designed to support the effective implementation of the Competence-Based Curriculum (CBC), ensuring learner-centered teaching practices that align with Rwanda's vision of empowering students to reach their full potential, actively participate in society, and pursue meaningful careers.

As Rwanda advances toward building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding students to connect classroom knowledge with real-life experiences, thus fostering individual growth and national development.

This student's book serves as a comprehensive resource for teaching Chemistry at senior two. The subject is an integral part of the CBC, aimed at equipping students with essential knowledge, skills, values, and attitudes necessary for becoming responsible and competent citizens. Through this subject, students are encouraged to develop problem-solving abilities and contribute positively to their communities.

In pursuit of quality education, the Government of Rwanda emphasizes the importance of using teaching and learning materials that are aligned with the curriculum.

I sincerely thank everyone who contributed to the refinement of this textbook. Special recognition goes to the REB team who coordinated the entire process. We thank all partners, teachers, and individuals who were involved in the process from start to the end. Every feedback was highly valued and helped to improve learning outcomes of our students.



Dr. MBARUSHIMANA Nelson

Director General of Rwanda Basic Education Board

ACKNOWLEDGMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Chemistry student's book for senior two. This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to the Rwanda Basic Education Board (REB) staff, subject experts, University lecturers and teachers for their valuable work in the adaptation of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process.



MURUNGI Joan

**Head of Curriculum, Teaching and Learning Resources
Department/REB**

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UNIT 1:

TRENDS IN PROPERTIES OF ELEMENTS IN THE PERIODIC TABLE

1	2																	18
1 H																		2 He
Period 2	3 Li	4 Be	Metals										5 B	6 C	7 N	8 O	9 F	10 Ne
Period 3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
Period 4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
Period 5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
Period 6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
Period 7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Uuq	115 Uup	116 Uuq	117 Uus	118 Uuo
Lanthanide series (-)	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinide series (-)	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Key unit competence

Analyse the trends and patterns in the properties of elements across periods and down groups to explain their periodic behavior.

Learning Objectives

By the end of the unit, you should be able to:

Knowledge and understanding

- Describe the properties of metals, non-metals, and metalloids
- Describe trends in reactivity of elements with acids, water, and halogens.
- Explain the trends of elements in the physical properties across a period and down a group

Skills

- Classify elements into metals, metalloids, and non-metals.
- Compare the physical properties of metals and non-metals using simple experiments.
- Compare the reactivity of metals across the period and down the group with the help of simple experiments.

Values and attitudes

- Respect the procedures during practical activities.
- Develop teamwork in group activities.
- Show curiosity and interest in exploring how trends in the periodic table explain element behavior.

Introduction

The Periodic Table is a simple way to organize elements so that we can easily see how their properties change in a pattern. It is organized, among others, into horizontal rows called **periods** and vertical columns called **groups**. As we move across a period from left to right or down a group from top to bottom, elements show clear trends in properties such as atomic size, reactivity, and metallic character. Understanding these trends helps learners to predict how different elements behave in chemical reactions and why they are grouped in the given way. This knowledge builds a strong foundation for studying the behavior of matter in everyday life and in science.

Key terms: Trends, periods, groups, metallic character.



Introductory activity

1. What is the meaning of trends in the periodic table of chemical elements?
2. Do you think that the study of periodic trends is important? Explain your answer.

1.1. Classification of elements into metals, metalloids and non-metals.



Learning activity 1.1

Observe Figure 1.1(a) of the periodic table and answer the following questions.

1. How many chemical elements are known in the modern periodic table?
2. These elements are classified into three categories: metals, metalloids and non-metals. Which are they?
3. In which part of the periodic table each class of elements appear in the periodic table?

	1																	18
Period 1	1																	2
	H																	He
Period 2	3	4											5	6	7	8	9	10
	Li	Be											B	C	N	O	F	Ne
Period 3	11	12											13	14	15	16	17	18
	Na	Mg											Al	Si	P	S	Cl	Ar
Period 4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Period 5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Period 6	55	56	*	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
	Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Period 7	87	88	**	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
	Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Uut	Fl	Uup	Lv	Uus	Uuo
Lanthanide series (*)	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71			
	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu			
Actinide series (**)	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103			
	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr			

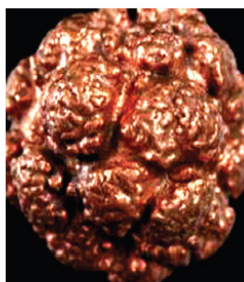
Figure 1.1(a): Periodic table

Currently, 118 chemical elements are known, and they are classified into three categories: **metals, metalloids, and non-metals**. Metals constitute most elements and are primarily located on the left and central regions of the periodic table (blue colour). Non-metals, on the other hand, are found on the right-hand side of the periodic table (yellow colour). Metalloids occupy a narrow diagonal band between metals and non-metals (green colour) and exhibit properties of both groups, metals and non-metals. This classification helps to understand the chemical behavior and physical properties of elements across the periodic table (Figure 1.2). In the periodic table, there is a regular variation in the properties of elements in groups and periods.

The majority of known elements (about 80%) are metals. All metals are solid at room temperature except mercury; gallium and caesium also have melting points close to room temperature, which is a liquid metal at room temperature and pressure. Metals are very important for the economy of every country.

The major metals in the earth's crust are aluminium, iron, calcium, sodium, potassium and magnesium. Aluminium is the most abundant metal in the earth's crust.

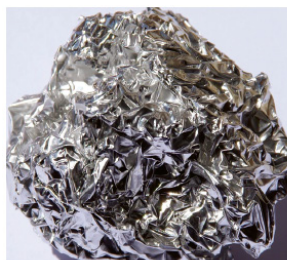
Examples of metals



Copper



Iron



Aluminium



Gold and silver

Figure 1.1(b): Metals

There are only 17 non-metals. Out of these, 10 non-metals are solid, 1 non-metal is liquid and the remaining 11 are gases. Non-metals occur as solids, liquids, and gases; bromine is the only liquid non-metal at room temperature.

Non-metals are small in number as compared to metals but they play an important role in our everyday life. In fact, life would not have been possible without the presence of non-metals. For example, oxygen (non-metal) is essential for breathing to maintain life. Another non-metal carbon is one of the most important elements for existence of life on the earth. This is because carbon compounds like proteins, carbohydrates, vitamins and fats are essential for growth and development of living organism. Non-metals are also the main components of vegetable oil, acids, fertilisers, and fungicides. The major non-metals in the earth's crust are oxygen, silicon, phosphorus

and sulphur. Oxygen is the most abundant non-metal in the earth's crust.

Non-metals are the major constituents of air, earth's crust and oceans. For example, oxygen and nitrogen are the major constituents of air.

Oxygen and silicon are the major constituents of earth's crust. Oxygen and hydrogen are the major constituents of oceans.

Examples of some non-metals



Sulfur



Carbon



Iodine

Figure 1.1(c): Non-Metals

Metalloids are the elements found along the stair-step line that distinguishes metals from non-metals. Metalloids have properties of both metals and non-metals. Some of the metalloids, such as silicon and germanium, are semiconductors. This means that they can carry an electrical charge under special conditions. This property makes metalloids to be useful in computers, calculators and other electronic devices.

The metalloids are : Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium and Polonium.

Metalloids tend to be economically important because of their unique conductivity properties. Metalloids have intermediate/semiconducting electrical properties; conductivity varies among metalloids and conditions.

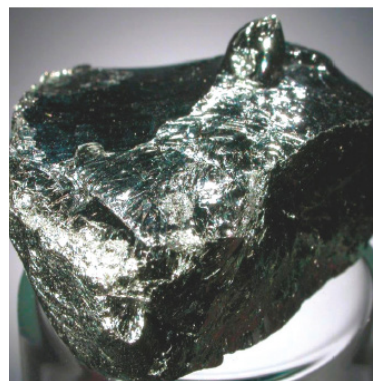
Examples of metalloids



Boron



Silicon



Germanium



Arsenic



Tellurium



Antimony



Polonium

Figure 1.1(d). Metalloids



Application activity 1.1

1. Consider the following list of substances or objects commonly found in the daily life: Cooking pot, Pencil lead, Electrical wire, Plastic ruler, Silicon chip (calculator/computer), Charcoal, Iron nail, Glass. For each item:
 - a) Identify the main element(s) constituting each object or material.
 - b) Classify each element as a metal, non-metal, or metalloid.
 - c) Justify your answer using, at least one property (e.g., shiny, conducts electricity, brittle, etc.).
2. Use the periodic table to classify the elements: Carbon, Chlorine, Iron, Sodium, Hydrogen, Magnesium, Fluorine, Nitrogen, Zinc, Strontium, Bromine, Barium, and Phosphorus.

[https://chem.libretexts.org/Bookshelves/General_Chemistry/Map%3A_Chemistry_-_The_Central_Science_\(Brown_et_al.\)/07%3A_Periodic_Properties_of_the_Elements/7.06%3A_Metals_Non_metals_and_Metalloids](https://chem.libretexts.org/Bookshelves/General_Chemistry/Map%3A_Chemistry_-_The_Central_Science_(Brown_et_al.)/07%3A_Periodic_Properties_of_the_Elements/7.06%3A_Metals_Non_metals_and_Metalloids)

1.2. Physical properties of metals and non-metals



Learning activity 1.2

Investigating physical properties of metals and non-metals

Observe the utensils used in our daily life like aluminium foil, casserole (sauce pan), charcoal, wood spoon, nail, plastic bottle, glass.

Describe these objects focusing on:

- Appearance (shiny or dull)
- Hardness
- Ability to bend or break
- Weight (heavy or light) Or get more information from the internet, if available.



Figure 1.2.(a): Daily life utensils

Guiding questions

- | | |
|---|---|
| a) Which of these objects are shiny? What does that tell us about metals? | c) Why do you think cooking pots are made of metals and not wood? |
| b) Which objects can be easily broken? Are they metals or non-metals? | d) Which materials are good conductors of heat? |

In our daily life, we use many materials made from different elements found in the environment. These elements can be grouped into metals and non-metals. Learning about their physical properties helps us to understand how and why different materials are used in everyday activities, from household tools to school equipment. This knowledge also supports us to make the better choice when selecting materials for different purposes in science and technology.

1.2.1. Physical properties of metals

Metals are materials we rely on every day, found in cars, buildings, tools, and many other objects around us. They are different from other materials because they share certain common features that make them highly useful. Before exploring what metals can do, it is important to understand the key physical properties that all metals have in common.

a) Metals are lustrous



Learning activity 1.2.1 (a)

Illustrating lustrous properties of metals

Take samples of iron, copper, aluminium, and magnesium. Note the appearance of each sample. Clean the surface of each sample by rubbing it with sandpaper, a file or steel wire and note its appearance again. Explain.

Metals have a natural shine, which means that they possess a bright, reflective surface. This lustrous appearance makes them useful in making jewelry and decorative items. For example, gold and silver are commonly used for jewelry because of their attractive shine.



Gold jewelry

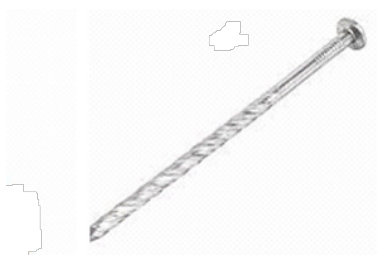


Figure 1.2.1(a): Lustrous objects.

A metal shows its true shine only when it is freshly cut or polished. Over time, exposure to air causes metals to lose their brightness due to corrosion. When we rub the corroded metal surface with sandpaper, the outer corroded layer is removed and the metal object becomes shiny once again.

b) Metals are hard and strong



Learning activity 1.2.1 (b)

Illustrating the variation in hardness among metals

Take small pieces of iron, copper, aluminium, and magnesium.

- Cut these metals with a sharp knife and note your observations.
- Hold a piece of sodium metal with a pair of tongs.

Caution:

- Always handle sodium metal with care.
- Dry it by pressing between the folds of a filter paper.
- Put it on a watch glass and try to cut it with a knife.
- What do you observe?

Most metals are hard, but hardness varies among metals. The hardness of metals varies from one metal to another. Metals such as iron and copper are hard, whereas aluminium is relatively soft and can be cut with a knife. In contrast, sodium and potassium are soft metals and can be easily cut with a knife.

In general, metals are strong, although some, like sodium and potassium, are soft. Metals can support large weights without breaking. For example, iron in the form of steel is very strong. Because of this strength, iron is widely used in making machines, chains, and vehicles.

c) Metals are malleable



Learning activity 1.2.1 (c)

Illustrating the malleability among metals

- Take pieces of aluminium, iron, zinc, lead, and copper.
- Place each metal on a block of iron and strike it four or five times with a hammer.
- Record the change in the shape of these metals.

Metals can be hammered into very thin sheets without breaking. This property is known as **malleability**. Malleability is the property of metals that allows them to be **hammered, pressed, or rolled into thin sheets without breaking**.

Aluminium is highly malleable, so it can be rolled into very thin sheets (foil). Gold and

silver are the most malleable metals and can be beaten into extremely thin sheets called foils.

Aluminium foil is widely used for packaging foods, medicines and cigarettes.



Figure 1.2.1.c. Materials made due to malleability of aluminium

d) Metals are ductile



Learning activity 1.2.1 (d)

Illustrating the ductility of a metal

- Take a thin piece or rod of copper.
- Gently pull or stretch the copper using your hands or simple tools.
- Observe whether the copper can be drawn into a thinner wire without breaking.
- Record your observation on how the shape of the copper changes.

Metals are ductile, which means they can be stretched and drawn into thin wires. Not all metals possess the same level of ductility. Gold and silver are the most ductile metals; for example, just one gram of gold can be stretched into a wire about 2 km long. Copper and aluminum are highly ductile, making their wires widely used for electrical wiring.

Ductility of metals is the property that allows a metal to be drawn out into thin wires without breaking.

Ductility is important because it makes metals useful in industries where **wires, cables, and flexible materials** are needed.



Figure 1.2.1. d: Copper wire.

e) Metals are good conductors of heat



Learning activity 1.2.1(e)

Illustrating thermal conductivity and melting among metals

Caution: Be careful while heating the objects.

- Take a steel spoon, a brass key, aluminium or copper wire (10 cm), and an iron rod.
- Light the burner.
- Hold one end of the iron rod in your hand.
- Keep the other end of each metal object to the flame of the burner for 3 to 4 minutes as shown in Figure 1.2.1(e) below. What do you feel?

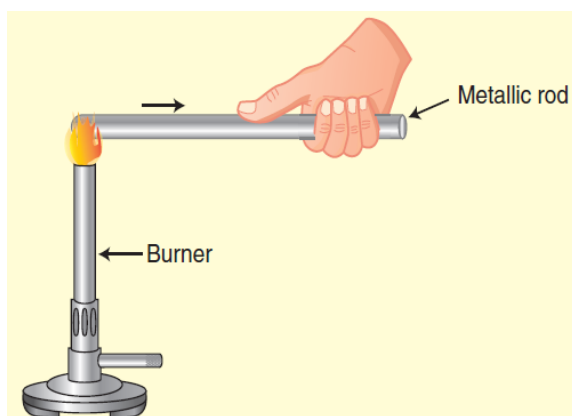


Figure 1.2.1(e): Thermal conductivity experimental set up

- State your observation in each case. Does the metal wire melt? Explain your observation.

Metals are good conductors of heat, which means heat moves quickly through them from a hot part to a cooler part.

However, not all metals conduct heat equally well:

- Silver and copper are among the best electrical and thermal conductors.
- Aluminium is also a very good conductor and is commonly used in cooking utensils.

Because of this property, metals are used to make pots, pans, and heat exchangers, where fast and even heating is important.

Most metals also have high melting points, meaning they can withstand high temperatures without melting. This makes them useful in machines, buildings, and industrial work.

The melting points and densities of some metals are shown in Table 1.1.

f) Metals are good conductors of electricity



Learning activity 1.2.1 (f)

Illustrating electrical conductivity among metals

Take a dry cell, a torch bulb fixed in a holder, and some connecting copper wires with crocodile clips. Connect them as shown in Figure 1.2.1 (f) to form an electric circuit. Leave a small gap between the ends of crocodile clips A and B so that the circuit is incomplete. Since no current flows in an open circuit, the bulb does not light.

Now complete the circuit by:

- Placing a piece of aluminium foil between the ends of crocodile clips A and B, as shown in Figure 1.2.1 (f).
- Replacing the aluminium foil with an iron rod between the same clips.

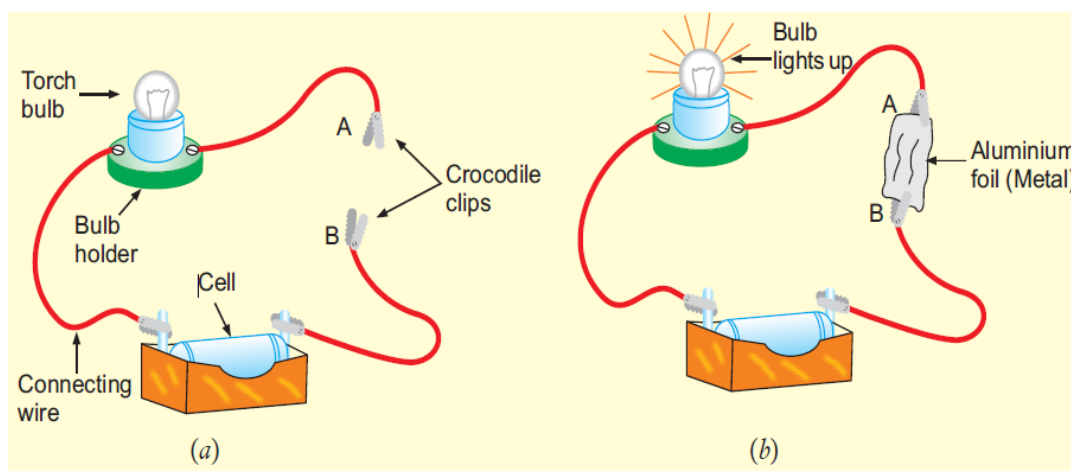


Figure 1.2.1. (f): Metal Electrical conductivity experimental set up

Explain what happens to the bulb?

When electricity is applied to a metal, the metal's **free electrons** start to move.

These moving electrons carry energy from one part of the metal to another, so electrons all along the metal begin to move. This movement of electrons is called an **electric current**. Not all metals conduct electricity equally well: Silver is the best electrical conductor, followed by copper, gold, aluminium, and tungsten. Copper is widely used for electrical wires because it conducts electricity very well and is cheaper than silver and gold.

For safety, electrical wires are covered with materials like PVC (plastic). PVC is an insulator, which means it does not allow electricity to pass through and it helps to prevent electric shocks.

g) Metals are sonorous



Learning activity 1.2.1.(g)

Illustrating sonority of metals

Hold a piece of metal in your left hand and strike it with another piece and listen to the sound.

- Hold a piece of plastic container and strike it with a wooden material and listen to the sound.

What is the difference between the sounds heard in (a) and (b), respectively?

Metals are sonorous, meaning they can produce a ringing sound when struck. This property of metals is called sonority, and it arises from their ability to vibrate and generate sound upon impact. Because of this, metals are commonly used to make musical instruments such as bells and the strings of a guitar.



Church bell



Guitar

Figure 1.2.1. (g): Sonorous materials

h) Metals have high densities

The density of a substance is defined as its mass per unit volume. Metals generally have high densities, which makes them heavy substances. However, some metals, such as aluminium, sodium, and potassium, have relatively low densities. The densities of selected metals are presented in Table 1.1.

Table 1.1: Melting points and densities of metals

Metals	Melting points	Densities (at room temperature)
Iron (Fe)	1538°C	7.8 g/cm ³
Copper (Cu)	1084°C	8.9 g/cm ³

Aluminium (Al)	660°C	2.7 g/cm ³
Sodium (Na)	98°C	0.9 g/cm ³
Potassium (K)	64°C	0.8 g/cm ³
Gallium (Ga)	30°C	5.9 g/cm ³
Caesium (Cs)	28°C	1.9 g/cm ³



Application activity 1.2.1

1. True or false questions

- | | |
|--|--|
| a) Metals are generally malleable. | d) Most metals usually have low melting points |
| b) Metals are sonorous. | e) Metals are good conductors of electricity |
| c) All metals are solid at room temperature. | |
- Why are the handles of metal cooking pans usually made of plastic or wood?
 - If silver is a better conductor of heat than copper and aluminium, why don't we use it for most pots and pans?
 - What is the specific role of plastics in an electrical circuit, and why is it necessary?

1.2.2. Physical properties of non-metals

a) Non-metals are poor conductors of heat and electricity



Learning activity 1.2.2(a)

Demonstrating that non-metals do not conduct electricity

Take a dry cell, a torch bulb fitted in a holder, and some copper connecting wires with crocodile clips. Connect them as shown in Figure 1.2.2(a) to form an electric circuit. Notice that there is a gap between the ends of crocodile clips A and B, so no current flows in the open circuit shown in Figure 1.2.2 (a).

Now, perform the following steps:

- Insert a piece of sulphur (a non-metal) between the crocodile clips A and B, as shown in Figure 1.2.2 (a).
- Replace the sulphur with a piece of dry wood between the ends of crocodile clips A and B.



Observe what happens in each case and record your findings.

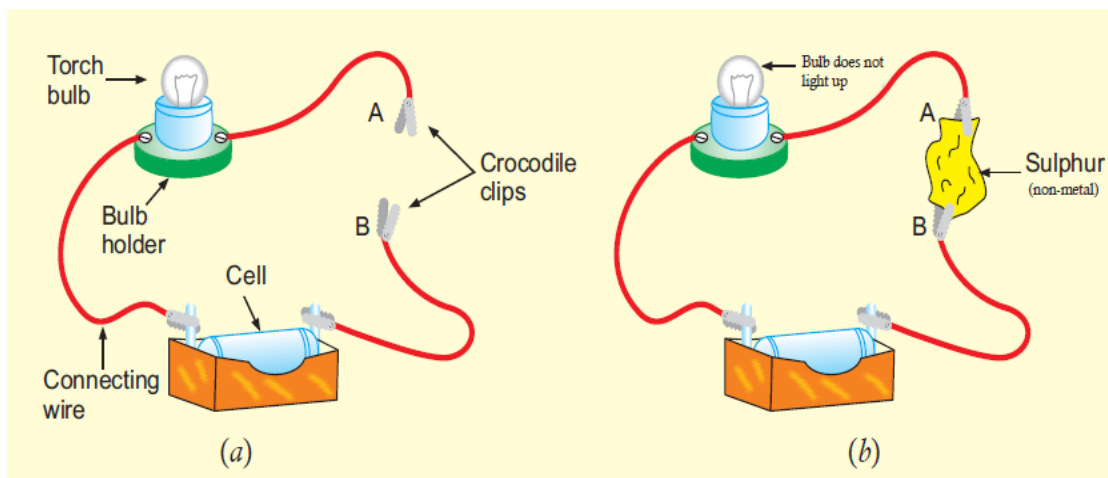


Figure 1.2.2.(a): Non-metal electrical conductivity experimental set up.

In both cases, we see that the bulbs do not light up at all. This means that neither sulphur nor a piece of dry wood allows electric current to pass through it, and no current flows in the circuit. This activity shows that non-metals do not conduct electricity. Non-metals are not conductors of electricity. The only exception is carbon (non-metal) in the form of graphite, which is a good conductor of electricity and is therefore used in making batteries and cells.

b) Non-metals are poor conductors of heat



Learning activity 1.2,2,(b)

Illustrating thermal conductivity of non- metals

Caution: Be careful while heating the objects.

- Take a stick (10 cm) made of wood.
- Light the burner.
- Hold one end of the stick in your hand.
- Keep the other end of the stick very near the flame of the burner for 5 minutes as shown in Figure 1.2.2(b). below. What do you feel?

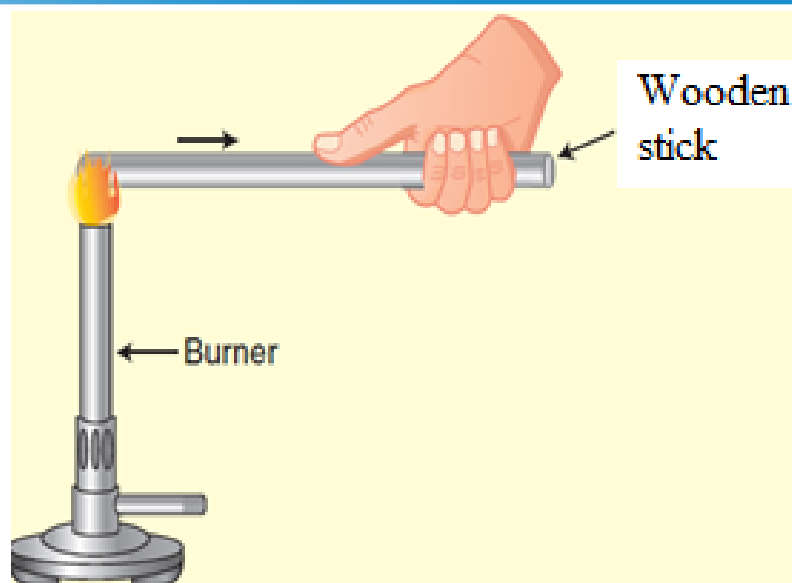


Figure 1.2.2 (b): Thermal conductivity of non-metals experimental setup.

Non-metals are poor conductors of heat. Most non-metals do not allow heat to pass through them easily. Graphite is an exception because it conducts electricity well.

Non-metals occur in all three physical states: solid, liquid, and gas. For example, carbon, sulphur, and phosphorus are solids; bromine is a liquid, while oxygen, hydrogen, and nitrogen exist as gases. Non-metals also exhibit a wide range of colors. Phosphorus can be red, black, or white; sulphur is yellow; chlorine is green; and some non-metals, such as oxygen, hydrogen, nitrogen and fluorine are colorless. Other important physical properties of non-metals include:

- a) **Non-metals are generally not lustrous:** Unlike metals, non-metals do not have a shiny surface. The only exception is iodine, which possesses a metallic luster.
- b) **Non-metals are neither hard nor strong:** Most solid non-metals are soft and can be broken easily, such as sulphur and phosphorus. The only exception is carbon in the form of diamond, which is extremely hard and is considered the hardest natural substance on Earth.
- c) **Non-metals are neither malleable nor ductile:** Non-metals are brittle, meaning they break into pieces when hammered or stretched. Consequently, they cannot be hammered into thin sheets or drawn into wires. This property of breaking easily is called *brittleness*, which is a characteristic feature of non-metals.
- d) **Non-metals have low melting and boiling points:** Most non-metals melt and boil at relatively low temperatures. The exceptions are graphite and diamond (allotropes of carbon), which have very high melting points. For example, graphite melts at about 3600°C .
- e) **Non-metals have low density:** They are generally light substances with comparatively low densities.
- f) **Non-metals are non-sonorous:** Unlike metals, non-metals do not produce a ringing sound when struck with an object.

Physical properties of metalloids

Metalloids have properties of both metals and non-metals.

Metalloids like silicon and germanium are semiconductors, which means that they conduct electricity poorly at room temperature, but their conductivity increases when heated or when tiny amounts of impurities are added. Metalloids exhibit a unique behavior by partially conducting electricity, which makes them essential in devices like computers and in the semiconductor industry.



Application activity 1.2

1. True or false questions
 - a) Metals are generally malleable.
 - b) Metals are sonorous.
 - c) All metals are solid at room temperature.
 - d) Most metals usually have low melting points
 - e) Metals are good conductors of electricity
2. Why are the handles of metal cooking pans usually made of plastic or wood?
3. If silver is a better conductor of heat than copper and aluminium, why don't we use it for most pots and pans?
4. What is the specific role of plastics in an electrical circuit, and why is it necessary?
5. Most non-metals are characterized as insulators. However, which non-metal is a notable exception to this rule, and how is it utilized in everyday **technology**?
6. If you were to strike a piece of solid phosphorus with a hammer, what would likely happen, and what specific physical property does this demonstrate?
7. Non-metals are generally described as being soft and having low melting points. Identify the "record-breaking" exceptions to these two properties.
8. What distinguishes the electrical conductivity of metalloids, such as silicon and germanium, from that of typical metals?

1.3. Trends in reactivity for metals and non-metals

In chemistry, elements do not behave in the same way. Some react very quickly, while others react slowly or hardly at all. This difference is called **reactivity**.

When studying elements in the periodic table, we often look at how their properties change in an organized way. This regular pattern of change is called a **trend**. A **trend** helps us to predict how elements will behave without testing each one. In this lesson, you will learn about the trends in reactivity.

1.3.1 Trend in reactivity of metals



Learning activity 1.3.1

Investigating reactivity trends in group I and group II metals

- Take small pieces of lithium, sodium, and potassium metals (Group I metals) and place them separately in labeled test tubes.
- Take small pieces of magnesium and calcium metals (Group II metals) and place them separately in labeled test tubes.
- Add equal amounts of cold water to each test tube
- Observe the speed of reaction based on the amount of gas produced in test tube.
- Keep the test tubes in a safe place.

Guiding questions

- i) Compare the rate of gas production in the Group I metals.
- ii) Compare the rate of gas production in the Group II metals.
- iii) Arrange the metals in order of increasing reactivity based on the observations.

- Group I metals: Potassium reacts fastest, then sodium, then lithium.
- Group II metals (magnesium, calcium): Calcium reacts faster than magnesium. Magnesium reacts very slowly with hot water and reacts readily with steam; it reacts negligibly with cold water.

Order of increasing reactivity (from least to most reactive): Magnesium < Calcium < Lithium < Sodium < Potassium

The chemical reactivity of metals increases as you move down a group in the periodic table. As you go down a group of metals in the periodic table, their atoms lose electrons more easily. This happens because the outermost electron is farther from the nucleus, so it is less tightly held. As a result, these metals become more reactive. Below is the reactivity trend in group I metals.

	Group 1	
Lithium	Li	Least reactive
Sodium	Na	
Potassium	K	
Rubidium	Rb	
Cesium	Cs	
Francium	Fr	Most reactive

Chemical reactivity of metals increases on going down in a group

On moving from left to right in a period, the chemical reactivity of elements first decreases and then increases.

Na	Mg	Al	Si	P	S	Cl
Very reactive		Least reactive			Very reactive	
Chemical reactivity decreases				Chemical reactivity increases		

In the third period of the periodic table, the reactivity of metals decreases as their atomic size becomes smaller and they lose electrons less easily. For example, sodium is very reactive, magnesium is less reactive, and aluminium is even less reactive. Silicon is less reactive than sodium, magnesium, and aluminium in this comparison; the stated trend should not be extended to phosphorus, sulfur, and chlorine as a single simple reactivity scale. After silicon, the nonmetals show the opposite trend: phosphorus is quite reactive, sulfur is more reactive, and chlorine is very reactive. This happens because the atomic size decreases and the tendency to gain electrons increases. Therefore, in the third period, chemical reactivity first decreases from sodium to silicon, then increases from phosphorus to chlorine.

1.3.2. Trend in reactivity of non-metals

The chemical reactivity of non-metals decreases as we move down a group in the periodic table. For example, in Group 17 (the halogens), reactivity decreases from fluorine to iodine as shown below. As we move down a group of non-metals, their

tendency to gain electrons decreases. This reduction in electron-attracting ability causes their chemical reactivity to decrease as well.

Group 17		
Fluorine	F	Most reactive Chemical reactivity of non-metals decreases on going down in a group Least reactive
Chlorine	Cl	
Bromine	Br	
Iodine	I	



Application activity 1.3

1. What is the general trend for metal reactivity as you move down a group in the periodic table?
2. What happens to the reactivity of elements as you move from left to right across a period?
3. Which is the least reactive metal in the third period?
4. Why does reactivity increase as you move across the right side of a period?

1.4. Chemical properties of metals

Metals are substances that readily lose electrons to form positive ions, making them highly reactive in many chemical processes. Their chemical behavior is largely determined by how easily they react with substances such as oxygen, water, and acids. In general, metals tend to form basic oxides, produce hydrogen gas when reacting with acids, and show different levels of reactivity depending on their position in the reactivity series. Understanding these properties helps explain everyday phenomena such as corrosion, combustion, and metal extraction.

1.4.1. Reaction of metals with water



Learning activity 1.4.1

Illustrating the reaction of metals with water

Apparatus and reagents: Sodium, potassium, calcium, aluminium, iron, magnesium, zinc, copper, beakers, water, Bunsen burner, test tube, retort stand, cork, delivery tube, beehive shelf, trough

Procedure:

Caution: Do not touch sodium and potassium with your bare hands. They cause severe burns.

- Collect samples of sodium, potassium, calcium, aluminium, iron, magnesium, zinc, and copper.
- Put small pieces of samples separately in beakers half-filled with cold water.
- Observe which metals react with cold water.

Did any metal produce fire in water?

Did any metal start floating after some time?

- Put the metals that do not react with cold water in beakers filled with hot water.
- Observe which metals react with hot water.
- For metals that did not react with hot water, arrange the apparatus as shown in the figure.

Experiment set up:

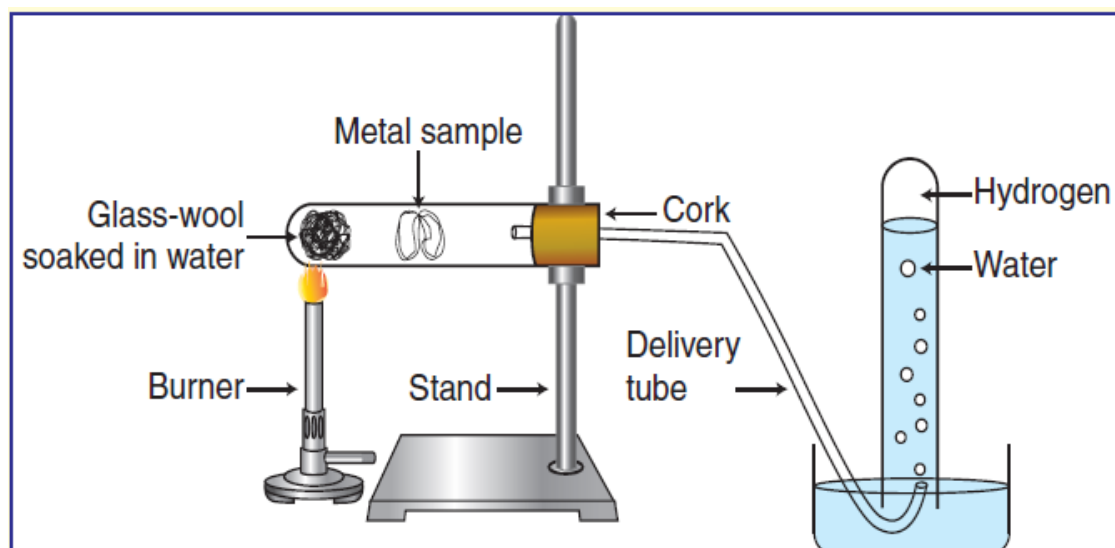


Fig.1.4.1: Reaction of metals with steam

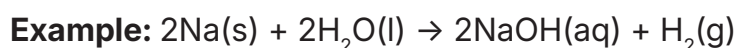
- Observe which metals react with steam.

Did any metal not react even with steam?

- Interpret all your observations.

Not all metals are equally reactive. Metals react with water to form a metal hydroxide and hydrogen gas, but the extent of the reaction varies. Some metals, such as sodium, potassium, and calcium, react with cold water. Others, like magnesium, react only with hot water. Metals such as aluminium, iron, and zinc do not react with either cold or hot water; instead, they react with steam to form a metal oxide and hydrogen gas. Some metals, including copper, gold, silver, and mercury, do not react even with steam. When a metal reacts with cold or hot water, the products formed are a metal hydroxide and hydrogen gas.

Metal + Water → Metal hydroxide + Hydrogen



When metals such as magnesium, aluminium, zinc, and iron react with steam (water gas), the products formed are metal oxides and hydrogen gas.

Metal + Steam → Metal oxide + Hydrogen



Note: Metal oxides are basic in nature. Their solutions in water turn red litmus paper blue. Some metal oxides react with water to form alkali. Some of them, such as ZnO and PbO, are amphoteric

Below are examples of the reactivity of selected metals with cold water, hot water, and steam

Table 1.4.1: Reactivity of selected metals with cold water, hot water, and steam

Metal	Type of Water / Steam	Chemical Equation	Observation / Notes
Sodium (Na)	Cold water	$2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)} + \text{Heat (Exothermic)}$	Very vigorous reaction. Heat causes hydrogen to catch fire. Sodium is very reactive .
Potassium (K)	Cold water	$2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)} + \text{Heat (Exothermic)}$	Very violent reaction. Hydrogen catches fire immediately. Potassium is more reactive than sodium .
Calcium (Ca)	Cold water	$\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$	Gentle reaction. Metal floats due to hydrogen bubbles. Calcium is less reactive than sodium .
Magnesium (Mg)	Cold water	No reaction	Magnesium does not react with cold water.
	Hot water	$\text{Mg(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$	Reaction occurs. Magnesium floats due to hydrogen bubbles.
	Steam	$\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$	Very fast reaction with steam.

Metal	Type of Water / Steam	Chemical Equation	Observation / Notes
Aluminium (Al)	Steam	$2\text{Al(s)} + 3\text{H}_2\text{O(g)} \rightarrow \text{Al}_2\text{O}_3\text{(s)} + 3\text{H}_2\text{(g)}$	Reacts only with steam, not cold or hot water.
Zinc (Zn)	Steam	$\text{Zn(s)} + \text{H}_2\text{O(g)} \rightarrow \text{ZnO(s)} + \text{H}_2\text{(g)}$	Reacts with steam only.
Iron (Fe)	Steam	$3\text{Fe(s)} + 4\text{H}_2\text{O(g)} \rightarrow \text{Fe}_3\text{O}_4\text{(s)} + 4\text{H}_2\text{(g)}$	Reacts with steam only.
Copper (Cu)	Cold/Hot water or Steam	No reaction	Copper does not react with water or steam. Very unreactive.

From the table, the reactivity trend of metals becomes clear as follows:

Across a period (left to right in the Periodic Table): Metal reactivity generally decreases from left to right. Highly reactive metals like sodium (Na) and potassium (K) react very vigorously with cold water, while metals such as copper (Cu) do not react at all with water or steam. This shows that metals on the left side of a period are more reactive than those on the right.

Down a family (group): Metal reactivity generally increases down a group. For example, potassium (K) is more reactive than sodium (Na), as potassium reacts more violently with water. This happens because atoms become larger down the group and lose their outer electrons more easily.

1.4.2. Reaction of metals with acids



Learning activity 1.4.2

Illustrating the reaction of metals with dilute acids

Caution:

- **Do not touch** dilute hydrochloric acid (HCl) or sulphuric acid (H₂SO₄) with bare/uncovered hands.
- Handle all acids carefully and use **eye protection**.

Apparatus and chemicals

Small pieces of magnesium, aluminium, zinc, copper, and iron, test tubes, sandpaper, dilute hydrochloric acid, and dilute sulphuric acid.



Procedure:

1. Clean the metal samples using sandpaper.
2. Place each piece of metal in a separate test tube.
3. Add 10 ml of dilute hydrochloric acid to each test tube.
4. Observe carefully the reaction.
 - Which metals reacted with dilute hydrochloric acid?
5. Repeat steps 2–4 using dilute sulphuric acid.
 - Did any metal not react with dilute sulphuric acid?
 - Record your observations and interpret them.

In Activity 1.4.2, you must have observed that magnesium produced hydrogen bubbles at the fastest rate. The reactivity of the metals decreases in the following order: $\text{Mg} > \text{Al} > \text{Zn} > \text{Fe}$. In the case of copper, no bubbles were observed, indicating that copper does not react with dilute hydrochloric acid. However, copper does not react with dilute sulfuric acid under ordinary conditions.

In general, metals react with dilute acids to form a metal salt and hydrogen gas. The intensity of the reaction varies: some metals react violently, while others react rapidly or slowly. For example, sodium reacts violently with dilute acids, whereas magnesium reacts rapidly. Aluminium, iron, and zinc react slowly with dilute acids, while gold and silver do not react at all.

When a metal reacts with dilute hydrochloric acid, the products formed are metal chlorides and hydrogen gas.

It is not advisable to react Group I metals with acids because the reaction is extremely vigorous and dangerous. It produces hydrogen gas rapidly, which can easily ignite or cause an explosion.

Metal + Hydrochloric acid (dilute) $\rightarrow$ Metal chloride (salt) + Hydrogen gas

Example: $\text{Mg(s)} + 2\text{HCl(dil)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

When a metal reacts with dilute sulphuric acid, the products formed are metal sulphate and hydrogen.

Metal + Sulphuric acid (dilute) $\rightarrow$ Metal sulphate (salt) + Hydrogen gas

Example: $2\text{Al(s)} + 3\text{H}_2\text{SO}_4\text{(l)} \rightarrow \text{Al}_2(\text{SO}_4)_3\text{(aq)} + 3\text{H}_2\text{(g)}$

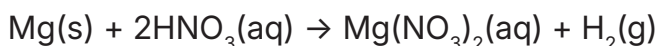
Below are examples of the reactivity of selected metals with acids

Table 1.4.2: Reactivity of selected metals with acids

Metal	Acid (Dilute)	Chemical Equation	Observation / Notes
Sodium (Na)	Hydrochloric acid (HCl)	$2\text{Na(s)} + 2\text{HCl(aq)} \rightarrow 2\text{NaCl(aq)} + \text{H}_2\text{(g)}$	Very violent reaction. Hydrogen gas is rapidly produced.
	Sulphuric acid (H_2SO_4)	$2\text{Na(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{Na}_2\text{SO}_4\text{(aq)} + \text{H}_2\text{(g)}$	Very vigorous reaction. Hydrogen gas evolved.
Magnesium (Mg)	Hydrochloric acid (HCl)	$\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$	Fast reaction with effervescence (bubbling).
	Sulphuric acid (H_2SO_4)	$\text{Mg(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)}$	Fast reaction producing hydrogen gas.
Aluminium (Al)	Hydrochloric acid (HCl)	$2\text{Al(s)} + 6\text{HCl(aq)} \rightarrow 2\text{AlCl}_3\text{(aq)} + 3\text{H}_2\text{(g)}$	Steady reaction after the oxide layer is removed.
	Sulphuric acid (H_2SO_4)	$2\text{Al(s)} + 3\text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{Al}_2(\text{SO}_4)_3\text{(aq)} + 3\text{H}_2\text{(g)}$	Moderate reaction producing hydrogen gas.
Zinc (Zn)	Hydrochloric acid (HCl)	$\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$	Moderate, steady effervescence.
	Sulphuric acid (H_2SO_4)	$\text{Zn(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{H}_2\text{(g)}$	Moderate reaction producing hydrogen gas.
Iron (Fe)	Hydrochloric acid (HCl)	$\text{Fe(s)} + 2\text{HCl(aq)} \rightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$	Slow reaction with gentle bubbles.
	Sulphuric acid (H_2SO_4)	$\text{Fe(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{H}_2\text{(g)}$	Slow reaction producing hydrogen gas.
Copper (Cu)	Hydrochloric acid (HCl)	$\text{Cu(s)} + \text{HCl(aq)} \rightarrow$ No reaction	No visible reaction. Copper does not react.
	Sulphuric acid (H_2SO_4)	$\text{Cu(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow$ No reaction	No visible reaction. Copper does not react.

Note: Hydrogen gas is generally not evolved when copper reacts with nitric acid because nitric acid is an oxidising acid; nitrogen oxides are formed instead. This is because nitric acid does not allow hydrogen gas to be released. Instead, any hydrogen that begins to form is quickly converted into water (H_2O). At the same time, nitric acid is converted into nitrogen oxide gases such as NO , NO_2 , or N_2O .

However, Magnesium reacts with very dilute nitric acid to form magnesium nitrate and hydrogen gas under appropriate conditions:



1.4.3. Reaction of metals with halogens

Metals react with halogens to form ionic halides. Metal halides are usually solid and conduct electricity in solution. Table 1.4.3 below shows examples of reactions of selected metals with chlorine

Table 1.4.3: Reactivity of selected metals with chlorine

Metal	Chemical Equation	Product Formed
Sodium (Na)	$2\text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NaCl(s)}$	Sodium chloride (NaCl).
Calcium (Ca)	$\text{Ca(s)} + \text{Cl}_2\text{(g)} \rightarrow \text{CaCl}_2\text{(s)}$	Calcium chloride (CaCl ₂).
Aluminium (Al)	$2\text{Al(s)} + 3\text{Cl}_2\text{(g)} \rightarrow 2\text{AlCl}_3\text{(s)}$	Aluminium chloride (AlCl ₃).
Iron (Fe)	$2\text{Fe(s)} + 3\text{Cl}_2\text{(g)} \rightarrow 2\text{FeCl}_3\text{(s)}$	Iron chloride (FeCl ₃).
Copper (Cu)	$2\text{Cu(s)} + \text{Cl}_2\text{(g)} \rightarrow \text{CuCl}_2\text{(s)}$	Copper(II) chloride (CuCl ₂).
Zinc (Zn)	$\text{Zn(s)} + \text{Cl}_2\text{(g)} \rightarrow \text{ZnCl}_2\text{(s)}$	Zinc chloride (ZnCl ₂).

1.4.4 Reaction of metals with oxygen



Learning activity 1.4.4

Illustrating the reaction of metals with oxygen

Materials:

Small pieces of potassium, sodium, magnesium, aluminium, zinc, copper, iron, and some iron filings.

Safety:

- Handle metals with tongs and avoid direct contact.

Procedure:

1. Using tongs, hold a small piece of potassium over a flame and note observations.
2. Repeat the process with the other metals.
3. If a product forms, carefully collect it.
4. Allow the metal and the products to cool down.
5. Try to dissolve in distilled water each product formed and test the solution by using blue litmus paper. Note your observation and give equations of reactions that occur.

Guiding questions:

- Which metals burn easily?
- How does the metal surface look after burning?
- Are the products soluble in water?
- Interpret your observations

Almost all metals react with oxygen to form metal oxides, but they do not all react at the same rate. Alkali metals such as potassium and sodium are highly reactive and can catch fire when exposed to air, which is why they are stored under kerosene oil to prevent accidents. At ordinary temperatures, metals like magnesium, aluminium, zinc, and lead develop a thin protective oxide layer on their surface, which prevents further oxidation. Iron does not burn easily, but iron filings burn vigorously when sprinkled into a flame. Copper does not catch fire, yet when heated, its surface turns black due to the formation of copper (II) oxide. Silver and gold are very unreactive and do not react with oxygen even at high temperatures. Table 1.4.4(a) and (b) below shows examples of reactions of selected metals with oxygen and reactions of metal oxides with water.

Table 1.4.4. a: Reactivity of selected metals with oxygen

Metal	Reactant (Gas)	Product	Balanced Equation
Sodium	Oxygen (O ₂)	Sodium oxide	$4\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Na}_2\text{O(s)}$
Potassium	Oxygen (O ₂)	Potassium oxide	$4\text{K(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{K}_2\text{O(s)}$
Magnesium	Oxygen (O ₂)	Magnesium oxide	$2\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$

The red litmus paper turns blue the metal oxide solution. Therefore, the concerned metal oxides are alkaline.

Table 1.4.4.b: Reactivity of selected metal oxides with water

Metal	Product formed	Solubility in water
Potassium	Potassium oxide (K ₂ O)	Very soluble → forms KOH (soluble)
Sodium	Sodium oxide (Na ₂ O)	Very soluble → NaOH (soluble)
Magnesium	Magnesium oxide (MgO)	Slightly soluble
Aluminium	Aluminium oxide (Al ₂ O ₃)	Insoluble
Zinc	Zinc oxide (ZnO)	Insoluble in cold water
Iron	Iron(III) oxide (Fe ₂ O ₃)	Insoluble
Copper	Copper oxide (CuO / Cu ₂ O)	Insoluble

Most easily burned metals: Potassium, Sodium, Magnesium

Surface after burning: Metals generally form a metal oxide, which can be powdery or coating

Solubility: Alkali metal oxides are soluble, forming hydroxides; others are mostly insoluble

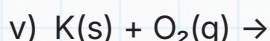
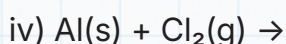
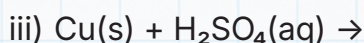
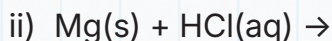
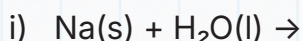
How do metals look like after heating

Metal	Observation when heated in air	Residue / Product formed
Potassium	Burns quickly with lilac flame	White powder (potassium oxide)
Sodium	Burns quickly with a yellow flame	White powder (sodium oxide)
Magnesium	Burns with bright white flame	White powder (magnesium oxide)
Aluminium	Burns when powdered or in thin ribbon	White powder (aluminium oxide)
Zinc	Burns with bluish-green flame at high temp	White coating (zinc oxide)
Iron	Red-hot, may form sparks	Reddish-brown powder (iron(III) oxide)



Application activity 1.4

1. Complete and balance the following equations of reactions



2. Which metals do not react with dilute acids?

3. How can you test if a given metal oxide is basic or acidic?

<https://flexbooks.ck12.org/cbook/ck-12-cbse-science-class-7-as-per-nep-2020/section/4.2/primary/lesson/chemical-properties-of-metal-and-non-metals/>

1.5. Chemical properties of non-metals

While non-metals are defined by their physical brittleness and lack of luster, their true diversity is revealed through their **chemical properties**. Unlike metals, which tend to lose electrons, non-metals are “electron seekers” that gain or share electrons to achieve stability. This fundamental drive governs how they react with oxygen, water, and other elements.

1.5.1 Reaction of non-metals with halogens (chlorine)



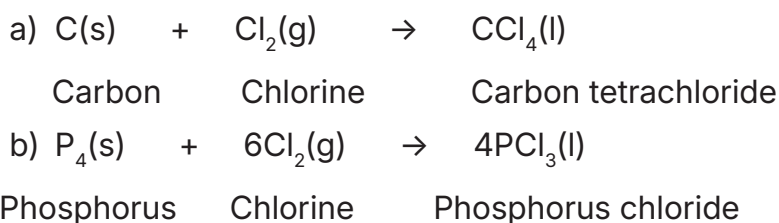
Learning activity 1.5.1

Describe the reactivity of non-metals with halogens and the physical states of the products formed.

Non-metals react with halogens to form covalent halides (chlorides, bromides, iodides, fluorides).

- Fluorides are predominantly gases
- Chlorides are gases or liquids
- Bromides are liquids or solids
- These chlorides are usually liquids.
- Iodides are predominantly solids

Examples:



1.5.2 Reaction of non-metals with oxygen



Learning activity 1.5.2

Reaction of non-metals with oxygen

Materials:

- Sulphur
- Carbon (charcoal)
- Bunsen burner
- Limewater
- Test tubes

Procedure:

1. Burn a small amount of carbon, **sulphur** in the air
(harmful fumes are produced; use a fume cupboard or perform in open air).
2. Collect the gas produced and dissolve it in water.
3. Test the solution with **blue litmus paper**.
4. Burn charcoal and pass the gas into **limewater**.

Record your observations and state the conclusion

Carbon burns slowly in air with a glowing flame to produce a colourless gas that turns limewater milky. This colourless gas is carbon dioxide.

Equation of the reaction is: $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ (carbon dioxide: acidic oxide)

Sulphur burns in air with a blue flame and produces a choking gas with a pungent smell (sulphur dioxide).

Equation of the reaction is: $\text{S(s)} + \text{O}_2 \rightarrow \text{SO}_2\text{(g)}$ (sulphur dioxide: acidic oxide)

The gases (CO_2 and SO_2) dissolve in water to form acidic solutions as these solutions turn blue litmus paper red.

Non-metals react with oxygen to form either acidic oxides or neutral oxides.



Application activity 1.5

1. What type of compounds are formed when non-metals react with chlorine?
2. What observations are made when sulphur burns in air?
3. What happens when carbon burns in air?
4. What happens when carbon dioxide or sulphur dioxide dissolves in water?
5. What is the general conclusion about the reaction of non-metals with oxygen?
6. Why do non-metal oxides turn blue litmus red?

1.6. Comparison among the physical and chemical properties of metals and non-metals



Learning activity 1.6

We have studied the physical and chemical properties of metals and non-metals. What are the main differences between them?

The main differences between physical and chemical properties of metals and non-metals are summarized in the tables below:

Table 1.6. a) Differences between the physical properties of metals and non-metals

Metals	Non-Metals
1. Malleable and ductile: Metals can be hammered into thin sheets or drawn into wires.	Brittles: Non-metal breaks easily; neither malleable nor ductile.

Metals	Non-Metals
2. Good conductors of heat and electricity.	Poor conductors of heat and electricity (except graphite, which conducts electricity).
3. Lustrous (shiny) and can be polished.	Non-lustrous (dull) and cannot be polished (except iodine, which is lustrous).
4. Solid at room temperature (except mercury, which is liquid).	Can be solid, liquid, or gas at room temperature.
5. Strong and tough.	Neither strong nor tough.
6. Sonorous – produce a ringing sound when struck.	Non-sonorous – do not produce a ringing sound.

Table 1.6. b) Differences between the chemical properties of metals and non-metals

Metals	Non-Metals
1. Form basic, amphoteric, or neutral oxides, depending on the metal	Form acidic or neutral oxides.
2. Displace hydrogen from water or steam.	Do not react with water or steam; it cannot displace hydrogen.
3. Displace hydrogen from dilute acids.	Do not react with dilute acids; cannot displace hydrogen.
4. Form ionic chlorides with chlorine.	Form covalent chlorides with chlorine.
5. Usually, do not combine with hydrogen; only a few reactive metals form ionic hydrides.	React with hydrogen to form stable covalent hydrides.



Application activity 1.6

A) Multiple Choice Questions (MCQs)

- Which of the following is a property of metals?
 - Brittle
 - Poor conductor of electricity
 - Ductile
 - Low melting point
- Which non-metal is a good conductor of electricity?
 - Sulphur
 - Carbon (graphite)
 - Chlorine
 - Nitrogen

3. The oxides of non-metals are generally:

- | | |
|---------------|------------|
| a) Basic | c) Acidic |
| b) Amphoteric | d) Neutral |

4. Which pair correctly matches a substance with its type?

- | | |
|------------------------|----------------------|
| a) Iron – Non-metal | c) Aluminium – Metal |
| b) Sulphur – Non-metal | d) Chlorine – Metal |

B) True or False Questions

5. Metals are malleable while non-metals are brittle.
6. Non-metals usually gain electrons during chemical reactions.
7. Metals do not react with dilute acids.
8. Non-metals form basic oxides.

C) Short Answer Questions

9. State two physical differences between metals and non-metals.
10. Give one chemical difference between metals and non-metals.
11. Why are metals good conductors of electricity?
12. State one exception to the statement *"All metals are solids at room temperature."*

<https://flexbooks.ck12.org/cbook/ck-12-cbse-science-class-7-as-per-nep-2020/section/4.2/primary/lesson/chemical-properties-of-metal-and-non-metals/>

1.7. Uses of metals and non-metals

Metals and non-metals are both essential in everyday life. Metals are mainly used where strength, conductivity, and durability are required, while non-metals are important in respiration, agriculture, sanitation, fuels, and chemical industries.



Learning activity 1.7

What are the main uses of metals and non-metals?

The main uses of metals and non-metals are summarized in the tables below:

Table 1.7: Uses of metals and non-metals

Uses of Metals	Uses of Non-metals	Uses of Metals	Uses of Non-metals
- Aluminium is extensively used for making airplanes and automobile engines. - Sodium is used in nuclear reactors. - Tin is used for coating iron containers for packaging food. - Mercury is used in thermometers. - Steel is used in making machines, buildings, vehicles, ships, tanks, and other structures.		Hydrogen and carbon are the essential constituents of the cells of all living beings. Without these elements, life is not possible. Furthermore, carbon in the form of coal is used for generating heat energy for houses as well as factories.	
Gold and silver are used for making ornaments.		- Nitrogen in the form of fertilisers is essential for the growth and development of plants. - Urea and calcium ammonium nitrate are common fertilisers used by the farmers. - Oxygen is absolutely essential for the respiration of all kinds of plants and animals. It is also used in the manufacture of steel and for artificial respiration on high altitudes, in space and deep sea diving. - Silicon is used for making microchips used in making all kinds of electronic goods. - Phosphorus is used in match industry and fertilisers. - Sulphur is used for making fire crackers, gun powder and sulphuric acid. - Chlorine is used in the disinfection of drinking water. - Iodine dissolved in alcohol [tincture iodine] is antiseptic in nature and is used in dressing wounds.	

Uses of Metals	Uses of Non-metals
	Helium is used in balloons. Argon is used in electrical bulbs and fluorescent tubes.



Application activity 1.7

1. State True or False

- | | |
|---|---|
| a) Mercury is used in thermometers | d) (Phosphorus is used in fertilisers |
| b) Gold and Silver are used for making electric wires | e) Chlorine is not used in the disinfection of drinking water |
| c) Helium is used in balloons | |

2. Gap filling:

- Sulphur is used for making _____
- Silicon is used for making _____
- Sodium is used in _____ reactors

3. Name the noble gas used in electric bulbs

4. Which element is used for coating iron containers for packaging food?

5. Name two metals used for making machines



1.8 End of unit assessment

A) Knowledge & Understanding

- Which of the following metals exists in the liquid state?

a) Sodium	b) Silver	c) Mercury	d) Neon
-----------	-----------	------------	---------
- When moving down a group, the metallic character of elements.....

a) increases	b) decreases	c) both	a) and b)	d) neither a) nor b)
--------------	--------------	---------	-----------	----------------------
- Silicon and germanium are examples of.....

a) metals	b) metalloids	c) non-metals	d) none of these
-----------	---------------	---------------	------------------
- Which of the following statement(s) is/are not correct.
 - The hardest substance known on earth is diamond.
 - Graphite – a non-metal – conducts electricity.



- c) The melting points of gallium and caesium are very high.
d) Brittleness is the characteristic property of a non-metal.
5. All non-metals are solids and gases except:
a) fluorine b) chlorine c) bromine d) iodine
6. Which metal is the best conductor of electricity?
a) Silver b) Copper c) Aluminium d) Iron
7. Name the gas produced when metals react with dilute acids.
a) Oxygen b) Hydrogen c) Nitrogen d) Chlorine
8. The formula of manganese nitrate is
a) $\text{Mn}(\text{NO}_3)_2$ b) $\text{Mn}_2(\text{NO}_3)$ c) $\text{Mn}_3(\text{NO})_2$ d) MnNO
9. Carbon reacts with oxygen to give
a) carbon dioxide b) carbon monoxide
c) both (a) and (b) d) none of these
10. What is meant by terms malleable and ductile? Explain with examples.

B) Skills

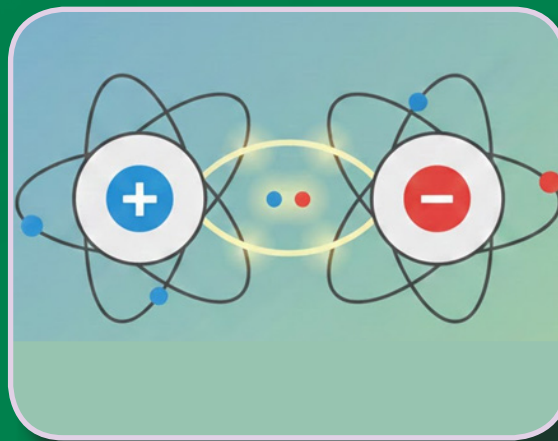
11. Complete and balance chemical equations:
i) $\text{K} + \text{H}_2\text{O} \rightarrow$
ii) $\text{Mg} + \text{steam} \rightarrow$
iii) $\text{Mg} + \text{water} \rightarrow$
iv) $\text{Ca} + \text{H}_2\text{SO}_4 \rightarrow$
v) $\text{Fe} + \text{Cl}_2 \rightarrow$
vi) $\text{P} + ? \rightarrow \text{PCl}_3$
12. Describe the electrical conductivity of metals and non-metals
13. Explain why copper does not react with water

C) Attitudes and Values

14. During a practical activity on the reactivity of metals with dilute acid, answer the following:
a) List two safety rules you followed in the laboratory.
b) Explain why it is important to follow these procedures.

UNIT 2:

CHEMICAL BONDING



Key unit competence

Relate the nature of bonding to the properties of substances.

Learning Objectives

By the end of the unit, you should be able to:

Knowledge and understanding

- Explain the nature of ionic, covalent, and metallic bonding.
- State the physical properties of ionic compounds and those of covalent compounds.
- Explain the physical properties of metals in terms of their structures.

Skills

By the end of the unit, you should be able to:

- Show the formation of bonds using dot and cross diagrams.
- Classify various chemical compounds as ionic or covalent.
- Use experiments to show the physical properties of metals, ionic, and covalent compounds.

Values and attitudes

- Develop a sense of orderliness and self-confidence in presentations of results.
- Respect the procedures while carrying out experiments.

Introduction

Atoms join together (bond) to become stable, like rare gases. Most atoms do not feel comfortable with their outer shells partly filled, so they share or transfer electrons to achieve a full outer shell. By bonding, atoms form molecules or compounds that make up everything around us, such as water, salt, and air.



Introductory activity

1. How do atoms get stability?
2. What happens when atom gains or loses electron(s)?
3. Which types of chemical bonds should be formed?

2.1. Ionic bonding

Atoms rarely exist alone in nature, they are almost always connected to other atoms. The way they connect depends on how they find stability. One of the most important ways is through **ionic bonding**.

2.1.1 Formation of ions from atoms



Learning activity 2.1.1

1. Write electronic configuration of the following atoms:
a) Cl ($Z=17$); (b) Ne ($Z=10$), (c) Mg ($Z=12$)
2. How many electron(s) in outermost shell for each atom above?
3. Explain instability of atoms, to describe the octet rule and predict if atom will gain or lose electrons.

Atoms are most stable when their outermost shell (valence shell) is full of electrons. A full outer shell usually contains 2 electrons for the first shell or 8 electrons for other shells. This is called a stable electronic configuration.

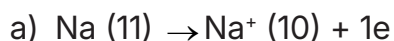
Atoms that do not have a full outer shell are unstable. To become stable, such atoms tend to gain, lose, or share electrons. **Formation of ions from atoms is the result of gaining, losing electron(s).**

Atoms with 1, 2, or 3 valence electrons usually lose 1, 2, or 3 electrons respectively to form positive ions. In this case the number of protons which are positively charged is superior of that of electrons which are negatively charged and atom becomes positive charged (positive ion called cation)

Metal ions carry positive by losing one or more electrons in their outermost shell.

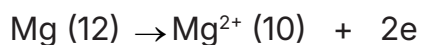
Examples:

a) Formation of sodium ion



2, 8, 1 2, 8

b) Formation of magnesium ion



2, 8, 2 2, 8

c) Formation of aluminium ion



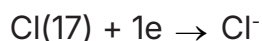
2, 8, 3 2, 8

Atoms with **5, 6, or 7** valence electrons usually gain electrons to form negative ions.

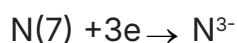
Non-metal ions usually carry negative charges by gaining one or more electrons in their outermost shell.

Examples:

Formation of chloride ion

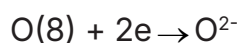


a) Formation of nitrogen ion



2, 5 2, 8

b) Formation of oxygen ion



2, 6 2, 8

Atoms with 4 valence electrons usually share electrons with other atoms.

The Octet Rule

The octet rule states that atoms tend to gain, lose, or share electrons in order to have eight electrons in their outermost shell

Having eight electrons in the outer shell makes an atom stable.

Exception

The first shell is stable with 2 electrons (for example, hydrogen and helium).

Notice that when a non-metal forms an anion, its name changes slightly. For example, Chlorine forms a chloride ion and Oxygen forms an oxide ion.

Polyatomic ions can exist in both negative and positive forms. However, NH_4^+ (ammonium) is the only positive polyatomic ion that is not metallic. Therefore,

Ammonium compounds are ionic compounds that contain the polyatomic NH_4^+ ion instead of a metal cation

Examples of polyatomic ions (radicals):

- Nitrate: NO_3^-
- Carbonate: CO_3^{2-}
- Phosphate: PO_4^{3-}
- Sulfate: SO_4^{2-}
- Hydroxide: OH^-

2.1.2. Formation of ionic bonding

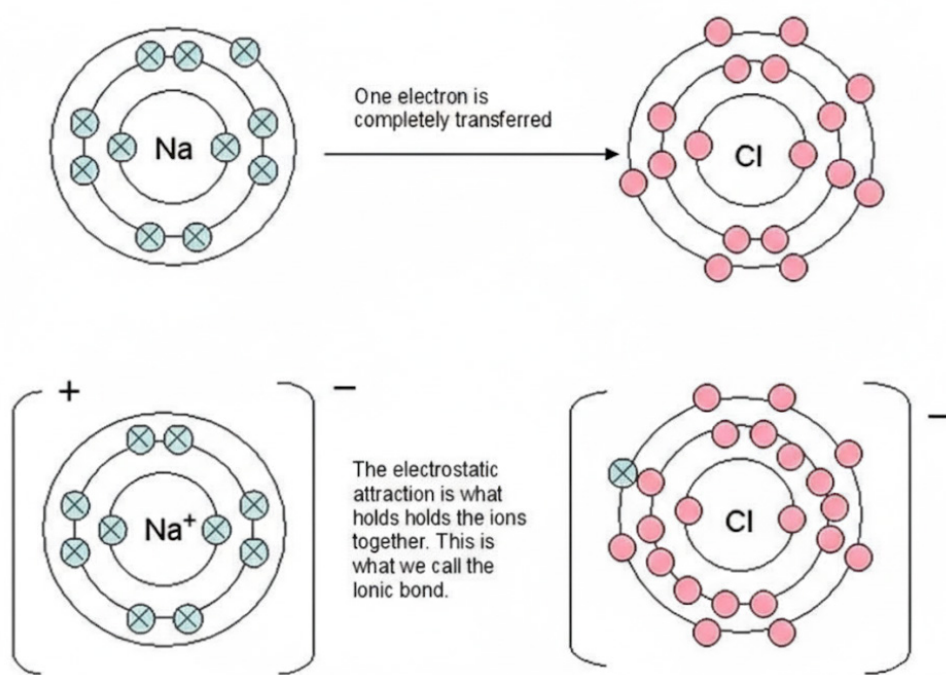


Learning activity 2.1.2

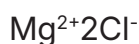
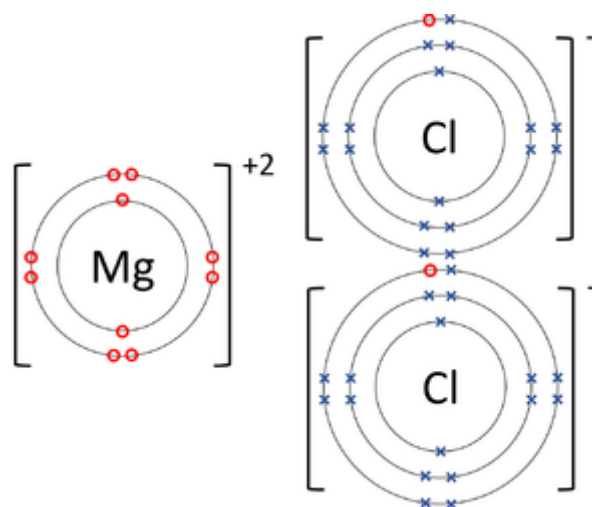
1. Why do metals tend to form positive ions while non-metals form negative ions?
2. What would happen when a positively charged particle meets a negatively charged particle?
3. What do you think holds a sodium ion (Na^+) and a chloride ion (Cl^-) together in table salt (NaCl)?

An ionic bond is a type of chemical bond formed by the electrostatic force between oppositely charged ions, usually a positive ion (cation) and a negative ion (anion), after electrons are transferred from one atom to another.

For example, in table salt (NaCl), sodium ions (Na^+) and chloride ions (Cl^-) are held together by ionic bonds. $\text{Na} \rightarrow \text{Na}^+$, $\text{Cl} + 1\text{e}^- \rightarrow \text{Cl}^- \Rightarrow \text{Na}^+ \text{Cl}^-$



- a) When a magnesium atom comes in contact with chlorine gas, it forms magnesium chloride (MgCl_2) as shown below:



Formation of magnesium chloride (MgCl_2)

In summary:

An ionic bond is usually formed between a metal atom and a non-metal atom. The metal atom loses electrons and becomes a positive ion (cation). The non-metal atom gains those electrons and becomes a negative ion (anion). The attraction between these opposite charges forms the ionic bond.

Some common ionic compounds, their formulae, and the ions present in them are shown in Table 2.1.2.

Table 2.1.2: Formulae and nomenclature of some ionic compounds

Nomenclature	Formula	Ions Present
Calcium oxide	CaO	Ca^{2+} and O^{2-}
Copper sulphate	CuSO_4	Cu^{2+} and SO_4^{2-}
Magnesium chloride	MgCl_2	Mg^{2+} and 2Cl^-
Potassium chloride	KCl	K^+ and Cl^-
Potassium hydroxide	KOH	K^+ and OH^-
Sodium carbonate	Na_2CO_3	2Na^+ and CO_3^{2-}
Sodium hydroxide	NaOH	Na^+ and OH^-

Exercise: Complete the table below:

Nomenclature	Formula	Ions Present
Aluminium oxide		
Ammonium chloride		

Nomenclature	Formula	Ions Present
Calcium hydroxide		
Calcium nitrate		

Answers

Nomenclature	Formula	Ions Present
Aluminium oxide	Al_2O_3	2Al^{3+} and 3O^{2-}
Ammonium chloride	NH_4Cl	NH_4^+ and Cl^-
Calcium hydroxide	$\text{Ca}(\text{OH})_2$	Ca^{2+} and 2OH^-
Calcium nitrate	$\text{Ca}(\text{NO}_3)_2$	Ca^{2+} and 2NO_3^-

2.1.3 Properties of ionic compounds



Learning activity 2.1.3

Illustrating physical properties of ionic compounds

1. Experiment to show electrical conductivity of solid ionic compounds

Materials

Solid sodium chloride, Battery, Bulb, Wires and electrodes

Procedure

1. Set up a simple electrical circuit with a battery and bulb.
2. Place solid sodium chloride between the two electrodes to complete the circuit.
3. Observe the bulb.

Note your observations and give a conclusion

Observation

- The bulb does not light.

Conclusion

- Solid ionic compounds do not conduct electricity because their ions are fixed in position and cannot move freely.

2. Experiment to show electrical conductivity of ionic compound solutions

Materials

Sodium chloride, Water, Beaker, Battery, bulb, wires, and electrodes.

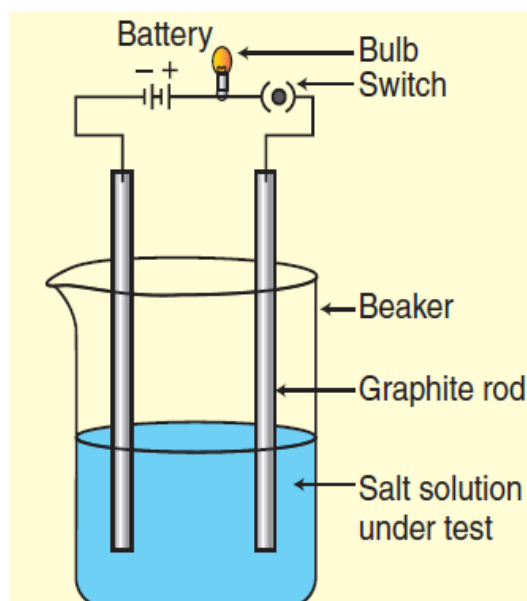
Procedure



Procedure

1. Dissolve sodium chloride in water, in a beaker.
2. Insert the electrodes into the solution.
3. Connect the electrodes to the electrical circuit.
4. Observe the bulb.

Experiment set up:



Note your observations and give a conclusion.

Observation

The bulb lights.

Conclusion

Ionic compounds conduct electricity in solution because the ions are free to move.

3. Experiment to show solubility in water of ionic compounds

Materials

Sodium chloride, Water, Petrol, Kerosene, Beaker, Spoon or stirrer

Procedure

1. Pour water into a beaker 1, petrol in beaker 2 and kerosene in beaker 3.
2. Add some sodium chloride in each beaker.
3. Stir the mixture.

Note your observations and give a conclusion



4. Experiment to illustrate melting and physical states of ionic compounds

Materials:

Sodium chloride (NaCl) or any other salt, Beakers, Bunsen burner, Metal spatula, Matchbox

Procedure:

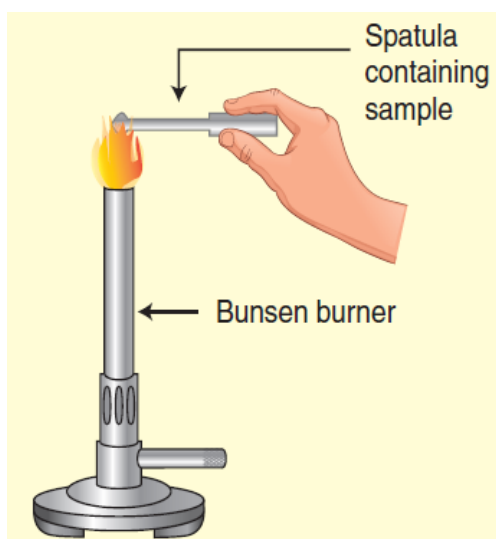
1. Observation of physical state

- Take a small sample of sodium chloride.
- Note its physical state (solid, liquid, or gas) at room temperature.

2. Testing melting point

- Place a small amount of the sample on a metal spatula.
- Heat it directly over a flame as shown in **Figure below**:

Does the salt melt?



Note your observations and give conclusion.

From the activity, you may have observed the following properties of ionic compounds:

- **Physical State:** Ionic compounds are usually crystalline solids at room temperature.
- **Electrical conductivity:** Solid ionic compounds do not conduct electricity because their ions are fixed in position and cannot move freely. But ionic compounds conduct electricity in solution because the ions are free to move.
- **Solubility:** Ionic compounds are generally soluble in water but insoluble in organic solvents like petrol and kerosene.
- **Melting and Boiling Points:** Ionic compounds have high melting and boiling points, They do not melt easily when heated.

- **Electrical Conductivity:** Ionic compounds conduct electricity when dissolved in water or when melted. When an ionic solid dissolves in water or melts, its crystalline structure breaks down, producing free-moving ions. These ions carry an electric charge, allowing the compound to conduct electricity.

Melting and boiling points of some ionic compounds are presented in Table below

Table 2.1.3: Melting and boiling points of some ionic compounds

Ionic compound	Melting point (K)	Boiling point (K)
NaCl	1074	1686
LiCl	887	1600
CaCl ₂	1045	1900
CaO	2850	3120
MgCl ₂	981	1685



Application activity 2.1

1. Which ions are probably formed from the following atoms F(9), Be(4), K (19), S(16), P(15), Mg (12), Ar (18)?
2. Give two examples of ionic compounds. Write their chemical formulae.
3. The overall charge on an ionic compound is zero. (True or False)
4. Name the ions present in calcium nitrate.
5. Ionic compounds are made up of a _____ and a _____.
6. Give an example of an ionic compound made up of only non-metals.
7. With the help of dot and cross, show the formation of CaCl₂.
8. Why do ionic compounds conduct electricity when dissolved in water?

2.2. Covalent bonding

In an ionic bond, a metal gives up an electron to a non-metal. But what happens when two atoms meet and neither one is willing to let go its electrons? This usually happens between two non-metals.

2.2.1. Formation of a covalent bond



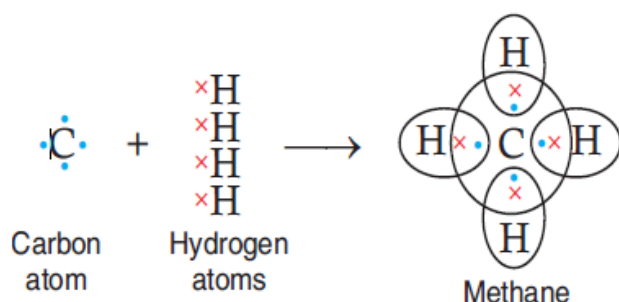
Learning activity 2.2.1

1. Is the oxygen molecule (O₂) an ionic compound? Explain based on the types of elements involved.
2. Beyond oxygen, what are some other examples of non-ionic (covalent) molecules, and what do they have in common?
3. How do you think bonds are formed between atoms in the oxygen molecule?

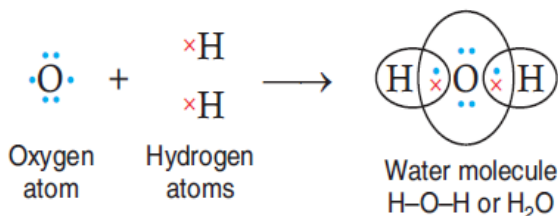
In the molecules of oxygen O_2 to form bond 2 atoms share electron. The bond between 2 atoms is not ionic bond, it is a **covalent bond**. A covalent bond is a chemical bond formed when two atoms share electrons. Compounds that contain covalent bonds are called **covalent compounds**. A covalent bond forms when both atoms involved need electrons to achieve a stable inert gas electron arrangement. Most non-metal atoms have 5, 6, or 7 electrons in their outermost shells, so they need additional electrons to become stable. They achieve this stability by sharing electrons with other non-metal atoms. Therefore, whenever a non-metal combines with another non-metal, a covalent bond is formed.

For example:

- i) Carbon shares four electrons with four hydrogen atoms to form methane.



- ii) As in a water molecule, two hydrogen atoms share electrons with the oxygen atom.



Some common covalent compounds, their formulae, and the elements present in them are shown in Table 2.2.1.

Table 2.2.1: Formulae and nomenclature of some covalent compounds

Nomenclature	Formula	Elements present
Methane	CH_4	C, H
Ammonia	NH_3	N, H
Alcohol (Ethyl alcohol)	C_2H_5OH	C, H, O
Hydrogen sulphide gas	H_2S	H, S
Carbon dioxide	CO_2	C, O
Carbon disulphide	CS_2	C, S
Carbon tetrachloride	CCl_4	C, Cl

Nomenclature	Formula	Elements present
Glucose	$C_6H_{12}O_6$	C, H, O
Sugar cane	$C_{12}H_{22}O_{11}$	C, H, O
Urea	$CO(NH_2)_2$	C, O, N, H

2.2.2 Properties of covalent compounds



Learning activity 2.2.2

Illustrating physical properties of covalent compounds

Materials needed: Cooking oil, ethanol, water, candle wax, pan, stove, electrodes, power source, and table salt (NaCl).

Procedure:

1. Solubility test

- Take a small sample of cooking oil.
- Try to dissolve it in water. What do you observe
- Next, try dissolving it in ethanol. What do you observe?
- Does cooking oil dissolve in water or ethanol? What does this tell you about its solubility?

2. Melting

- Take a candle wax and observe it while burning.
- Note how long it takes for the wax to melt.
- What does this tell you about the melting of candle wax?

3. Electrical conductivity test

- Place two electrodes in the boiling water to complete a circuit. Observe what happens.
- Add a small amount of table salt (NaCl) to the water and repeat the test.
- Why does pure water not conduct electricity, but salt water does?

Guiding questions:

- Based on these observations, what can you say about the solubility, melting and boiling points, and electrical conductivity of covalent compounds?
- How do covalent compounds differ from ionic compounds in terms of these physical properties?

From the activity above, the following properties of covalent compounds can be noted.

Properties of covalent compounds

1. **State:** Covalent compounds can be solids, liquids, or gases.

Examples: Alcohol and cooking oil (liquids), Methane and chlorine (gases), Glucose and wax (solids).

2. **Melting and boiling points:** They usually have low melting and boiling points.
3. **Solubility:** Most do not dissolve in water but dissolve in organic solvents. Some, like sugar, urea, and glucose, dissolve in water.
4. **Electrical conductivity:** They do not conduct electricity because they have no free ions.

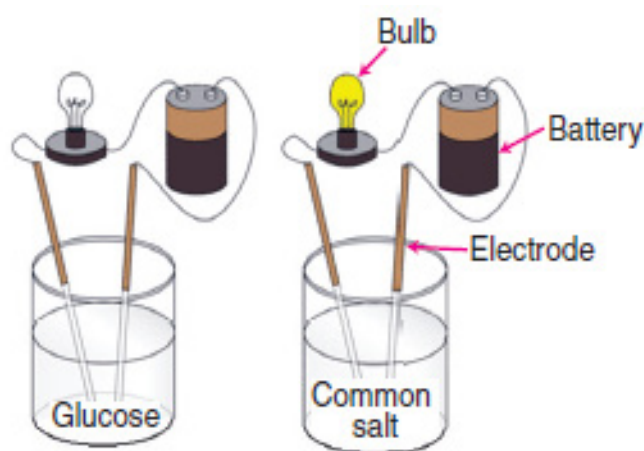


Fig 2.2.2: Electrical conductivity set up

Properties of ionic and covalent compounds are summarized in Table 2.2.2.

Table 2.2.2: Differences between ionic compounds and covalent compounds

Property	Ionic compounds	Covalent compounds
Physical state	Usually, crystalline solids	Usually liquids, gases, or solids with low melting and boiling points
Melting and boiling points	High; non-volatile	Generally low
Electrical conductivity	Conduct electricity when dissolved in water or melted	Most do not conduct electricity
Solubility	Usually soluble in water; insoluble in organic solvents (e.g., alcohol, ether, acetone)	Usually insoluble in water (except compounds like glucose, sugar, urea); soluble in organic solvents
Volatility	Non-volatile	Often volatile

2.2.3. Giant covalent structures (graphite and diamond).



Learning activity 2.2.3

Describe giant covalent molecules, their properties and uses

While most covalent substances form small, separate molecules, some form giant covalent structures, also called macromolecular structures. A giant covalent is a three-dimensional network of atoms held together entirely by strong covalent bonds. Unlike simple molecules, these substances do not exist as discrete molecules. They have very high melting points, extreme hardness, and are generally insoluble in most solvents.

Carbon and its allotropes giant structures

Carbon is an element capable of forming giant covalent structures due to its small atomic size, tetravalency, and ability to catenate (form chains and networks by bonding to itself). Carbon has two main allotropes that are giant covalent structures: diamond and graphite.

- **Diamond:** Each carbon atom forms four covalent bonds with other carbon atoms in a three-dimensional tetrahedral network. This giant covalent structure makes diamond extremely hard, gives it a very high melting point, and prevents it from conducting electricity.
- **Graphite:** Each carbon atom forms three covalent bonds with other carbon atoms, forming flat layers of hexagonal rings. The layers are held together by weak van der Waals forces, allowing them to slide over each other, making graphite soft and useful as a lubricant. Graphite can conduct electricity due to delocalized electrons between the layers.

Properties and uses of diamond and graphite are summarized in the tables 2.2.3.

Table 2.2.3: Properties and uses of diamond and graphite

Diamond	Graphite
• Cutting and drilling tools (e.g. glass cutters, rock drills). • Making jewellery. • Abrasives for polishing hard materials.	• Making pencil nibs. • As a lubricant. • Making electrodes in batteries and electrolysis. • Used in nuclear reactors (moderator).



Application activity 2.2

1. What do you mean by a covalent bond?
2. Give two examples of covalent compounds. Also write their chemical formulae.
3. When a _____ combines with another _____, covalent bond is formed.
4. Choose the covalent compound(s).
a) CH_4 b) H_2S c) CS_2 d) All of these
5. Using cross and dot diagram, show the formation of carbon dioxide.
6. Carbon tetrachloride and urea are covalent compounds. (True or False)
7. In the formation of covalent bonding, _____ of electrons takes place in the _____ shells of atoms.
8. Some covalent compounds are solid. (True or False)
9. Most covalent compounds are _____ in water but _____ in organic solvents.
10. Name two covalent compounds which are soluble in water.
11. Why most covalent compounds do not conduct electricity?
12. Melting and boiling points of covalent compounds are
a) high
b) low
c) between 500°C and 1000°C
d) cannot be determined.
13. a) State two physical properties of diamond.
b) State two physical properties of graphite.
14. a) State two uses of diamond and explain one reason for each use.
b) State one use of graphite and explain why it is suitable.
15. a) Why is diamond very hard?
b) Why graphite conducts electricity.

2.3. Metallic bonding

The force that holds metal atoms together is called a metallic bond. Metallic bonding is different from both covalent and ionic bonding, as neither of these can fully explain the unique properties of metals. For example, metals are excellent conductors of electricity even in their solid state, whereas most ionic and covalent compounds cannot conduct electricity as solids. (except for graphite).

2.3.1 Formation of metallic bond



Learning activity 2.3.1

1. Metals are made of atoms, just like other substances. What do you think holds these atoms together in a metal?
2. Why do metals conduct electricity in solid and liquid states, but salt only conducts when dissolved or melted?

A **metallic bond** is the type of chemical bonding that holds metal atoms together in a metal.

In simple terms, metal atoms release some of their outer (valence) electrons, which become free to move throughout the entire structure. This creates a “sea of electrons” surrounding positively charged metal ions.

In a pure metal, identical metal atoms release their valence electrons, which become delocalized and move freely throughout the structure as a “sea of electrons.” These electrons are not associated with any one atom. A metallic bond is the strong electrostatic attraction between the positively charged metal cores and the delocalized free electrons. A **metallic bond** is the type of chemical bonding that holds metal atoms together in a pure metal. When different types of metal atoms are mixed, they form what is called alloy.

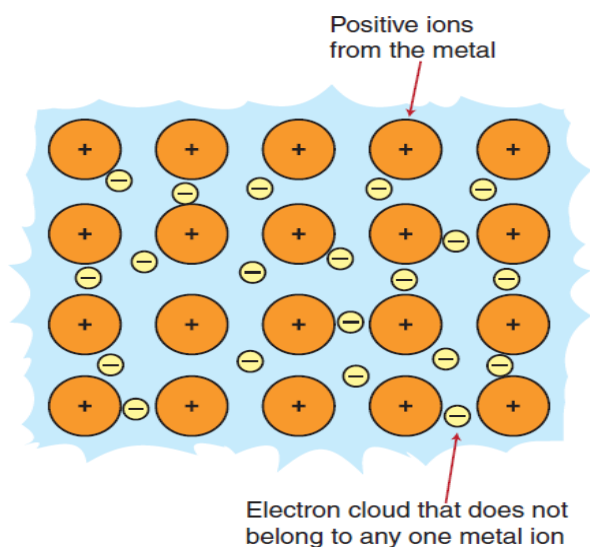


Figure 2.3: Formation of metallic bond.

2.3.2 Physical properties of metals



Learning activity 2.3.2

Illustrating the properties of metals (Ductility, malleability, heat and electric conductivity)

Refer to the activities in 1.2 section in unit 1

The metallic bond gives metals their unique physical properties. Because the metal atoms share a "sea of delocalized electrons," the atoms are held together strongly while allowing the electrons and layers of atoms to move. This special bonding explains why metals conduct heat and electricity and why they can be shaped easily without breaking. The properties of metal due to the metallic bond can be summarized as follows.

1. **Good conductors of heat and electricity:** Metals allow heat and electric current to pass through them easily because of the presence of freely moving electrons. Silver and copper are among the best conductors of heat; copper conducts heat better than aluminium.
2. **Lustrous (shiny):** Metals have a shiny surface because they reflect light strongly.
3. **Strong and hard:** Most metals are strong and hard. For example, iron (in the form of steel) is very strong and is widely used in the construction of bridges, buildings, and vehicles. However, not all metals are strong; sodium and potassium are soft metals.
4. **Ductile:** Metals can be drawn or stretched into thin wires. Gold and silver are among the most ductile metals.
5. **Malleable:** Metals can be hammered and rolled into thin sheets without breaking.



Application activity 2.3

1. Metallic bond is neither a covalent bond nor an electrovalent bond. (True or False)
2. The force which binds various metal atoms together is called _____.
3. Make a 2D structure of electron sea model.
4. Write a short note on formation of metallic bonding.
5. Name the metal which is the best conductor of electricity.
6. Aluminium is a better conductor of heat than copper. (True or False)
7. Metals are _____. This means that they can be hammered into thin sheets.
8. Why are cooking utensils made of metals?
9. Which of the following statement(s) is/are correct or not for metals?
 - a) Metals such as sodium and potassium are not strong.
 - b) Iron is used in the construction of buildings.
 - c) Gold and Silver are among the best ductile metals.
 - d) All of these.

2.4. End of Unit assessment

2.4.1. Project title:

Construction of a Simple Apparatus to Demonstrate Electrical Conductivity in Metals, Non-Metals, and in solutions.

Context

Electrical conductivity is an important property of materials and substances. Some materials, such as metals, allow electric current to flow easily, while others, such as non-metals, do not. Similarly, some solutions conduct electricity because they contain charged particles (ions), whereas others do not because they lack free-moving ions. Understanding these differences helps students explain many everyday applications of electricity and the behavior of materials.

In this project, Senior 2 students will design and construct a simple electrical conductivity tester using readily available materials. The apparatus will be used to investigate and compare the conductivity of metals, non-metals, ionic solutions, and covalent solutions.

Challenge

How can we construct a simple, low-cost apparatus that can accurately demonstrate and compare the electrical conductivity of different materials and solutions?

Task

Working in groups, students will:

1. Design and construct a simple electrical conductivity testing apparatus using available and adequate materials.
 - Test a variety of materials including: Metals, Non-metals, Ionic solutions and Covalent solutions.
2. Record observations on whether the bulb/LED lights up and compare the brightness where applicable.
3. Analyze results and explain the conductivity behavior of each material or solution.
4. Present the apparatus, findings, and conclusions to the class.

Guiding Questions

1. What is electrical conductivity?
2. Why do metals conduct electricity better than most non-metals?
3. What particles carry electric current in metals?
4. Why do ionic solutions conduct electricity?
5. Why do covalent solutions generally fail to conduct electricity?
6. How can the conductivity tester indicate whether a substance conducts electricity?

7. Which tested material is the best conductor and why?
8. How does the brightness of the bulb/LED relate to conductivity?
9. What improvements can be made to the apparatus to increase its accuracy or safety?
10. How are the results of this investigation useful in everyday life and industry?

Expected Outcome

Students will successfully construct a working conductivity tester and use it to distinguish between conductors and non-conductors. You will find out which materials generally good conductors and which ones are generally poor conductors and why. You will also find out solutions conduct electricity and which ones do not usually conduct electricity and why

Suggested Assessment Criteria

- Quality and functionality of the apparatus.
- Correct use of scientific procedures.
- Accuracy of observations and data recording.
- Scientific explanation of results.
- Teamwork, creativity, and presentation skills



2.4.2. End of unit assessment questions

A) Knowledge & Understanding

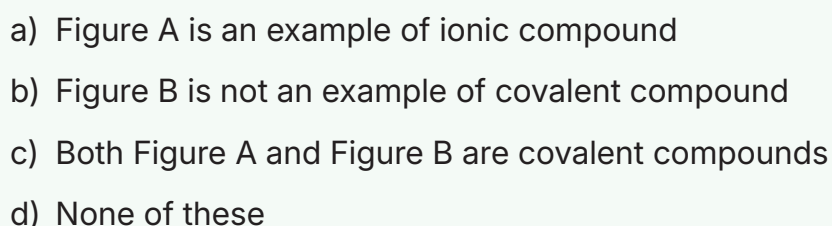
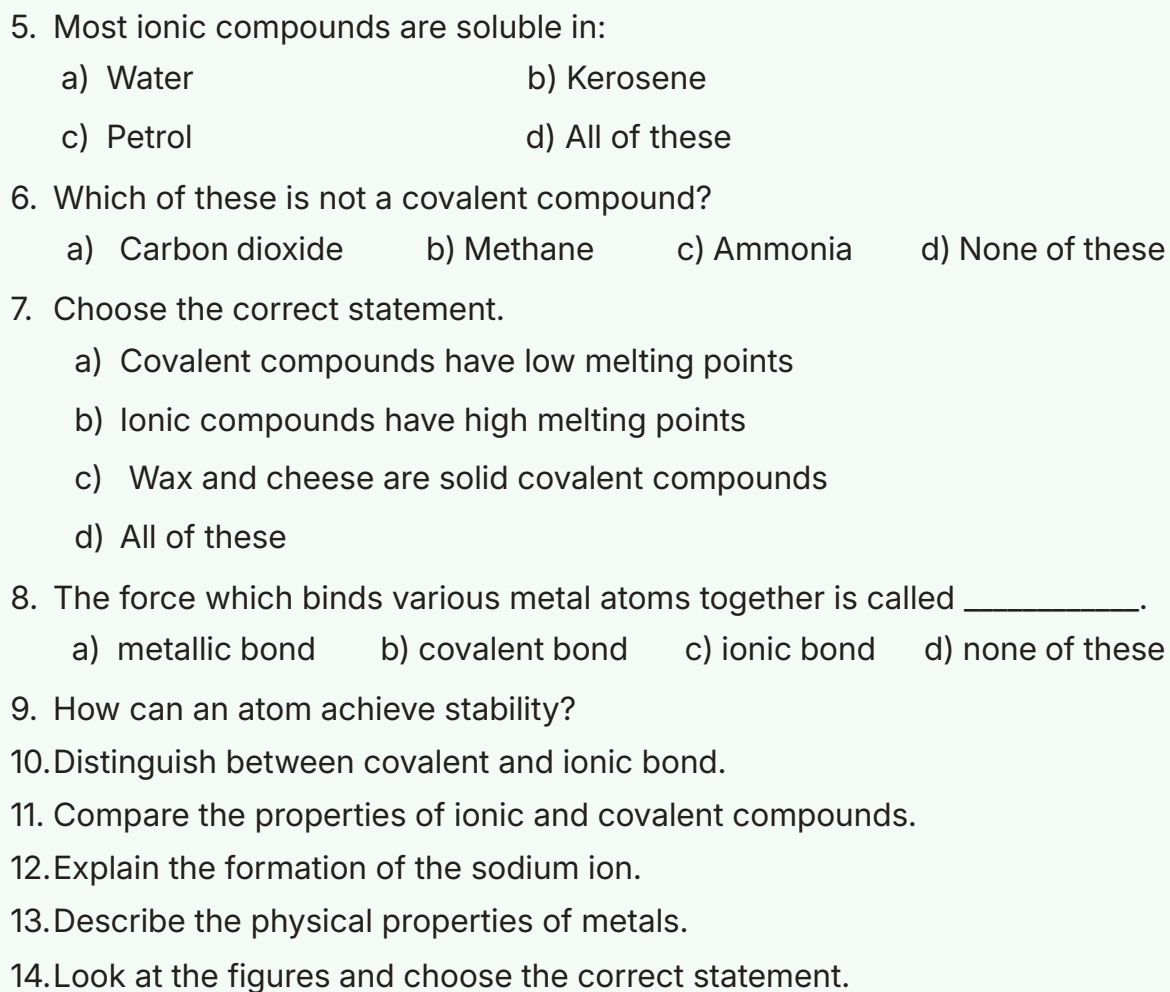
1. Sulphur, phosphorus and chlorine are all non-metals. Which one of them gains **one electron** to achieve a noble gas configuration?

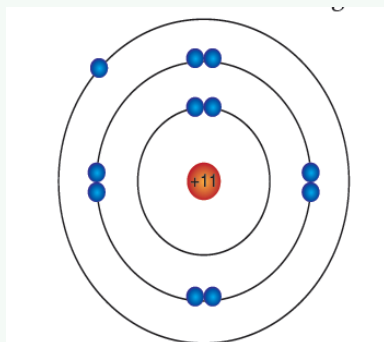
a) Phosphorus	b) Sulphur
c) Chlorine	d) Chlorine and sulphur
2. The electronic configuration of the **sodium ion (Na^+)** is:

a) 2,8,1	b) 2,8
c) 2,8,8	d) 2,8,2
3. The electronic configuration of the **chloride ion (Cl^-)** is:

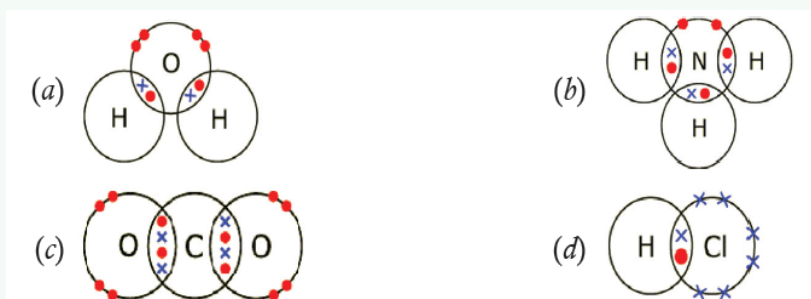
a) 2,8	b) 2,8,7
c) ,8,8	d) 2,8,3
4. Choose the ionic compound:

a) Calcium chloride	b) Sodium hydroxide
c) Copper sulphate	d) All of these





15. The following figure illustrates the electronic configuration of an element. In which group and period does this element belong? Explain your answer.



16. Which of the following depicts a non-metal with one valence electron and a non-metal with six valence electrons?

B. Skills

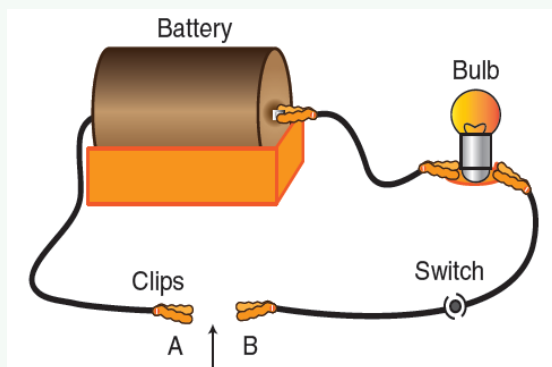
17. Give five examples of each

a) Ionic compounds

b) Covalent compounds

18. Compare the conductivity of distilled water with sodium chloride solution.

19. Which of the following materials does not make the circuit complete when inserted in between the crocodile clips?



a) Aluminium foil

b) Copper wire

c) Both (a) and (b)

d) Sulphur



C. Attitudes and Values

20. Why is it important to record your experimental results in a clear table instead of writing them in a rough form?
21. Why is it important to follow the teacher's instructions exactly during a chemical bonding experiment?

UNIT 3:

CATEGORIES OF CHEMICAL REACTIONS



Key unit competence

Classify different types of chemical reactions based on reactant behavior and energy changes, and relate chemical reactions to real-life applications.

Learning Objectives

By the end of this unit you should be able to:

Knowledge and understanding

- Explain the difference between a decomposition reaction and a combination reaction.
- Explain single displacement, double displacement (precipitation and neutralisation), and combustion reactions.

Skills

- Apply the rules of balancing equations to write balanced chemical reactions.
- Use experiments to show precipitation reactions.
- Classify chemical reactions as endothermic and exothermic using simple experiments.
- Write balanced ionic equations.

Values and attitudes

- Develop a team spirit, sense of responsibility when performing experiments.

Introduction

In chemistry, substances can change into new substances through chemical reactions. These reactions happen every day around us, for example, when iron rusts, food cooks, or fuels burn etc. Scientists group similar reactions into categories to make them easier to understand and remember. In this lesson, you will learn the main categories of chemical reactions and how to identify them using simple examples from everyday life. This will help you understand how matter changes and why these changes are important in science and daily life.



Introductory activity

Everyday chemical changes

Think about what happens in the following daily life situations:

- Milk is left at room temperature on a hot day.
- An iron pan or iron nail is left in a humid environment.
- Grapes are left to ferment.
- Food is cooked.
- Food is digested in our bodies.
- We breathe (respiration).

Guiding questions:

1. Which of these changes produce new substances?
2. Which ones are fast or slow?
3. Are the changes took place useful or harmful?

3.1.Types of chemical reactions

3.1.1 Combination reactions



Learning activity3.1.1

Reaction between iron and sulphur

Safety Precautions

- Always wear protective gloves if available.
- Handle the test tube with tongs while heating.
- Wear eye protection at all times.
- Avoid direct contact with the mixture or inhaling fumes.



Materials

Iron powder (filings), Bunsen burner, Test tube, Sulphur powder, Pair of tongs.

Procedure

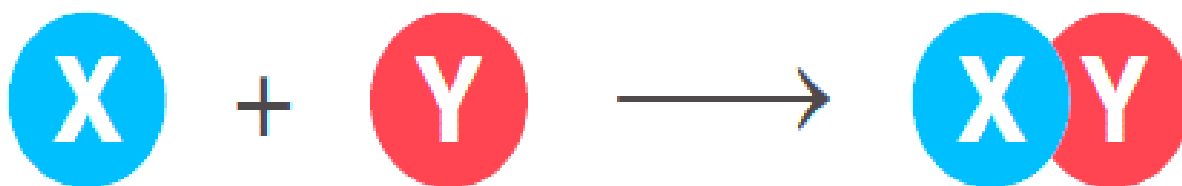
1. Prepare a mixture of iron powder and sulphur powder in a 7:4 ratio by mass.
2. Observe and note the appearance of the pure elements and the mixture.
3. Take about 0.5 g of the mixture and place it in a hard glass test tube.
4. Heat the test tube at the base of the mixture: start gently, then increase to a stronger heat, using a blue flame throughout.
5. Allow the test tube to cool completely.
6. Observe the product formed and try to write the chemical equation for the reaction.

Guiding questions

1. What changes do you observe in the mixture when it is heated?
2. Write the chemical equation of this reaction.

In this activity, when a mixture of iron filings and sulphur is heated, the sulphur melts and reacts with the iron to form a new black compound called iron (II) sulphide (FeS). During this process, **two** different elements combine to produce a **single** product. This type of chemical change is known as a **combination reaction** or **synthesis reaction**.

A combination reaction is a reaction in which two or more reactants join together to form one product. The general form of a combination reaction involving two reactants is:



In this equation, X and Y may be two elements, two compounds, or an element and a compound. However, the product **XY** is always a compound. In the next section, some common examples of combination reactions are summarized in the table 3.1.1 below:

Table 3.1.1: Common examples of combination reactions

Type of Combination	Chemical equation	
Element + Element → Compound	$\text{Fe (s)} + \text{S (s)} \rightarrow \text{FeS (s)}$	Two elements (Fe and S) combine to form one compound (FeS). This is a classic example of a combination reaction.
	$2\text{Na (s)} + \text{Cl}_2 \text{ (g)} \rightarrow 2\text{NaCl (s)}$	Elements combine to form sodium chloride (table salt), widely used in food and industry.
	$\text{H}_2 \text{ (g)} + \text{Cl}_2 \text{ (g)} \rightarrow 2\text{HCl (g)}$	Hydrogen chloride formed here is used industrially to produce hydrochloric acid.
Compound + Element → Compound	$2\text{CO (g)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{CO}_2 \text{ (g)}$	Carbon monoxide burns in oxygen to form carbon dioxide; important in combustion processes.
	$\text{PCl}_3 \text{ (l)} + \text{Cl}_2 \text{ (g)} \rightarrow \text{PCl}_5 \text{ (s)}$	A compound reacts with an element to form a more complex compound used in chemical synthesis.
Compound + Compound → Compound	$\text{CaO (s)} + \text{CO}_2 \text{ (g)} \rightarrow \text{CaCO}_3 \text{ (s)}$	Produces calcium carbonate, widely used in cement, construction, and chalk production.
	$\text{NH}_3 \text{ (g)} + \text{HCl (g)} \rightarrow \text{NH}_4\text{Cl (s)}$	Forms ammonium chloride, used in fertilizers and dry cells.

3.1.2. Decomposition reactions



Learning activity 3.1.2

Action of heat on baking soda (NaHCO_3 : Sodium hydrogen carbonate)

Materials;

Baking soda, spoon, heat source (stove or candle), small pan.

Procedure

1. Place a l 20 grams of baking soda in a pan.
2. Heat gently.

Weigh the remaining mass.

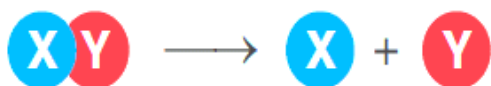
Guiding questions

1. Interpret the difference in mass

When baking soda or Sodium hydrogen carbonate, is heated, it breaks down (decomposes) into a new solid called sodium carbonate, water vapor, and carbon dioxide gas. This shows a chemical change because new substances are formed.

Reaction: $2\text{NaHCO}_3(\text{s}) \xrightarrow{\text{heat}} \text{Na}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$: Decomposition reaction

Decomposition reaction: A decomposition reaction is a chemical reaction in which a single compound breaks down into two or more simpler substances, which can be either elements or compounds. The general equation for a decomposition reaction with two products is:



Here, the reactant XY is always a compound, while the products X and Y may be elements, compounds, or a combination of both. In other words, decomposition reactions are the opposite of combination reactions, where two or more substances join to form a single compound. These decomposition reactions usually require an energy source such as heat, light, or electricity to break the bonds in the compound.

Some common examples of decomposition reactions are summarized in Table 3.1.2 below.

Table 3.1.2: Common examples of decomposition reactions

Type of Decomposition	Equation	Comments / Applications
Compound → Element + Element	$2\text{H}_2\text{O}(\text{l}) \xrightarrow[\text{Decomposition}]{\text{Electricity}} 2\text{H}_2(\text{gas}) + \text{O}_2(\text{gas})$ Water Hydrogen Oxygen	Water decomposes into hydrogen and oxygen by electricity.
Compound → Compound + Element	$2\text{H}_2\text{O}_2(\text{l}) \xrightarrow{\text{Light / MnO}_2} 2\text{H}_2\text{O}(\text{gas}) + \text{O}_2(\text{gas})$ Hydrogen peroxide Water Oxygen	Light breaks down hydrogen peroxide into water and oxygen. MnO_2 makes it faster. The oxygen produced can be collected.
Compound → Compound + Element	$2\text{NaNO}_3(\text{s}) \xrightarrow[\Delta]{380-500^\circ\text{C}} 2\text{NaNO}_2(\text{s}) + \text{O}_2(\text{g})$ Sodium nitrate Sodium nitrite Oxygen	Sodium nitrate decomposes to sodium nitrite and oxygen at high temperature.

Type of Decomposition	Equation	Comments / Applications
Compound → Compound + Element	$2\text{KClO}_3 \xrightarrow[\text{Decomposition}]{\text{Heat}} 2\text{KCl}(s) + 3\text{O}_2(g)$ Potassium chlorate Potassium chloride Oxygen	Potassium chlorate decomposes under heat to potassium chloride and oxygen. The reaction requires MnO_2 (catalyst) to lower the heat needed.
Compound → Compound + Compound	Calcium carbonate → Calcium oxide + Carbon dioxide	Calcium carbonate (limestone) breaks into quicklime and carbon dioxide. Quicklime is used in glass and cement.

3.1.3. Single displacement reactions



Learning activity 3.1.3

Reaction of iron nails with copper (II) sulphate

Materials: 3 iron nails, test tubes, sandpaper, thread, copper (II) sulfate solution (about 10 ml in test tube)

Procedure:

- Take three iron nails and clean them thoroughly by rubbing with sandpaper to remove any rust or dirt.
- Fill the test tube with 10 ml of copper (II) sulfate solution each.
- Immerse the nail in the test tube for 20 minutes and keep the third nail aside for comparison. Copper(II) sulphate solution is not very dilute.
- After 20 minutes, remove the nails from the test tube.
- Compare the color of the iron nails dipped in the solution and that of the ones kept aside.
- Record your observations.

The nail kept aside remains shiny, metallic iron.

Guiding questions

1. Which element was replaced in the solution?
2. Which new substance is formed on the nails?
3. How does this reaction demonstrate the concept of a single replacement reaction?
4. Why does the blue color of the solution become lighter?

1. Iron is **more reactive** than copper: Iron **displaces copper** from copper (II) sulphate solution.
2. Copper metal is deposited on the iron nails (reddish-brown coating).
3. Iron goes into solution as **iron (II) sulphate (FeSO₄)**, which is **green**, explaining the colour change.

Equation of the reaction that occurs: $\text{Fe (s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{Cu (s)}$.

This reaction is a simple displacement reaction. Iron displaces Copper

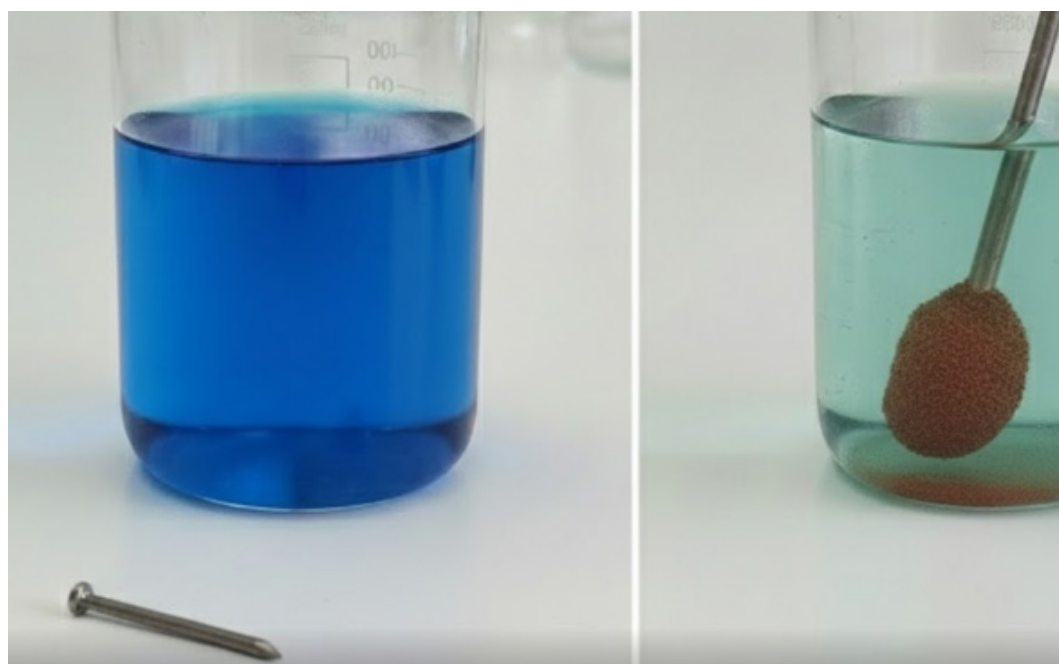
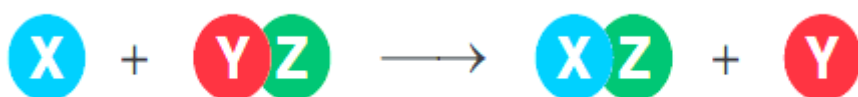


Figure 3.1.3: simple displacement reaction. Iron displaces Copper

Single replacement reactions are chemical reactions in which one element replaces another element in a compound. The general form of this reaction is:



In the reactants, **X** is an element and **YZ** is a compound. In the products, **XZ** is the new compound formed, and **Y** is the element that has been displaced.

In single replacement reactions, a more reactive metal replaces a less reactive metal from its compound. To help predict which metal can replace another, scientists arranged common metals in order of their reactivity. This list is called the reactivity series.

Reactivity Series of Metals (from most reactive to least reactive):

$\text{K} > \text{Na} > \text{Ca} > \text{Mg} > \text{Al} > \text{Zn} > \text{Cr} > \text{Fe} > \text{Ni} > \text{Sn} > \text{Pb} > \text{H} > \text{Cu} > \text{Ag} > \text{Au}$

A metal higher in the series can displace a metal below it from a solution of its compound. For example, iron (Fe) can replace copper (Cu) because iron is more reactive than copper.

Example	Chemical Equation	Observation	More Reactive Metal
1	$\text{Fe(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{Cu(s)}$	Blue CuSO_4 fades to green FeSO_4 ; red-brown copper coats iron	Iron displaces copper
2	$\text{Cu(s)} + 2\text{AgNO}_3\text{(aq)} \rightarrow \text{Cu(NO}_3)_2\text{(aq)} + 2\text{Ag(s)}$	Silver (grey-white) deposits on copper; the solution turns blue	Copper displaces silver
3	$\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$	Blue CuSO_4 becomes colourless; red-brown copper coats zinc	Zinc displaces copper
4	$\text{Mg(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{Cu(s)}$	Blue CuSO_4 fades; copper deposits on magnesium	Magnesium displaces copper
5	$\text{Pb(s)} + \text{Cu(NO}_3)_2\text{(aq)} \rightarrow \text{Pb(NO}_3)_2\text{(aq)} + \text{Cu(s)}$	Green $\text{Cu(NO}_3)_2$ becomes colourless; copper coats lead	Lead displaces copper
6	$\text{Mg(s)} + \text{CuO(s)} \rightarrow \text{MgO(s)} + \text{Cu(s)}$	Black CuO turns to brown copper; magnesium oxide forms	Magnesium displaces copper
7	$\text{Fe}_2\text{O}_3\text{(s)} + 2\text{Al(s)} \rightarrow \text{Al}_2\text{O}_3\text{(s)} + 2\text{Fe(l)}$	Aluminium reduces iron(III) oxide; molten iron is produced	Aluminium displaces iron

<https://courses.lumenlearning.com/suny-introductory-chemistry/chapter/types-of-chemical-reactions-single-and-double-displacement-reactions/>

3.1.4. Double displacement (precipitation and neutralisation)



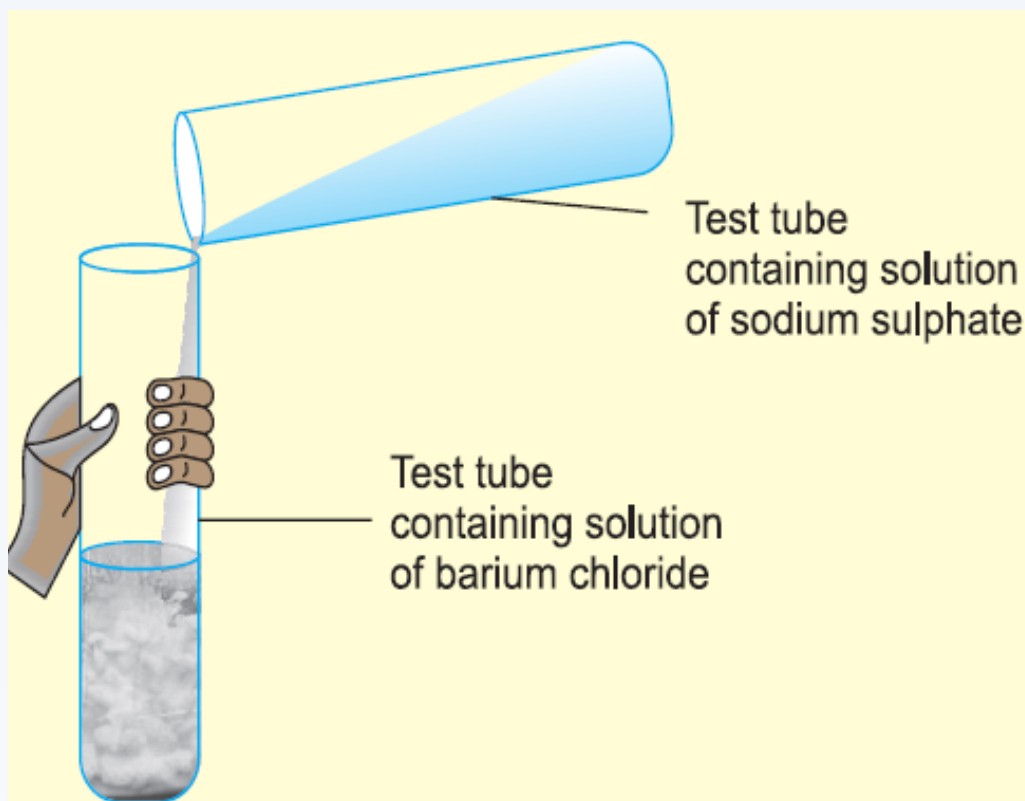
Learning activity 3.1.4

a) Illustration of precipitation reaction

Materials: Sodium sulphate solution (Na_2SO_4), Barium chloride solution (BaCl_2), Test tubes

Procedure:

1. Pour 3 ml of sodium sulphate solution into a test tube.
2. In a second test tube, measure 3 ml of barium chloride solution.
3. Carefully mix the two solutions as shown in the figure below.
4. Shake the test tube gently and observe the changes.



Mixing of sodium sulphate and barium chloride

Guiding questions

1. What do you observe when the two solutions are mixed?
2. Write the equation of this reaction

b) Illustrating a neutralization reaction

Materials needed: Dilute sodium hydroxide (NaOH) solution, Dilute hydrochloric acid (HCl), Phenolphthalein indicator, Test tube.

Procedure:

1. Pour 2 ml of dilute NaOH solution into a test tube.
2. Add 2–3 drops of phenolphthalein solution to the NaOH and note the colour that appears.
3. Add dilute HCl solution drop by drop to the colored NaOH solution. What happens to the colour as the acid is added?
4. After the colour disappears, add a few drops of colorless NaOH solution again and note any colour change.

The neutralisation and precipitation reactions are double displacement reactions.

Double displacement reactions are chemical reactions in which two compounds exchange their ions to form two new compounds. These reactions are also called metathesis reactions or exchange reactions. The general form of a double displacement reaction is:



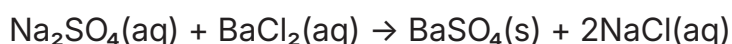
Where A and B are positive ions (cations) and X and Y are negative ions (anions)

- a) Double displacement reactions often occur in aqueous solutions and typically result in the removal of ions from the solution in one of two ways: by precipitation reaction or by neutralization reaction

a) Precipitation reaction

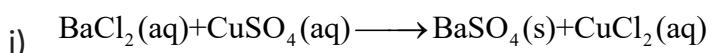
When two solutions containing soluble compounds are mixed, an insoluble substance (called precipitate) may form, making the mixture cloudy. A reaction that produces a precipitate is called a precipitation reaction. These reactions are useful for detecting the presence of certain ions or metals.

Examples from Activity 3.: When sodium sulphate solution reacts with barium chloride solution, a white precipitate of barium sulphate is formed as shown below:

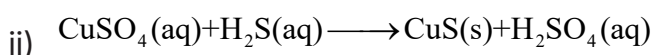


Sodium	Barium	Barium	Copper(II)
Sulphate	chloride	sulphate	chloride

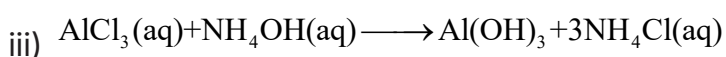
Other examples of precipitation reactions are:



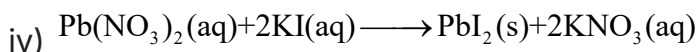
White ppt



Black ppt



White ppt

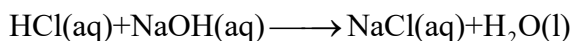


Yellow ppt

b) Neutralisation reaction

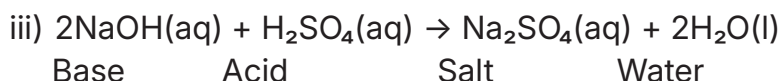
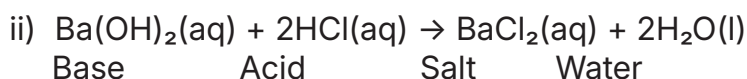
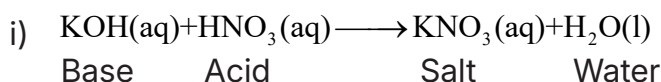
When an acid reacts with a base, the hydrogen ion (H^+) from the acid combines with the hydroxide ion (OH^-) from the base to form water. This reaction is called a neutralisation reaction. It is also a type of double displacement reaction because the cation of the base and the anion of the acid combine to form a salt.

Example from Activity II (Section 3.1.3.b): When hydrochloric acid reacts with sodium hydroxide:



Here, phenolphthalein is pink in a basic medium. Adding HCl makes the solution acidic and colourless. Adding NaOH again neutralises the excess acid, restoring the pink colour.

Other examples of neutralisation reactions are:



<https://ns1.almerja.com/more.php?idm=70854>

3.1.5. Combustion reactions



Learning activity 3.1.5

Burning of magnesium ribbon

Materials: Magnesium ribbon (2–3 cm), Bunsen burner, Tongs, Heat-resistant mat, Safety goggles if any, Test tube (optional, for collecting gases).

Safety precautions:

1. Wear safety goggles (if any) to protect your eyes from bright light and sparks.
2. Handle the magnesium ribbon with tongs to avoid burns.
3. Never look directly at the bright flame; the light from burning magnesium can damage eyes.
4. Ensure the area is well-ventilated.

Procedure:

1. Cut a small piece of magnesium ribbon (about 2–3 cm).
2. Hold the ribbon with tongs over an evaporating dish (Figure below).
3. Light the Bunsen burner and bring the flame close to the ribbon.
4. Observe what happens as the magnesium catches fire.
5. Allow the reaction to complete and cool the residue.
6. Test the residue by adding a small amount of water or using moist red litmus paper.



Figure 3.1.5: Burning magnesium ribbon

Guiding questions

1. What do you notice when the magnesium ribbon is brought near the flame?
2. What is the solid product formed after burning magnesium, and how can you test its properties?
3. Write the chemical equation for the reaction.
4. What type of chemical reaction is occurring?

Magnesium is a reactive metal that combines rapidly with oxygen in air when heated. The white powder formed is magnesium oxide (MgO). Magnesium reacts with oxygen to form a product, MgO . This reaction is a combustion reaction. This reaction is exothermic because it releases energy.

The magnesium ribbon burns with a very bright, dazzling white flame, and white smoke is produced. After the flame goes out, a white powdery ash remains. Heat and light energy are released.

A combustion reaction is a chemical reaction in which a substance burns in oxygen and releases energy as heat and light.

In simple terms, combustion happens when a fuel reacts with oxygen at high temperature, producing new substances (often gases like carbon dioxide and water vapor) and giving off heat and light.

Examples of combustion reactions:

- a) Burning of coal (C): $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
- b) Burning of natural gas (methane): $\text{CH}_4\text{(g)} + 2\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(l)}$
- c) Combustion of glucose: $\text{C}_6\text{H}_{12}\text{O}_6\text{(aq)} + 6\text{O}_{2\text{(g)}} \rightarrow 6\text{CO}_{2\text{(g)}} + 6\text{H}_2\text{O(l)}$

<https://chemistrytalk.org/combustion-reactions/>



Application activity 3.1

1. All combustion reactions release heat. (True or False)
2. Combustion reaction is also referred to as _____.
3. Which types of reactions represented by the chemical equations below?
 - i) $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$
 - ii) $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$
 - iii) $\text{HNO}_3 + \text{KOH} \rightarrow \text{KNO}_3 + \text{H}_2\text{O}$
 - iv) $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$
 - v) $\text{NH}_3 \rightarrow \text{N}_2 + \text{H}_2$
4. The insoluble substance formed during a chemical reaction is called _____.
5. What happens when barium chloride solution is added to copper sulphate solution?
6. Complete the following equation:
 $\text{CuSO}_4\text{(aq)} + \text{Mg(g)} \rightarrow ? + ?$
7. Why does blue colour of the copper sulphate solution fade when a piece of iron object is placed into it? Could the same observation take place if Silver is used instead of Iron?
8. When a copper strip is placed in a solution of zinc sulphate, copper sulphate is formed. (True or False)
9. Write a balanced chemical reaction of the following:
 - a) When copper oxide is heated with magnesium powder.
 - b) When iron oxide is heated with aluminium powder.
10. Complete the reaction: $\text{H}_2\text{(g)} + \text{Cl}_2\text{(g)} \rightarrow ?$ Is this a combination reaction? Explain.

3.2. Classification of chemical reactions as endothermic and exothermic reactions



Learning activity 3.2

Differentiate between exothermic and endothermic reactions

Safety precautions

- Wear eye protection if any throughout the activity, as some chemicals are irritants.
- Handle magnesium powder and ribbon carefully; both are highly flammable.
- Use the thermometer carefully to avoid breakage and to measure temperature changes accurately.
- Do not touch chemicals with bare hands; wash immediately if spills occur.

Materials required: Beaker, Thermometer, Sodium hydroxide solution, Dilute hydrochloric acid, Sodium hydrogen carbonate solution, Spatula, Citric acid.

Procedure:

1. Set up the apparatus as shown in the diagrams.

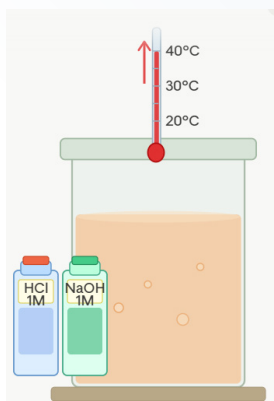
2. Experiment A:

- i) Measure 10 cm^3 of sodium hydroxide solution into the beaker. Record its initial temperature.
- ii) Add 10 cm^3 of dilute hydrochloric acid and gently stir using the thermometer. Record the highest or lowest temperature reached.

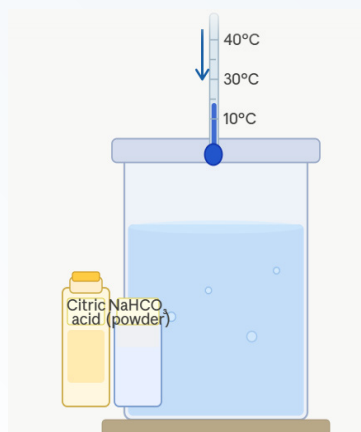
3. Experiment B

- i) Measure 10 cm^3 of citric acid solution into the beaker. Record its initial temperature.
- ii) Add 5 g of sodium hydrogencarbonate (NaHCO_3). Record the highest or lowest temperature reached.

Experimental set up



Experiment A



Experiment B



Complete the table below.

Reaction	Temperature before mixing (°C)	Temperature after mixing (°C)	Increase or decrease of temperature	Exothermic or Endothermic
1. Sodium hydroxide solution + dilute hydrochloric acid				
Citric acid + Sodium hydrogencarbonate				

Guiding questions

1. In which reactions does the temperature increase or decrease?
2. Does the temperature change indicate that the reaction is exothermic (gives out heat) or endothermic (takes in heat)?

The reaction between sodium hydroxide and hydrochloric acid releases heat (the temperature increases). This reaction is called **exothermic reaction**. But the reaction between citric acid and sodium hydrogencarbonate absorbs heat (the temperature decreases). This reaction is called **endothermic reaction**.

3.2.1. Exothermic reactions (heat released)

Exothermic reactions are reactions that release heat energy to the surroundings. The temperature increases and the energy change is negative.

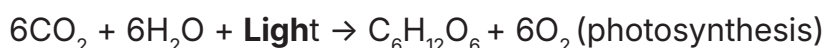
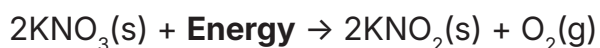
Examples of exothermic reactions:

- $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + \text{Energy}$ (Combustion reaction)
- $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{Heat}$
- $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu} + \text{Heat}$
- $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$ (Respiration reaction)

3.2.2. Endothermic Reactions (Heat Absorbed)

Endothermic reactions are reactions that absorb heat energy from the surroundings. The temperature decreases and the energy change is positive

Examples of endothermic reactions:



N.B: During a chemical reaction, energy is always involved.



Application activity 3.2

Classify the following reactions into exothermic and endothermic.

1. $\text{C}_5\text{H}_{12} + 8\text{O}_2 \rightarrow 5\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$
2. $\text{CaCO}_3 + \text{energy} \rightarrow \text{CaO} + \text{CO}_2$

3.3. Ionic equations

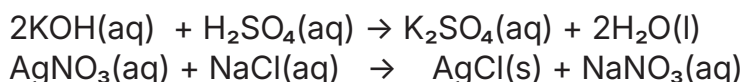


Learning activity 3.3

1. What happens when table salt (NaCl) dissolves in water?
2. Predict the ions present in each aqueous solution of Na_2SO_4 , KCl, H_2SO_4 , and NH_4Cl .

Most chemical equations are written as **condensed (molecular) equations**. In this form, all reactants and products are written with their state symbols.

Examples of molecular equations:



There are other equations that show reaction in terms of the ions present in solution. These types of equations are called **ionic equations**.

A chemical equation is written based on the principle that:

- i) Only soluble substances (strong electrolytes = their solutions are good conductors of electricity) are written in ionic form.
- ii) Insoluble substances, solids, liquids, and gases are **NOT** written into ions.

Strong electrolytes include:

- Soluble ionic compounds (NaCl, KI, CuSO_4 , etc)
- Strong acids (HCl, HNO_3 , H_2SO_4 , etc.)
- Strong bases (NaOH, KOH, $\text{Ba}(\text{OH})_2$, etc.)

The following solubility rules predict the solubility of many ionic compounds

Table 3.3: Solubility of salts

	Exceptions
1	All salts of Na^+ , K^+ , and NH_4^+ are soluble.
2	All salts of NO_3^- , ClO_3^- are soluble.
3.	All salts of Ag^+ , Pb^{2+} , and Hg^{2+} are insoluble.

4.	Most chloride, bromide, and iodide salts are soluble.	Most chloride, bromide, and iodide salts of Ag^+ and Pb^{2+} are generally insoluble.
5.	Most salts containing CO_3^{2-} , O^{2-} , S^{2-} , OH^- , PO_4^{3-} , CrO_4^{2-} , $\text{Cr}_2\text{O}_7^{2-}$, and SO_3^{2-} are insoluble.	Salts of group 1 and NH_4^+ containing CO_3^{2-} , O^{2-} , S^{2-} , OH^- , PO_4^{3-} , CrO_4^{2-} , $\text{Cr}_2\text{O}_7^{2-}$, and SO_3^{2-} are soluble.
6.	All SO_4^{2-} salts are soluble.	CaSO_4 , SrSO_4 , and BaSO_4 are insoluble.

3. Steps to write the ionic equation

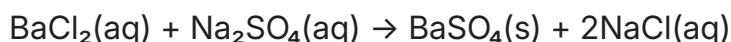
a) Write the complete (total) ionic equation

All soluble ionic compounds are written as separate ions.

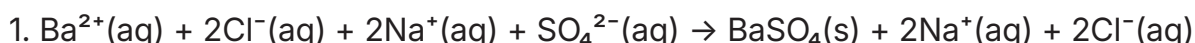
Example 1:

Reaction between BaCl_2 and Na_2SO_4

Molecular equation:



Complete ionic equation:



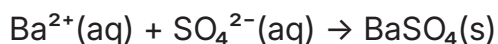
b) Spectator ions

Ions that appear unchanged on both sides of the equation are called **spectator ions** and they are cancelled out :

In the example above Na^+ and Cl^- are spectator ions.

After removing spectator ions, we get the **net ionic equation**, which shows only the species that actually react:

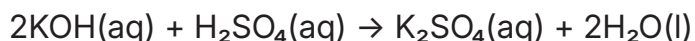
Net ionic equation:



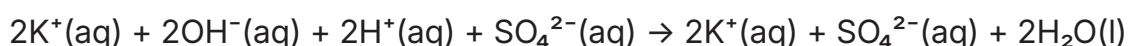
This form clearly shows the precipitation reaction.

Example 2: Reaction between 2KOH and H_2SO_4

Molecular equation:

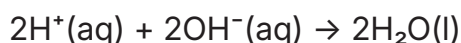


Complete ionic equation:



Spectator ions are K^+ and SO_4^{2-} . After removing spectator ions, we get the net ionic equation, which shows only the species that actually react.

Net ionic equation:





Application activity 3.3

- Write a balanced net ionic equation:
 - $\text{Cu(s)} + 2\text{AgNO}_3\text{(aq)} \rightarrow \text{Cu(NO}_3)_2\text{(aq)} + 2\text{Ag(s)}$
 - $\text{Fe(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{Cu(s)}$
 - $\text{BaCl}_2\text{(aq)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)} + \text{HCl(aq)}$
- Write the net ionic equation for all reactions of strong acids with strong bases that form salt and water.

<https://www.thoughtco.com/endothermic-and-exothermic-reactions-602105>



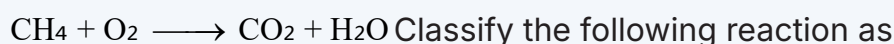
3.4 End of unit assessment

A) Knowledge & Understanding

- Which type of reaction occurs when two or more substances combine to form a single product?
 - Combination
 - Decomposition
- Which of the following reactions is generally exothermic?
 - Combustion
 - Photosynthesis
- In a double displacement reaction, what is formed?
 - Precipitate or water
 - Only gas
- Which statement describes a decomposition reaction?
 - A single compound breaks into two or more simpler substances
 - Two elements combine to form a compound
- What do ionic equations show?
 - Only the ions that participate in a chemical reaction
 - All ions present in the solution, including spectators
- Describe the main rule for writing ionic equations.

B) Skills

- Write the net ionic equation for the reaction between silver nitrate (AgNO_3) and sodium chloride (NaCl) solutions.
- Balance the following combustion reaction:



endothermic or exothermic: $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)} + \text{heat}$



9. Identify the type of reaction: $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$
10. Classify the following reactions as combination, decomposition, single displacement, double displacement, or combustion:
- a) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
- b) $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$
11. Write the net ionic equation for the reaction between calcium hydrogen carbonate and hydrochloric acid.

C) Attitudes & Values

12. Why is it important to wear safety goggles and gloves when performing chemical reactions in the lab?
- a) To protect yourself from harmful chemicals
- b) Because the teacher told you to
13. How can understanding chemical reactions help in daily life?
- a) Helps in safely using and storing chemicals
- b) Makes chemistry more complicated

UNIT 4:

INORGANIC SALTS, PROPERTIES, PREPARATION AND IDENTIFICATION OF IONS



Key unit competence:

- Analyze the properties and preparation methods of inorganic salts.
- Relate the properties of inorganic salts to real life application.

Learning objectives:

By the end of the unit, you should be able to:

Knowledge and understanding

- Demonstrate understanding of an inorganic salt.
- Distinguish between soluble and insoluble salts.
- State the physical properties of salts
- Demonstrate understanding of solubility.
- Describe factors that affect solubility.
- Explain the concept of unsaturated, saturated, and supersaturated solutions.
- Describe different methods of preparing soluble and insoluble salts.
- Name the sources and uses of salts in daily life.

Skills

- Classify salts into soluble and insoluble salts by dissolving them in water
- Use an experiment to demonstrate the electrical conductivity of salt solutions.
- Apply the rules to name inorganic salts
- Prepare different soluble and insoluble salts using suitable chemicals.
- Use experiments to show the effect of temperature on the solubility of different salts.
- Test the presence of ions in a solution.

Values and attitudes

- Develop a team spirit and sense of responsibility during experiments.
- Appreciate the uses of salts in daily life, such as sodium chloride as a table salt.

Introduction

In our daily life, we touch, taste, and use salts without even noticing them. Some salts are white and crystalline, others are blue, green, or even pink. Some dissolve easily in water, while others form solid precipitates. But what really are these salts made of? How are they prepared? And how can we identify the ions present in them? In this lesson, we will explore inorganic salts, their properties, how they are prepared in the laboratory, and simple tests used to identify their ions. Through experiments and observations, you will discover how chemists use colour changes, precipitates, and reactions to detect hidden ions just like chemical detectives. By the end of this lesson, you will understand why inorganic salts are important in the laboratory, in industry, and in everyday life, and you will be able to identify different ions using simple chemical tests.

Key term: Inorganics salts



Introductory activity

Salts in our daily life

You are already familiar with acids, bases, and table salt. Also you are familiar with some chemical reactions and their types.

1. What do you think a salt is?
2. Which particles make up salts?
3. How can salts be prepared in the laboratory?

4.1. Definition, nomenclature and chemical formulae of simple salts.

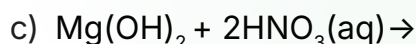
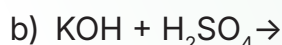
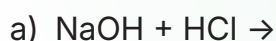


Learning activity 4.1

1. Complete the table below:

Formula	Name	Valence
SO_4^{2-}		
	Nitrate	
	Phosphate	
CO_3^{2-}		
	Chlorine atom	
	Hydroxide	

Complete the following equations showing the physical states of the compounds. Apply rules for writing chemical formula of products formed using valences from unit 5 senior one



4.1.1. Definition of salt

A salt is a compound formed when the hydrogen ions of an acid are totally or partially replaced by a metal ion or an ammonium ion (NH_4^+). This process occurs through a **neutralisation reaction**, in which an acid reacts with a base to form a salt and water.

In Chemistry, a salt is a compound made of positively charged ions (cations: metal ions or NH_4^+) from a base and negatively charged ions (anions: Cl^- , SO_4^{2-} , PO_4^{3-} , ...) from an acid.

There are 2 types of salts, **normal salts** or **neutral salts** and **acid salts**. A normal salt is formed when all the hydrogen ions of an acid are replaced by a metal or ammonium ion. All chloride and nitrate salts are normal salts. However, **acid salt** is produced when the hydrogen ions of an acid are partially replaced.

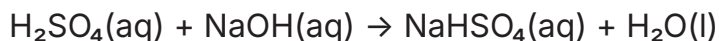
i) Normal salts, where all hydrogen atoms of the acid are replaced.

Example:



Zinc sulphate contains no hydrogen atom that can be replaced; it is therefore classified as a **normal salt**.

ii) **Acid salts**, where only part of the hydrogen atoms of the acid are replaced.

Example:

Sodium hydrogen sulphate still contains one replaceable hydrogen atom. For this reason, it is known as an acid salt.

Common acids and their corresponding salts are summarized in the table 4.1. below:

Table 4.1: Acids and their corresponding salts

Acid Name	Acid Formula	Radical (ion)	Name of Radical	Example of neutral salts	Example of acid salts
Hydrogen chloride	HCl	Cl^-	Chloride	NaCl	-
Hydrogen nitrate	HNO_3	NO_3^-	Nitrate	KNO_3	-
Hydrogen sulphate	H_2SO_4	SO_4^{2-}	Sulphate	Na_2SO_4	NaHSO_4
Hydrogen sulphite	H_2SO_3	SO_3^{2-}	Sulphite	K_2SO_3	KHSO_3
Hydrogen carbonate	H_2CO_3	CO_3^{2-}	Carbonate	MgCO_3	$\text{Mg}(\text{HCO}_3)_2$
Hydrogen phosphate	H_3PO_4	PO_4^{3-}	Phosphate	Na_3PO_4	NaH_2PO_4
Hydrogen ethanoate (Acetic) acid	CH_3COOH	CH_3COO^-	Ethanoate (Acetate)	CH_3COONa	-
Hydrogen fluoride	HF	F^-	Fluoride	NH_4F	-
Hydrogen bromide	HBr	Br^-	Bromide	NH_4Br	-
Hydrogen iodide	HI	I^-	Iodide	CaI_2	-
Hydrogen nitrite	HNO_2	NO_2^-	Nitrite	$\text{Mg}(\text{NO}_2)_2$	-
Hydrogen oxalate	$\text{H}_2\text{C}_2\text{O}_4$	$\text{C}_2\text{O}_4^{2-}$	Oxalate	CaC_2O_4	$\text{Ca}(\text{HC}_2\text{O}_4)_2$

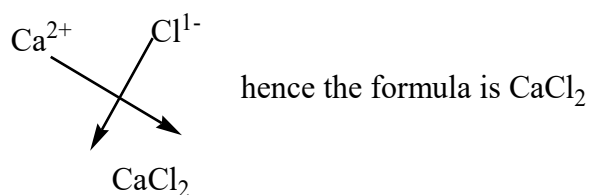
4.1.2. Chemical formula and Nomenclature of salts

A) Rules for writing chemical formulae of inorganic salts using valences

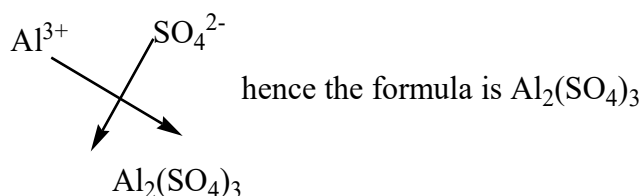
To write a chemical formula, write the symbols, note the valences, cross them over, reduce if possible, use brackets for radicals, and make sure the charges balance.

Examples:

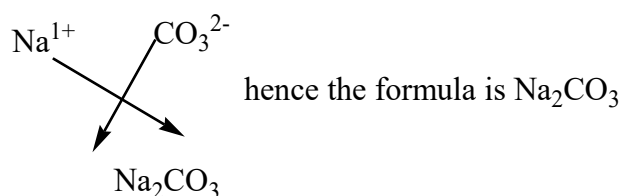
i) Calcium chloride



ii) Aluminium sulphate



iii) Sodium carbonate



B) Rules for naming inorganic salts from their chemical formulae

1. Naming compounds made of two elements (binary compounds)

- Write the name of the metal first.
- Write the name of the non-metal second, changing its ending to "**-ide**".

Examples:

- $\text{NaCl} \rightarrow$ sodium chloride
 - $\text{CaS} \rightarrow$ calcium sulphide
2. Naming compounds containing radicals (polyatomic ions)
- Write the name of the positive ion (metal or ammonium) first.
 - Write the name of the radical second (the radical name does not change).

Examples:

- $\text{Na}_2\text{SO}_4 \rightarrow$ sodium sulphate
- $\text{CaCO}_3 \rightarrow$ calcium carbonate
- $\text{NH}_4\text{NO}_3 \rightarrow$ ammonium nitrate

3. Compounds with variable valence metals

If a metal can have more than one valence, show its valence using roman numerals in brackets.

Examples:

$\text{FeCl}_2 \rightarrow$ iron (II) chloride

$\text{FeCl}_3 \rightarrow$ iron (III) chloride

The names of salts are derived from the metal or ammonium ion and the parent acid from which they are formed. When naming salts, the name begins with the metal or ammonium ion names, followed by the corresponding acid radical names.

Below are some examples of salts (formula) and their names.

Table 4.1.2: Names of some salts

Name of Radical	Example of salts	Name of salts (neutral and acid salts)
Chloride	NaCl	Sodium chloride
Nitrate	KNO_3	Potassium nitrate
Sulphate	Na_2SO_4	Sodium sulphate
Hydrogen sulphate	NaHSO_4	Sodium hydrogen sulphate
Sulphite	K_2SO_3	Potassium sulphite
Hydrogen sulphite	KHSO_3	Potassium hydrogen sulphite
Carbonate	MgCO_3	Potassium carbonate
Phosphate	Na_3PO_4	Sodium phosphate
Ethanoate (Acetate)	CH_3COONa	Sodium ethanoate/ Sodium acetate
Fluoride	NH_4F	Ammonium fluoride
Bromide	NH_4Br	Ammonium bromide
Iodide	CaI_2	Calcium iodide
Nitrite	$\text{Mg}(\text{NO}_2)_2$	Magnesium nitrate
Oxalate	CaC_2O_4	Calcium oxalate

- Acids with only one replaceable H^+ , such as HCl and HNO_3 , can form only normal/neutral salts because they have only one ionizable hydrogen atom.
- Acids with more than one replaceable H^+ such as Sulfuric acid (H_2SO_4) and Phosphoric acid (H_3PO_4) can form both acidic salts and normal salts.



Application activity 4.1

1. Name the following inorganic salts
 MgCl_2 , $\text{Ca}(\text{NO}_3)_2$, NH_4Br , K_3PO_3 , NaHCO_3
2. Write the formulae of the following inorganic salts
 - i) Aluminium hydrogen phosphite: $\text{Al}_2(\text{HPO}_3)_3$
 - ii) Ammonium phosphate: $(\text{NH}_4)_3\text{PO}_4$
 - iii) Sodium carbonate: Na_2CO_3
3. Complete the following table

Salt	Corresponding Acid	Corresponding Base
MgCl_2	HCl	$\text{Mg}(\text{OH})_2$
$\text{Ca}(\text{NO}_3)_2$	HNO_3	$\text{Ca}(\text{OH})_2$
NH_4Br	HBr	NH_4OH
K_3PO_3	H_3PO_3	KOH
NaHCO_3	H_2CO_3	NaOH

3. Identify normal (neutral) salts and acid salts from the list below: MgCl_2 , NaHCO_3 , $\text{Ca}(\text{NO}_3)_2$, NH_4Br , K_3PO_3

4.2. Physical properties of mineral salts



Learning activity 4.2

Appearance and solubility of inorganic Salts

Material required

Magnesium sulphate, calcium sulphate, sodium carbonate, lead (II) nitrate, potassium nitrate, silver chloride, sodium chloride, lead (II) chloride, barium sulphate, test tube, weighing balance, beaker, conductivity tester, or ammeter



Procedure

1. Observe each of the salts given, then provide the missing information in the table.

Salt	Formula	Physical state	Colour
Magnesium sulphate			
Calcium sulphate			
Sodium carbonate			
Lead (II) nitrate			
Potassium nitrate			
Silver chloride			
Sodium chloride			
Lead (II) chloride			
Barium sulphate			

Take 0.5 g of each of the salts provided in a test tube and add about 5cm³ of cold water.

Caution: Do not warm. Close the test tube with your thumb and shake thoroughly. Put the test tube in a test tube rack. Observe then record your observations. If there is any solid left, add a little more water and shake again.

2. Repeat procedure 2 above but this time using cold and hot water. Use a Bunsen burner to heat water. Note what happens.

Record your results in the following table by ticking in appropriate.

Salts	Soluble		Insoluble	
	Cold water	Warm water	Cold water	Warm water
Magnesium sulphate				
Calcium sulphate				
Sodium carbonate				
Lead (II) nitrate				
Potassium nitrate				
Silver chloride				
Sodium chloride				
Lead (II) chloride				
Barium sulphate				

Inorganic salts are generally solid substances at room temperature. Their appearance, especially colour, and their behaviour in water (solubility) are important properties used in their identification and classification.

4.2.1. Physical states and appearance of common mineral salts

Most inorganic salts appear as white crystalline solids. For instance, Magnesium sulphate, calcium sulphate, lead (II) nitrate, potassium nitrate, silver chloride, sodium chloride, and lead (II) chloride are white solids at room temperature. Carbonate salts may occur as crystalline solids or powders.

Some salts are coloured due to the presence of transition metal ions. Their colour helps in their identification.

Table 4.2.1 Examples of coloured salts

Salt	Colour
Hydrated copper (II) sulphate	Blue
Hydrated iron (II) sulphate	Green
Potassium permanganate	Purple
Potassium dichromate	Orange

4.2.2 Solubility of mineral salts in water

When salts are added to water, they may be soluble, sparingly soluble, or insoluble. Some are soluble in cold and warm water (Magnesium sulphate, Sodium carbonate, Lead (II) nitrate, Potassium nitrate, Sodium chloride, ...). Other inorganic salts like Silver chloride (AgCl) and barium sulphate are insoluble in water; calcium sulphate is sparingly soluble. Lead (II) chloride (PbCl_2) is soluble in warm water and insoluble in cold water. Based on these observations, the solubility rules of inorganic salts can be summarized as follows.

Solubility rules of inorganic salts

1. All nitrate salts are soluble in water.
2. All salts of sodium, potassium, and ammonium are soluble in water.
3. Most carbonate salts are insoluble in water, except those of sodium, potassium, and ammonium, which are soluble.
4. Most sulphate salts are soluble in water; however, lead (II) sulphate and barium sulphate are insoluble, while calcium sulphate is only sparingly soluble.
5. Most chloride, iodide, and bromide salts are soluble in water, except their silver and lead(II); however, lead (II) chloride dissolves in hot water.

4.2.3 Electric conductivity of inorganic salts

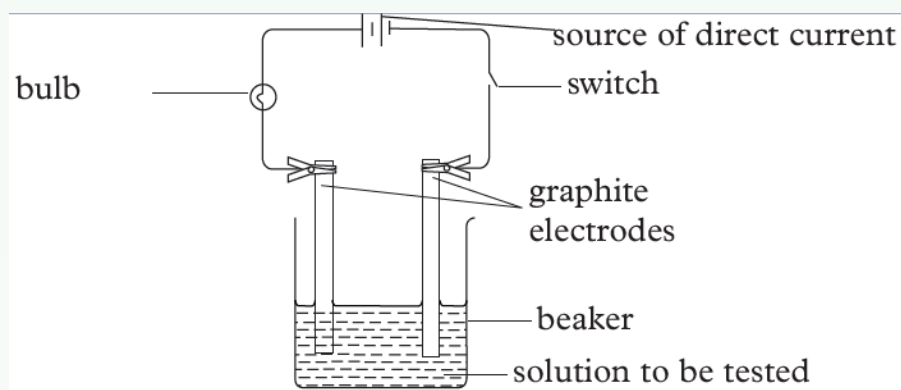


Learning activity 4.2.3

Experiment to demonstrate the electrical conductivity of inorganic salts

Prepare solutions of the various salts listed in Experiment 4.3 in small beakers.

1. Set up the electrical conductivity apparatus as shown below.



2. Switch on the current and observe whether the bulb lights up. Based on the observation, describe the conductivity of the solution.
3. Rinse the beaker and graphite electrodes thoroughly with distilled water before testing the next solution.
4. Repeat the test for each solution and record your observations in a table.

Salts	Observation on the bulb
Magnesium sulphate	
Sodium carbonate	
Sodium hydrogen carbonate	
Lead (II) nitrate	
Potassium nitrate	
Sodium chloride	
Magnesium chloride	

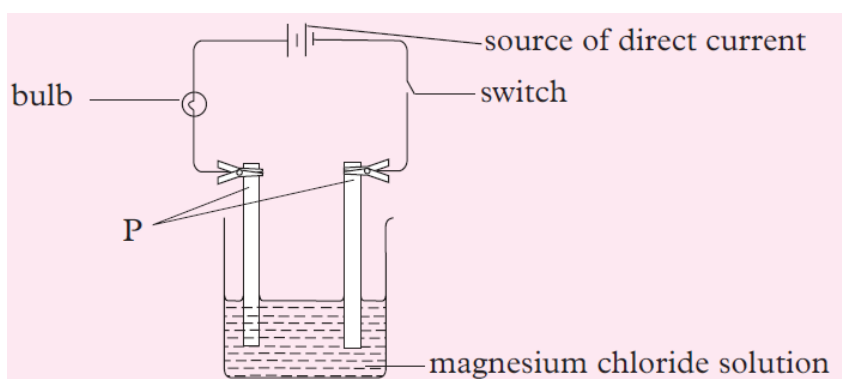
All soluble salts i.e. those that dissolve in water to form solutions, conduct electricity. When the switch is closed, the bulb lights up, indicating that the solution completes the electrical circuit.

In aqueous solutions, salts ionize to produce positive and negative ions, which move freely and carry an electric charge, allowing the current to flow through the solution.



Application activity 4.2

1. Why is $\text{CaCO}_3 + \text{dilute H}_2\text{SO}_4$ not advisable for preparing CaSO_4 ?
2. Comment on the solubility of calcium sulphate and lead (II) chloride.
3. The electrical conductivity of magnesium chloride was tested using the following set-up.



- a) Name suitable materials for rods P.
- b) State the observation made when the switch was closed.
- c) What does this experiment tell you about the electrical conductivity of aqueous magnesium chloride?

4.3. Understanding solubility and type of solution



Learning activity 4.3

Exploring solubility and types of solutions

Materials: Water, inorganic salt (e.g. sodium chloride or potassium nitrate), Test tubes or transparent cups, spatula or teaspoon, stirring rod, heat source.

Procedure

1. Add a small amount of the salt to water and stir until it dissolves completely.
2. Continue adding small portions of the salt while stirring.
3. Stop when some solid salt remains undissolved at the bottom.
4. Gently heat the saturated solution and add a little more salt until it dissolves.
5. Allow the solution to cool slowly without disturbance.



Figure 4.3. a) Illustration of solubility and type of solutions

Guiding questions

1. What happens to the salt when a small amount is added to water?
2. At what point does the salt stop dissolving, even after stirring?
3. Why does heating allow more salt to dissolve?
4. What do you observe when the hot solution is allowed to cool?
5. How do you distinguish between unsaturated, saturated and supersaturated solutions?

4.3.1. Solute, solvent, and solubility

- **A solution** is a homogeneous mixture of two or more substances.

The components of a solution are:

- **Solvent:** a substance, usually a liquid, that is present in the greatest amount and dissolves another substance (solute) to form a solution.

Water is the most common solvent because it dissolves many substances such as salt and sugar. A solution where water is a solvent is called **aqueous solution**

- **Solute:** The substance that is dissolved in a given solvent.

In a solution, the solute is evenly distributed throughout the solvent, forming a clear mixture.

The ability of a solute to dissolve in a solvent is described by **solubility**.

- **Solubility** is the maximum mass of a solute that can dissolve in 100 g of a solvent at a specified temperature to form a saturated solution.

$$\text{Solubility} = \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 100$$

Example:

72 g of saturated solution of potassium nitrate contains 30 g of salt. Calculate the solubility of the salt.

Solution: Mass of water = mass of solution – mass of salt = 72 – 30 = 42g

42 g of water dissolve 30 g of KNO_3

100 g of water dissolves $\frac{30 \times 100}{42} = 71.43$ g of KNO_3

The solubility of KNO_3 in 100 g of water = 71.43 g

4.3.2 Types of solutions

Depending on the amount of solute dissolved in the solvent at a given temperature, solutions are classified as **unsaturated**, **saturated**, or **supersaturated**.

- A) **Unsaturated solution:** A solution that contains less solute than the solvent can dissolve at a given temperature. More solute can still dissolve in it.
- B) **Saturated solution:** A solution that contains the maximum amount of solute that can dissolve in the solvent at a particular temperature. Any extra solute added remains undissolved.
- C) **Supersaturated solution:** is a solution that contains more than the maximum amount of solute that is capable of being dissolved at a given **temperature**. Such a solution is unstable.

Table 4.3.2: Differences between unsaturated and saturated solutions

Unsaturated Solution	Saturated Solution
Contains less solute than it can dissolve at a given temperature	Contains the maximum amount of solute it can dissolve at a given temperature
Can dissolve more solute	Cannot dissolve any more solute

Becomes more unsaturated when the temperature increases

Becomes unsaturated when the temperature increases



Application activity 4.3

1. Fill in the blank space
 - a) An unsaturated solution contains _____ solute than it can dissolve at a given temperature.
 - b) A saturated solution contains the _____ amount of solute it can dissolve at a given temperature.
 - c) An unsaturated solution _____ dissolve more solute.
 - d) A saturated solution _____ dissolve any more solute.
2. Is a liquid mixture of kerosene and water a solution? Justify your answer.
3. Fill in the blank space with solution, solvent, or solute.
 - a) A _____ is a homogeneous mixture of two or more substances.
 - b) The component of a solution that is usually a liquid and present in the greatest amount is the _____.
 - c) The substance that is dissolved in a given solvent is called the _____.
 - d) Water is the most common _____ because it dissolves many substances like salt and sugar.
 - e) In a solution, the _____ is evenly distributed throughout the solvent, forming a clear mixture.
 - f) Solubility is the maximum mass of a _____ that can dissolve in 100 g of a _____ at a specified temperature.
4. 45 g of saturated aqueous solution contain 10 g of salt at room temperature.
 - a) Calculate the solubility of the salt.
 - b) How many additional grams of the salt must be added to make the solution supersaturated at room temperature?

4.4. Factors influencing solubility of different salts



Learning activity 4.4

Investigating factors influencing solubility of different salts

Materials

Water (cold and hot), Sodium chloride (NaCl), Potassium nitrate (KNO_3), Calcium sulphate (CaSO_4), Iodine crystals (I_2), Carbon tetrachloride (CCl_4) or kerosene (as a safer classroom alternative, beakers or test tubes, stirring rod and heat source).

Procedure:

1. Add a small amount of sodium chloride to cold water and stir. Observe whether it dissolves.
2. Repeat using potassium nitrate and calcium sulphate.
3. Heat the solutions gently and add more of each salt while stirring.
4. In a separate test tube, add iodine to water and shake. Then add iodine to CCl_4 (or kerosene) and observe.

Guiding questions

1. Which salt dissolves more when the water is heated compared to cold water?
2. What difference do you observe when the same solute is added to hot water and to cold water?
3. What happens when iodine is added to water and when it is added to carbon tetrachloride (or kerosene)?
4. Why does hot water dissolve more solute than cold water?

From these observations, it becomes clear that some substances dissolve easily while others do not, and that heating can change how much solute dissolves. These results show that solubility is not the same for all substances but depends on certain factors. This leads us to study the main factors that influence the solubility of different salts.

1. Nature of Solute

The phrase “like dissolves like” explains why polar solutes (a substance with a slight positive and negative charge due to uneven electron sharing) dissolve in polar solvents, and non-polar solutes dissolve in non-polar solvents.

Examples:

- Ionic salts like NaCl , KCl , and KNO_3 dissolve readily in water (a polar solvent).
- Non-polar substances like iodine (I_2) and sulfur (S_8) do not dissolve in water but dissolve in non-polar solvents such as CCl_4 or CS_2 .

2. Temperature

Temperature has a direct effect on solubility. For most ionic compounds, solubility increases with temperature. Some compounds, however, dissolve better at lower temperatures.

General rule:

- Solubility of solids and liquids increases with temperature.
- Solubility of gases decreases with temperature.

Example:

When a solution is heated, more solute can usually dissolve. For example, potassium chloride dissolves easily in water, and even more KCl can be dissolved when the solution is heated. Heating weakens the attractive forces holding the ions together in the solid crystal, allowing more particles to separate and mix with the solvent. Below is the graph showing how solubility varies with temperature.

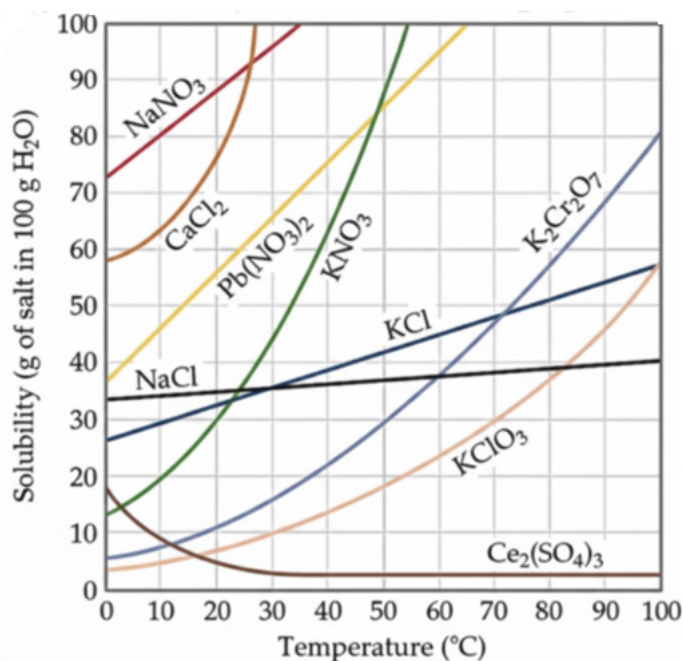


Figure 4:4 Solubility vs temperature for different salts

Interpretation with some examples

- Potassium nitrate shows a **sharp increase** in solubility as temperature rises. For example, at about 20°C, around 30 g dissolves in 100 g of water, and at 60°C, substantially more can dissolve. This means hot water dissolves much more potassium nitrate than cold water.
- Sodium chloride has an almost **flat curve**. Its solubility changes very little with temperature. About 35–40 g dissolve in 100 g of water over a wide temperature range.
- Cerium (III) sulfate behaves differently. Its solubility **decreases** as temperature increases. This means it dissolves better in cold water than in hot water.

- Potassium chlorate increases gradually with temperature. More of the salt dissolves in hot water, but the increase is not as rapid as potassium nitrate.

General conclusion

Most salts become more soluble as temperature increases, but the rate of increase differs from one salt to another. A few salts become less soluble when heated. This information is useful in crystallization and preparation of salts in the laboratory.



Application activity 4.4

1. What is the meaning of this statement "substances with similar polarity are more likely to dissolve in each other"?
2. The solubility of gases always increases with an increase in temperature: true or False? Explain.
3. What are the factors that affect solubility?
4. Consider the graph above:
 - a) What is the solubility of $K_2Cr_2O_7$ in 100g of water at $100^\circ C$?
 - b) What is the solubility of NaCl in 100g of water at $60^\circ C$?
 - c) Which substance whose elevation of temperature decreases solubility?

4.5. Preparation of normal salts



Learning activity 4.5

1. What products are formed when an acid reacts with a base?
2. Can an acid react with substances other than bases and still form a salt?
3. Do you think all salts can be prepared using the same method? Why or why not?
4. If a salt does not dissolve in water, how do you think it can be prepared?

Normal salts can be prepared in the laboratory using different chemical reactions. The common methods include reactions of acids with metals, metal oxides, bases, and non-metal oxides. We have two main types of salts: **soluble salts** and **insoluble salts**.

Soluble salts are prepared by first carrying out a chemical reaction to form a solution. After the reaction, the water evaporates so that crystals of the salt are formed. This process is called **crystallization**.

Insoluble salts, on the other hand, are prepared through a **precipitation reaction**. In this method, two solutions are mixed to form a solid that does not dissolve in water.

The mixture is then filtered. The liquid that passes through the filter paper is called the **filtrate**, while the solid left on the filter paper is called the **residue**. The residue is washed to remove impurities, dried properly, and then collected as the final product.

4.5.1 Preparing salts from the reaction of an acid with an active metal



Learning activity 4.5.1

Preparation of zinc sulphate $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (White Vitriol)

Materials: Dilute sulphuric acid (H_2SO_4), Zinc granules, 250 cm³ beaker, Filter funnel and filter paper, dish, Heat source, Glass rod.

Procedure:

1. Measure about 40 cm³ of dilute sulphuric acid into a 250 cm³ beaker.
2. Add about 10 g of zinc granules slowly to the acid.
3. Observe the reaction. Bubbles of hydrogen gas are produced.
4. Allow the reaction to continue until no more gas is evolved, showing that the reaction is complete.
5. Filter the mixture to remove any excess unreacted zinc.
6. Transfer the clear filtrate into a dish and heat gently to concentrate the solution until it reaches the crystallisation point.
7. Remove the dish from the heat and allow the solution to cool.
8. On cooling, needle-shaped crystals of zinc sulphate (white vitriol) are formed.
9. Separate the crystals and dry them between folds of filter paper.

Observation:

Colourless, needle-shaped crystals of zinc sulphate are obtained. Zinc sulphate, a **water-soluble salt**, can be prepared in the laboratory by reacting zinc metal with dilute sulphuric acid, followed by crystallization.

Chemical equation: $\text{Zn (s)} + \text{dil. H}_2\text{SO}_4 \text{ (aq)} \rightarrow \text{ZnSO}_4 \text{ (aq)} + \text{H}_2 \text{ (g)}$

Examples: a) $\text{Fe(s)} + 2\text{HCl(aq)} \rightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

b) $2\text{Na(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{Na}_2\text{SO}_4\text{(aq)} + \text{H}_2\text{(g)}$

4.5.2. Preparing salts from the reaction of acid with carbonates and hydrogen carbonates



Learning activity 4.5.2 (a)

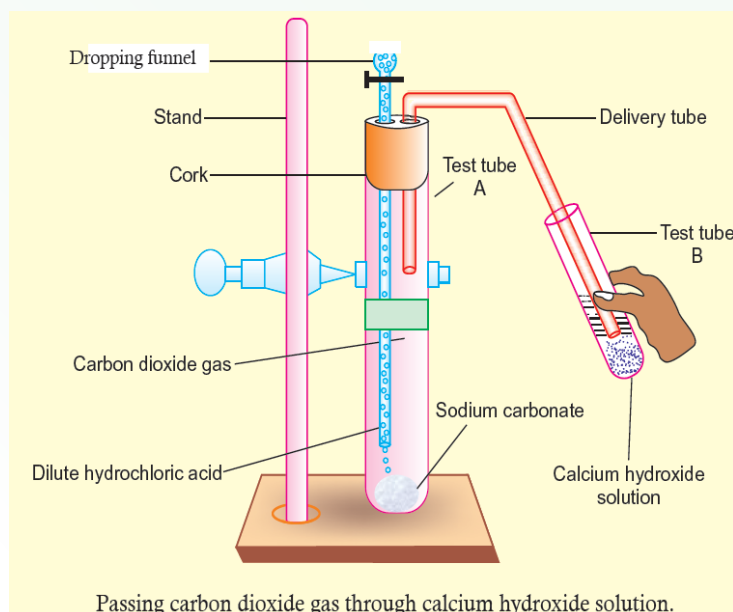
a) Preparation of sodium chloride

Materials:

- Sodium carbonate (Na_2CO_3)
- Calcium hydroxide solution ($\text{Ca}(\text{OH})_2$)
- Dilute hydrochloric acid (HCl)
- Sodium hydrogen carbonate (NaHCO_3)
- Two test tubes (A and B)
- Dropper

Procedure:

1. Take two test tubes and label them A and B.
2. Put about 0.5 g of sodium carbonate (Na_2CO_3) into test tube A.
3. Put about 10 ml of calcium hydroxide solution, $\text{Ca}(\text{OH})_2$, into test tube B.
4. Add about 2 ml of dilute hydrochloric acid (HCl) to test tube A.
5. Observe what happens in test tube A.
6. Pass the gas produced from test tube A into test tube B.
7. Record your observations.
8. Repeat the experiment using sodium hydrogen carbonate (NaHCO_3) instead of sodium carbonate.

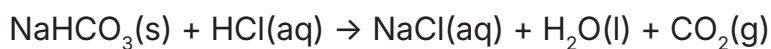


Passing carbon dioxide gas through calcium hydroxide solution.

9. a) What gas is produced when sodium carbonate or sodium hydrogen carbonate reacts with dilute hydrochloric acid?
 - i) Write balanced reaction equation

The reactions occurring in Activity 4.5.2. are written as:

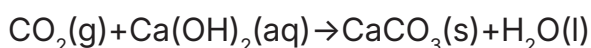
Test tube A:



Test tube B:

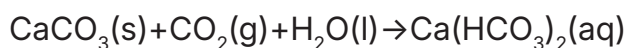
In the test tube B

Carbon dioxide reacts with limewater ($\text{Ca}(\text{OH})_2$) to form calcium carbonate, an insoluble white precipitate, and water.



Observation

- The clear limewater turns **milky or cloudy** due to the formation of a white precipitate of calcium carbonate.
- If excess carbon dioxide is passed through the solution the milkiness disappears because the precipitate dissolves to form soluble calcium hydrogen carbonate:



Learning activity 4.5.2 (b)

b) Preparation of calcium chloride

Materials required

- | | |
|----------------------------------|---|
| • Measuring cylinder | • Hydrochloric acid (50 cm ³) |
| • 250 cm ³ beaker (2) | • Calcium carbonate powder (6 g) |
| • Filter funnel | • Filter paper |
| • Conical flask | • Evaporating dish |
| • Bunsen burner | • Wire gauze |
| • Glass spatula | • Plastic vial |
| • Tripod stand | • Electronic top pan balance |

Procedure

1. Take 50 cm³ dilute hydrochloric acid in a beaker.
2. Using an electronic top pan balance, weigh 6.0 g of calcium carbonate in a small plastic bottle.
3. Add a spatula of calcium carbonate from the plastic bottle into the acid in the beaker.



4. Stir the mixture until all the calcium carbonate has dissolved before adding more calcium carbonate.
5. Continue to add more calcium carbonate to the acid until no more can dissolve and some of it is left at the bottom of the beaker.
6. Filter off the excess calcium carbonate with a piece of filter paper and a funnel.
7. Collect the filtrate and place it into an evaporating dish. Obtain crystals of calcium chloride from the solution by evaporation followed by crystallization.

Guiding questions

1. What do you observe when calcium carbonate is added to dilute hydrochloric acid?
2. Why is calcium carbonate added in small portions until no more dissolves?
3. Why is the mixture filtered before evaporation?
4. Why must the solution be concentrated by evaporation before crystals of calcium chloride can form?

When calcium carbonate is added to dilute hydrochloric acid, effervescence is observed. This bubbling is due to the release of carbon dioxide gas (CO_2), showing that a chemical reaction is taking place. As more calcium carbonate is added, it dissolves until the acid is completely used up; any excess solid remains unreacted. After filtration, a clear, colourless solution of calcium chloride (CaCl_2) is obtained. On evaporation and cooling, white crystals of calcium chloride are formed, confirming successful preparation of the salt.

Chemical equation:



Example:



4.5.3 Preparation of salts from reaction of acids with metal oxides



Learning activity 4.5.3

Preparation of copper sulphate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (Blue Vitriol)

Safety Precautions

1. Handle dilute sulphuric acid with care; avoid contact with skin and eyes.
2. Do not overheat the acid or the solution; warm gently to prevent splashing.

Materials:

Dilute sulphuric acid, copper (II) oxide (black solid), beaker (250 cm^3), glass rod, filter paper, funnel, evaporating dish, tripod stand, wire gauze, Bunsen burner.

Procedure:

1. Measure about 50 cm^3 of dilute sulphuric acid into a 250 cm^3 beaker and warm it gently.
2. Add small portions of black copper (II) oxide to the warm acid while stirring.
3. Continue adding copper (II) oxide until no more dissolves, indicating that the acid has been completely used.
4. Filter the mixture to remove any excess, unreacted copper (II) oxide.
5. Transfer the blue filtrate into an evaporating dish and heat gently to concentrate the solution.
6. When a hot saturated solution is obtained, remove it from heat and allow it to cool undisturbed.

Guiding questions

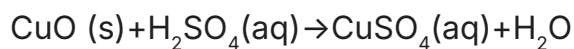
1. What happens when copper (II) oxide is added to dilute sulphuric acid?
2. Why is copper (II) oxide added little by little until no more dissolves?
3. Why is the mixture filtered after the reaction?
4. Why is the blue solution heated to concentrate it before cooling?
5. What does the formation of blue crystals show about the method used to prepare the salt?

When copper (II) oxide is added to dilute sulphuric acid, it dissolves and forms a blue solution, showing that a reaction has occurred to produce copper (II) sulphate. Adding the oxide little by little ensures that the acid is completely neutralized, and any excess solid remains unreacted.

Filtration removes the excess copper(II) oxide, leaving a clear blue solution of copper(II) sulphate. Heating the solution concentrates it to a saturated solution, which allows crystals to form on cooling.

The appearance of blue crystals confirms that copper(II) sulphate has been successfully prepared using the reaction of an acid with an insoluble base, followed by crystallization.

Chemical equation



Note: All basic metal oxides react with acids but there is not always a colour change. Magnesium oxide is a white powder. The salt produced when it reacts with acid is also white.

4.5.4. Preparing salts from reaction of acids with bases



Learning activity 4.5.4

Preparation of sodium sulphate

Materials:

Dilute sodium hydroxide solution, dilute sulphuric acid solution, phenolphthalein indicator, test tube, dropper.

Procedure:

1. Place about 2 ml of dilute sodium hydroxide solution in a clean test tube.
2. Add two drops of phenolphthalein indicator to the solution and observe the colour.
3. Add dilute sulphuric acid drop by drop to the solution while gently shaking the test tube.
4. Stop adding the acid when the pink colour just disappears.
5. Add a few drops of dilute sodium hydroxide solution to the mixture again and observe any change.

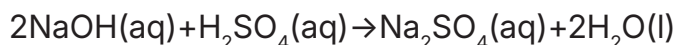
Guiding Questions

1. What colour is observed when phenolphthalein is added to sodium hydroxide solution?
2. What happens to the colour when dilute sulphuric acid is added drop by drop?
3. Why does the pink colour disappear at the point of neutralisation?
4. Why does the pink colour reappear when sodium hydroxide is added again?

When phenolphthalein is added to sodium hydroxide solution, the solution turns pink, showing that it is alkaline. As dilute sulphuric acid is added drop by drop, the pink colour gradually disappears, indicating that the alkali is being neutralised by the acid. At the point of neutralisation, the solution becomes colourless. When sodium

hydroxide is added again, the pink colour reappears, showing that the solution has become alkaline once more.

Equation:



This method demonstrates the preparation of salts by neutralisation, where an acid reacts with an alkali to form a salt and water. An indicator is used to show when neutralisation is complete.

General equation:

Acid + Alkali $\rightarrow$ Salt + Water

Many other soluble salts can be prepared, for example:

- Sodium chloride: $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
- Potassium nitrate: $\text{KOH} + \text{HNO}_3 \rightarrow \text{KNO}_3 + \text{H}_2\text{O}$

This method is suitable for preparing salts of sodium, potassium, and ammonium, which are all soluble in water.

4.5.5 Preparing salts from precipitation.



Learning activity 4.5.5

Preparation of lead carbonate

Materials Required

- Distilled water, Sodium carbonate (Na_2CO_3) and Lead nitrate ($\text{Pb}(\text{NO}_3)_2$)
- Test tubes and Glass rod

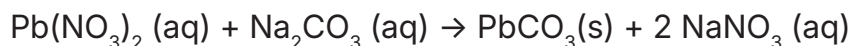
Procedure

1. Dissolve a small amount of sodium carbonate in water to make a solution.
2. Dissolve a small amount of lead nitrate in water to make a solution.
3. Pour about 20 ml of sodium carbonate solution into a clean test tube.
4. Add 20 ml of lead nitrate solution to the test tube slowly while stirring gently with a glass rod.
5. Observe any changes in the mixture, such as the formation of a precipitate.

Guiding Questions

1. What do you observe when the two solutions sodium carbonate and lead nitrate are mixed?
2. Why does a white precipitate form immediately?
3. How can you identify the salt formed in this reaction?
4. Why is this method called the reaction of two solutions to prepare salts?

When sodium carbonate and lead nitrate solutions are mixed, a precipitation reaction occurs. The insoluble salt, lead carbonate (PbCO_3), forms as a white solid, while the other product, sodium nitrate, remains dissolved in solution.



This activity illustrates how soluble salts can react to form an insoluble salt by precipitation method.

4.5.6. Preparation of salts from a reaction of metal oxides and non-metal oxides, and from a reaction of a base and a non-metal oxide.

a) From reaction of metal oxides and non-metal oxides



Learning activity 4.5.6.a

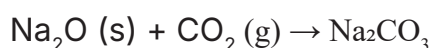
1. Give the formula and names of at least 2 metal oxides and 2 non-metal oxides
2. Complete the following equations:
 - a) $\text{Na}_2\text{O} (\text{s}) + \text{CO}_2 (\text{g}) \rightarrow$
 - b) $\text{CaO} (\text{s}) + \text{SO}_3 (\text{g}) \rightarrow$
 - c) $\text{MgO} (\text{s}) + \text{CO}_2 (\text{g}) \rightarrow$

Salts can also be prepared by the reaction of metal oxides with non-metal oxides. This works because metal oxides are basic while non-metal oxides are acidic, allowing them to combine and form a salt.

Examples:

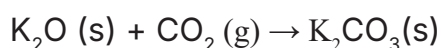
1. Sodium carbonate (Na_2CO_3)

Sodium oxide reacts with carbon dioxide at **500–550°C** to form sodium carbonate:



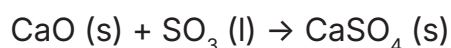
2. Potassium carbonate (K_2CO_3)

Potassium oxide reacts with carbon dioxide at **400–450°C** to form potassium carbonate:



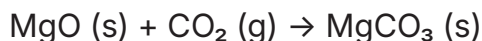
3. Calcium sulphate (CaSO_4)

Calcium oxide reacts with sulphur trioxide to form calcium sulphate:



4. Magnesium carbonate (MgCO_3)

Magnesium oxide reacts with carbon dioxide at high temperature to form magnesium carbonate:



Note: Reactions between metal oxides and carbon dioxide generally require very high temperatures to proceed effectively.

b) From a reaction of base and non-metal oxide



Learning activity 4.5.6.b

Preparation of Calcium Carbonate

Materials Required

- Calcium hydroxide solution (lime water)
- Carbon dioxide gas source
- Test tube
- Glass tubing or delivery tube
- Glass rod

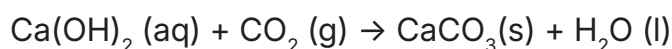
Procedure

1. Pour calcium hydroxide solution (lime water) into a clean test tube.
2. Pass carbon dioxide gas slowly through the lime water using a delivery tube.
3. Observe any changes in the solution and record your observations.

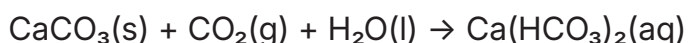
Guiding questions

1. What happens to lime water when carbon dioxide is passed through it?
2. Why does a white precipitate form?
3. What change occurs if excess carbon dioxide is added?
4. How does this activity illustrate the reaction of a base with a non-metal oxide to form a salt?

Passing carbon dioxide through calcium hydroxide solution produces calcium carbonate, an insoluble salt, which appears as a white precipitate, demonstrating that a base reacts with a non-metal oxide to form a salt.

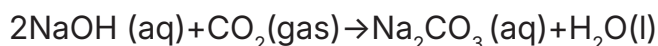


If excess carbon dioxide is added, calcium carbonate reacts further to form soluble calcium hydrogen carbonate, showing the importance of the amount of non-metal oxide in such reactions.

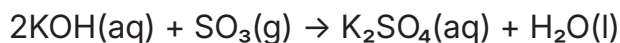


Here other examples of reactions between a metal oxide (basic oxide) and a non-metal oxide (acidic oxide):

1. Reaction between sodium hydroxide and carbon dioxide:



2. Reaction between potassium hydroxide and sulfur trioxide:

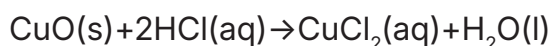


General Observation

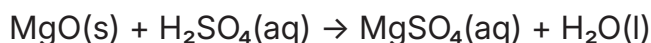
A base reacts with a non-metal oxide to form a salt and water. Non-metal oxides usually behave as acidic oxides.

c) From reactions between acids and metal oxides

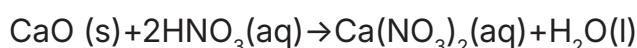
1. Reaction between hydrochloric acid and copper (II) oxide:



2. Reaction between sulfuric acid and magnesium oxide:



3. Reaction between nitric acid and calcium oxide:

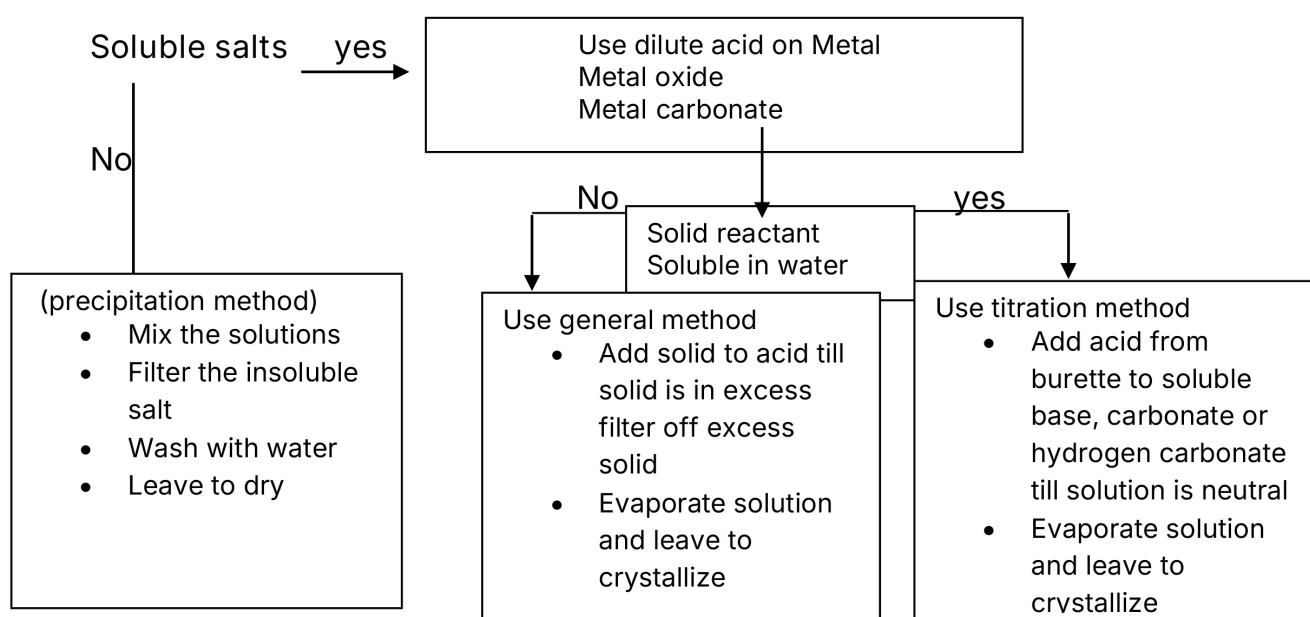


General observation

An acid reacts with a metal oxide to form a salt and water. This is a type of **neutralisation reaction** as metal oxides are generally basic

Methods of preparation of salts (summary)

The method chosen to prepare a certain salt depends on whether the salt is soluble or insoluble. Preparation procedures can be summarized as shown in the flow chart shown below:





Application activity 4.5

1. What are the products formed when an acid reacts with a base?
2. What is the difference between a soluble salt and an insoluble salt?
3. Name two methods used to prepare soluble salts.
4. What gas is produced when zinc reacts with dilute sulphuric acid?
5. Why is the solution heated during the preparation of soluble salts?
6. What gas is produced when calcium carbonate reacts with hydrochloric acid?
7. What observation shows that carbon dioxide has been passed through limewater?
8. Why is excess copper (II) oxide filtered off during the preparation of copper(II) sulphate?
9. What colour change occurs when sulphuric acid is added to sodium hydroxide containing phenolphthalein?
10. What is a precipitation reaction?
11. Complete the following equations:
 - a) $\text{Na}_2\text{O} + \text{CO}_2 \rightarrow$
 - b) $\text{MgO} + \text{CO}_2 \rightarrow$
 - c) $\text{CaO} + \text{SO}_3 \rightarrow$

4.6. Uses and sources of salts



Learning activity 4.6

Uses and sources of salts

1. Name some common names known today?
2. Give and explain sources of salts like sodium chloride (NaCl), sodium carbonate, and sodium hydrogen carbonate.

Salts are substances that are found everywhere around us underground, on the earth's surface, and even in dried-up seabed. Some salts have even come from outer space in meteors!

The main sources of salts are the seas and oceans, where they are dissolved in water. There are many types of salts, and they can be produced in different ways:

- White salt is made by evaporating salty water (brine) in special containers.
- Rock salt, which is used to make roads slippery in winter, is mined from ancient deposits.
- In some countries, sunlight is used to evaporate seawater and produce salt.

4.6.1 Sodium Chloride (NaCl)

Sources

Sodium chloride, or common salt, is present in seawater along with other salts. It can also be found as solid rock salt in many parts of the world. Large crystals of rock salt are often brown due to impurities.

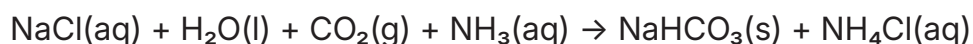
Uses of NaCl

- Food seasoning and preservation
- Preparation of chemicals
- De-icing roads in cold countries
- Production of chlorine and sodium hydroxide in industry

4.6.2 Washing Soda (Sodium Carbonate, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$)

A) Sources of sodium carbonate

Washing soda is produced from sodium chloride using the Solvay (Ammonia-soda) process. In this process, carbon dioxide is bubbled through brine saturated with ammonia, forming sodium hydrogen carbonate:



- Sodium hydrogen carbonate precipitates and is filtered.
- On heating, it decomposes to soda ash (Na_2CO_3):
$$2\text{NaHCO}_3 \xrightarrow{\text{heat}} \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$$
- Dissolving soda ash in water and crystallizing produces washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$).

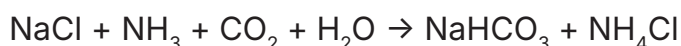
B) Uses of sodium carbonate

- Manufacture of glass, borax, soap, and caustic soda.
- Used in paper, paint, and textile industries.
- Softens hard water.
- Used in laundry and as a laboratory reagent.

4.6.3 Baking Soda (Sodium Hydrogencarbonate, NaHCO_3)

A) Sources of sodium hydrogen carbonate

Sodium hydrogencarbonate is a primary product of the **Solvay process**:



It can also be prepared by passing CO_2 through a sodium carbonate solution, causing NaHCO_3 to crystallize:



B) Uses of sodium hydrogen carbonate:

- **Baking**

- It is used in baking cakes and bread.
- It releases carbon dioxide gas when heated, making dough rise.

- **Antacid**

- It is used to relieve indigestion and excess stomach acid.

- **Fire extinguishers**

- Used in some fire extinguishers because it releases carbon dioxide, which helps put out flames.

- **Cleaning agent**

- Used for cleaning and removing stains or odors.

- **Laboratory use**

- Used to prepare carbon dioxide gas in laboratories.

- **Neutralising acids**

- Used to neutralise acidic substances in industries and homes.



Application activity 4.6

1. Write two uses of sodium carbonate.
2. What is the molecular formula of washing soda?
3. The chemical name of baking soda is_____.
4. Sodium chloride is present as an ingredient in antacids. (True or false)
5. The crystals of sodium chloride obtained from rocks are generally white in colour. (True or False)

4.7. Identification test of ions (cations and anions)

Many substances around us contain tiny charged particles called **ions**. These ions are present in salts, medicines, fertilizers, drinking water, foods, and many industrial products.

Knowing how to identify ions is very important in everyday life and in many fields. In medicine, ion tests help doctors and laboratory technicians detect substances in blood and urine. In agriculture, farmers use ion testing to determine soil nutrients needed for healthy crop growth. In industries, chemists test ions to control the quality of products such as fertilizers, soaps, beverages, and pharmaceuticals. Water treatment plants also use ion tests to check whether water is safe for drinking.

In this lesson, we will learn simple chemical tests used to identify different cations and anions. Through colour changes, precipitates, and gas tests, you will discover how chemists identify unknown substances and solve chemical problems like scientific detectives.



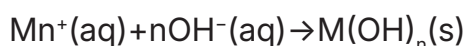
Learning activity 4.7

1. What should you do first before testing ions in a substance?
2. Why is it important to dissolve the substance in water?
3. What are reagents used for in ion testing?
4. What kinds of changes should you observe during the test?
5. How can a precipitate help identify an ion?
6. Why must observations be made carefully?
7. After observing the reaction, how can you determine which ion is present?
8. What conclusion can you make if no visible change occurs?
9. Why are ion tests important in laboratories, medicine, agriculture, and industries?

4.7.1. Identification of cations

Cations are positively charged ions. Ions can be identified using dilute sodium hydroxide solution (NaOH) and dilute aqueous ammonia (NH₃(aq)), as these chemicals can produce a precipitate or other observable products.

Most metal cations react with hydroxide ions (OH⁻) to form insoluble metal hydroxides, which appear as precipitates.

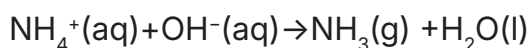


If an ion does not produce a precipitate, it can be identified using a flame test.

Ammonium ions (NH₄⁺)

Ammonium ions do not form a precipitate with either aqueous sodium hydroxide or aqueous ammonia and it can not make flame test as other positive ions. Its test then is special.

When a mixture of ammonium ions and any hydroxide is heated, ammonia gas is released.



Tests for ammonia gas:

- Has a sharp, characteristic smell
- Turns moist red litmus paper blue
- Test of ammonia with hydrochloric acid: Ammonia gas can be identified by bringing a glass rod dipped in concentrated hydrochloric acid near the gas: dense white fumes of NH₄Cl are produced.

Equation of the reaction that occurs: $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$

1. Reactions of metal cations with NaOH and aqueous ammonia

The reactions of metal cations with sodium hydroxide solution (NaOH) and aqueous ammonia (NH₃(aq)) are important qualitative analysis tests used to identify metal ions (cations) in unknown solutions. These tests are based on the formation of characteristic precipitates and their behavior in excess reagent.

When a few drops of NaOH or aqueous ammonia are added to a solution containing metal ions, insoluble metal hydroxides may form. The colour of the precipitate and whether it dissolves in excess reagent help to identify the cation present.

Examples:

- **Calcium ions (Ca²⁺)**

Form a white precipitate with aqueous sodium hydroxide, but no precipitate with aqueous ammonia. The precipitate does not dissolve in excess sodium hydroxide.

- **Zinc ions (Zn²⁺)**

Form a white precipitate with both aqueous sodium hydroxide and aqueous ammonia.

The precipitate dissolves in excess of both solutions.

- **Magnesium ions (Mg²⁺)**

Form a white precipitate with both aqueous sodium hydroxide and aqueous ammonia.

The precipitate does not dissolve in excess of either solution.

- **Copper(II) ions (Cu²⁺)**

Form a blue precipitate with both aqueous sodium hydroxide and aqueous ammonia.

The precipitate dissolves in excess ammonia to give a dark blue solution.

- **Iron(II) ions (Fe²⁺)**

Form a green precipitate with both aqueous sodium hydroxide and aqueous ammonia.

The precipitate does not dissolve in excess of either solution.

- **Iron(III) ions (Fe³⁺)**

Form a brown precipitate with both aqueous sodium hydroxide and aqueous ammonia.

The precipitate does not dissolve in excess of either solution.

- **Aluminium ions (Al³⁺)**

Form a white precipitate with both aqueous sodium hydroxide and aqueous ammonia.

The precipitate dissolves in excess sodium hydroxide but does not dissolve in excess aqueous ammonia.

Figures 4.7 a) and 4.7 b) summarize the reactions of cations with aqueous sodium hydroxide and aqueous ammonia, respectively.

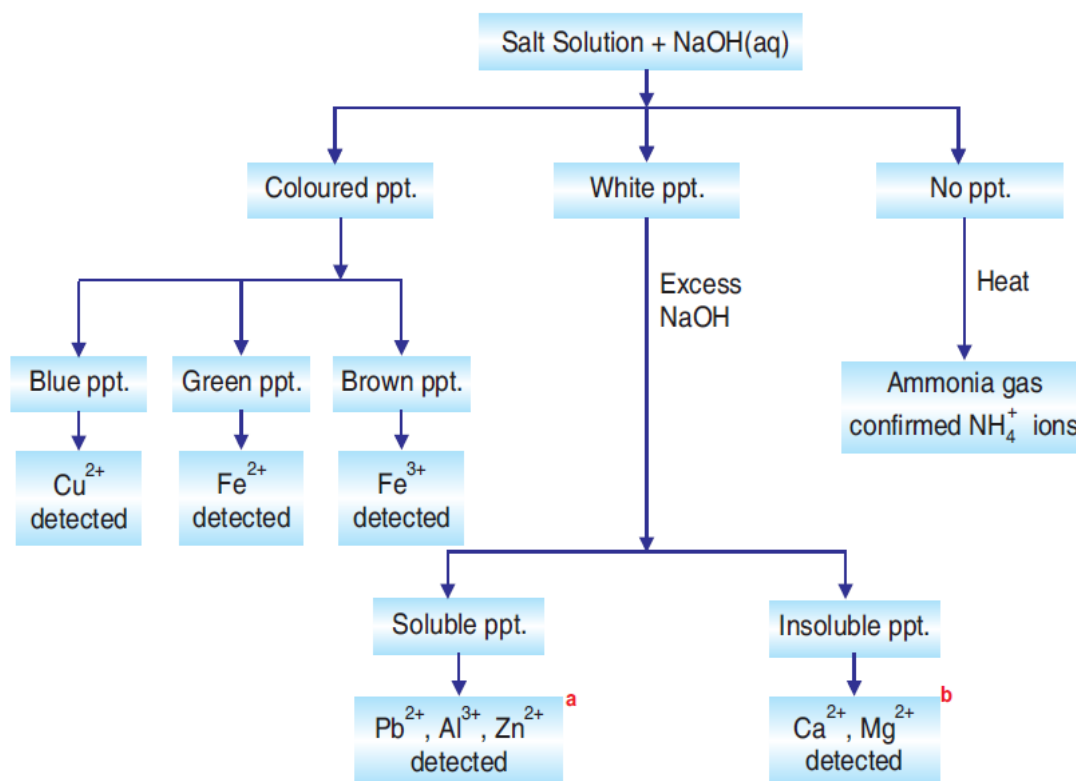


Figure 4.7 a) Identification of cations using aqueous sodium hydroxide

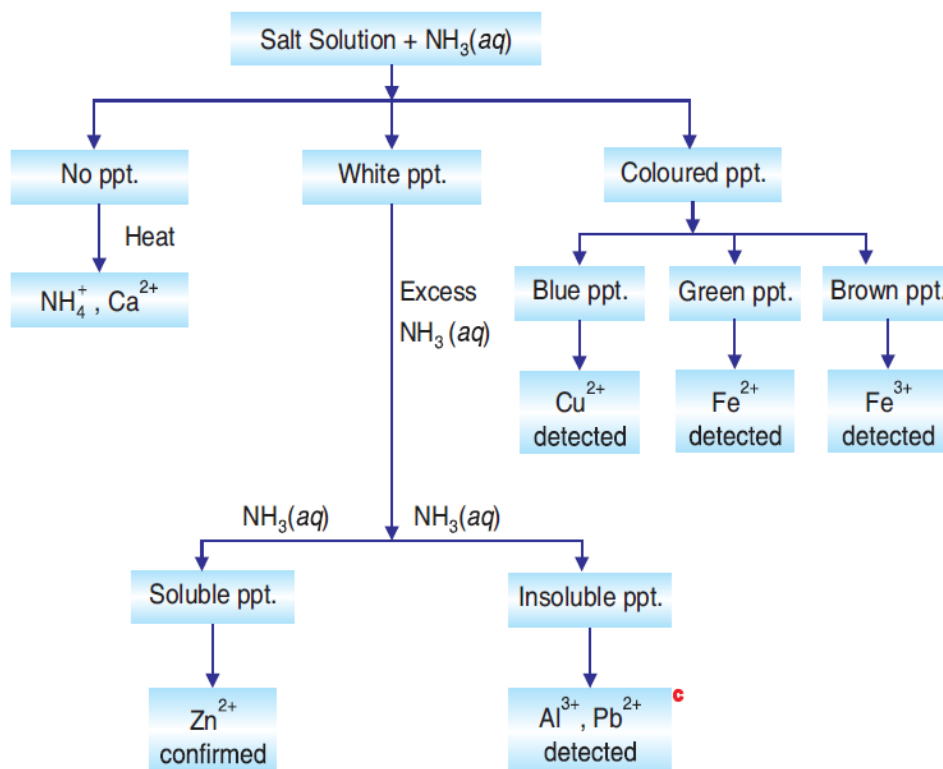


Figure 4.7 b): Identification of cations using aqueous ammonia

Distinguishing between similar cations

In some cases, different cations produce similar results when tested with sodium hydroxide or aqueous ammonia. To identify such ions accurately, additional confirmatory tests using specific reagents are carried out. These tests help to distinguish between cations that form similar precipitates in the main identification tests.

i) Distinguishing between Al^{3+} , Zn^{2+} and Pb^{2+}

The following tests are used to distinguish between aluminium (Al^{3+}), zinc (Zn^{2+}) and lead(II) (Pb^{2+}) ions.

- Test with aqueous potassium iodide (KI):
 - If a yellow precipitate is formed, the cation present is Pb^{2+} .
 - If no precipitate is formed, the cation may be Al^{3+} or Zn^{2+} .
- Test with aqueous hydrogen sulphide (H_2S):
 - Formation of a white precipitate with H_2S is not a suitable confirmatory test for Zn^{2+} ; use appropriate confirmatory tests for Al^{3+} , Zn^{2+} and Pb^{2+} .
- Test with aqueous ammonium hydroxide (NH_4OH):
 - A white precipitate that does not dissolve in excess ammonium hydroxide confirms Al^{3+} .

ii) Distinguishing between Al^{3+} and Pb^{2+}

Aluminium and lead (II) ions can be distinguished using aqueous potassium iodide (KI).

- Formation of a yellow precipitate confirms the presence of Pb^{2+} .
- No precipitate indicates the presence of Al^{3+}

iii) Distinguishing between Ca^{2+} and Mg^{2+}

Calcium and magnesium ions are distinguished using aqueous ammonia ($\text{NH}_3(\text{aq})$).

- Formation of a white precipitate that does not dissolve in excess ammonia confirms Mg^{2+} .
- No precipitate confirms the presence of Ca^{2+} .



Application activity 4.7(a)

1. NH_4^+ ions form a precipitate with aqueous sodium hydroxide. (True or False)
2. Fe^{2+} ions form _____ precipitate with aqueous sodium hydroxide.
3. How will you identify Cu^{2+} ions in a sample of solution?
4. The precipitate formed by Al^{3+} ion does not dissolve in
 - a) water, (b) excess aqueous sodium hydroxide
 - b) excess aqueous ammonia, (d) All of these.
5. Fe^{3+} ions form a brown precipitate with _____ and _____.

4.7.2 Identification of anions

Anions are negatively charged ions. They are identified using specific chemical reagents that produce characteristic precipitates or gases.

The common reagents used in the identification of anions include:

- Dilute hydrochloric acid (HCl)
- Aqueous barium nitrate ($\text{Ba}(\text{NO}_3)_2$)
- Aqueous barium chloride (BaCl_2)
- Aqueous silver nitrate (AgNO_3)
- Dilute nitric acid (HNO_3)
- Dilute aqueous ammonia ($\text{NH}_3(\text{aq})$)
- Red litmus paper

A) Identification of specific anions

1. Chloride ions (Cl^-)

When aqueous silver nitrate is added to a solution containing chloride ions, a white precipitate is formed.

- The precipitate does not dissolve in excess dilute nitric acid.
- It dissolves in excess aqueous ammonia.
- On standing in sunlight for about 10–15 minutes, the white precipitate darkens

2. Iodide ions (I^-)

Iodide ions form a yellow precipitate when the solution is first acidified with dilute nitric acid and then treated with aqueous silver nitrate or aqueous lead(II) nitrate.

- The precipitate is insoluble in excess aqueous ammonia.
- It dissolves in hot water, forming a colourless solution.

3. Nitrate ions (NO_3^-)

We can conduct a variety of experiments based on our preferences and the availability of lab equipment.

- Boil the suspected nitrate with sodium hydroxide solution and fine aluminium powder. The aluminium powder converts the nitrate ion into ammonia gas, the gas that changes the moist red litmus paper blue.
- Introduce iron(II) sulfate solution followed by concentrated sulfuric acid to perform the classic 'brown ring' test.
- Boil the suspected nitrate with concentrated sulfuric acid and copper metal (or a piece of paper representing carbon). The brown gas (NO_2) confirms that the substance is a nitrate.

4. Carbonate ions (CO_3^{2-})

Carbonate ions react with dilute hydrochloric acid to produce carbon dioxide gas.

- Effervescence (bubbling) of a colourless and odourless gas is observed.
- The gas turns limewater milky.
- When excess carbon dioxide is bubbled through limewater, the resulting milkiness will disappear.
- Carbonate ions turn moist red litmus paper blue.

5. Sulphite ions (SO_3^{2-})

- Sulphite ions form a white precipitate with barium chloride or barium nitrate; the precipitate dissolves in dilute acid with release of sulfur dioxide.
- The precipitate dissolves in both dilute hydrochloric acid and dilute nitric acid.

6. Sulphate ions (SO_4^{2-})

- Sulphate ions form a white precipitate with both aqueous barium chloride and aqueous barium nitrate.
- The precipitate does not dissolve in dilute hydrochloric acid or dilute nitric acid.

Figure 4.7 c) summarizes the reactions of anions with different chemical reagents used for their identification.

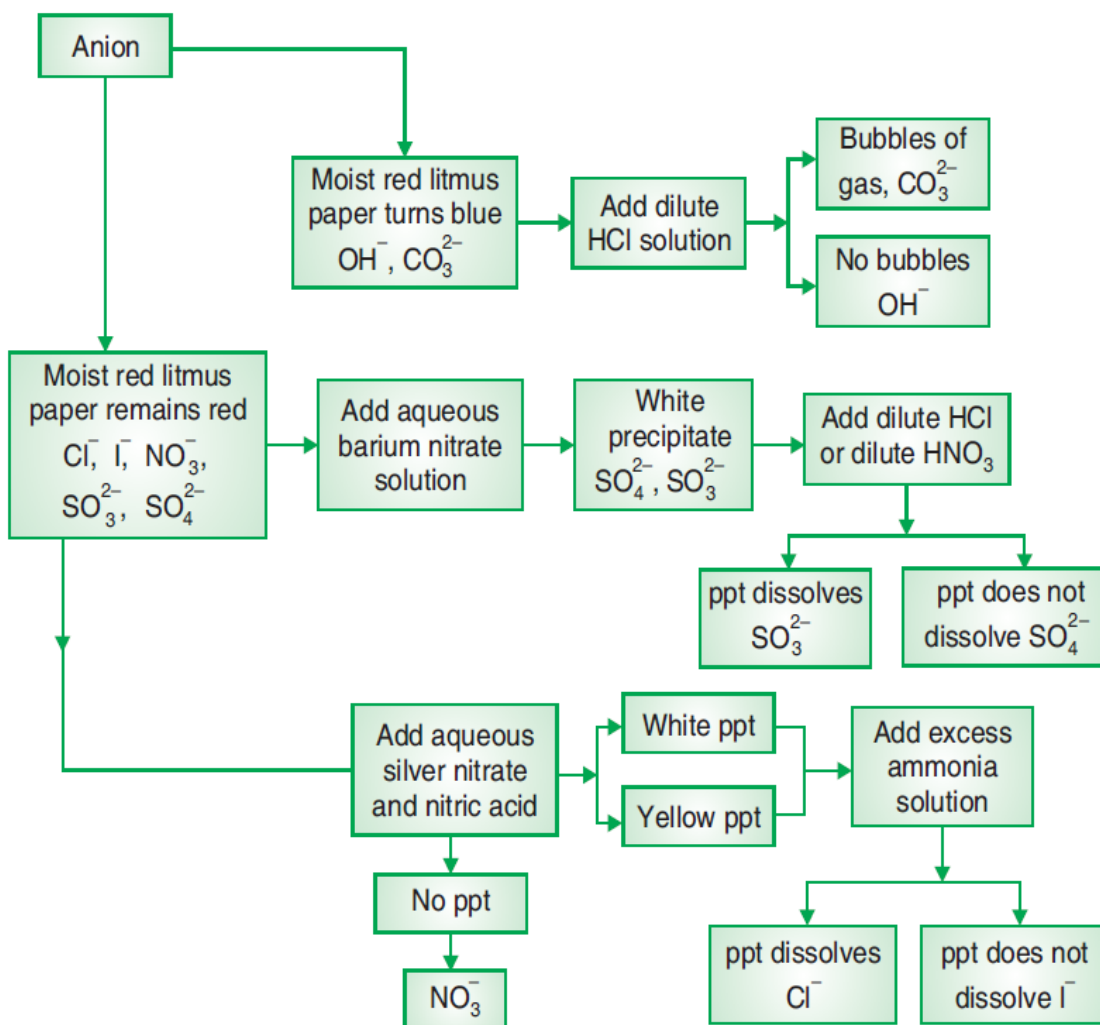


Figure 4.7 c): Identification of anions



Application activity 4.7 b)

1. What are anions?
2. Which reagent is commonly used to test chloride ions?
3. What observation shows the presence of chloride ions?
4. How are iodide ions identified?
5. What gas is produced when carbonate ions react with dilute hydrochloric acid?
6. How can carbon dioxide gas be confirmed?
7. What observation confirms sulphate ions in a solution?
8. How can nitrate ions be identified using sodium hydroxide and aluminium powder?

For more details visit the link:

<https://www.savemyexams.com/a-level/chemistry/aqa/17/revision-notes/4-as-practical-skills/4-2-organic-and-inorganic-chemistry-practicals/4-2-2-identifying-anions-and-cations/>



4.8 End of unit assessment

A) Knowledge and understanding

1. Which of the following is an example of an acid salt?

- | | |
|---------------------|-----------------------|
| A) NaCl | C) NaHSO ₄ |
| B) KNO ₃ | D) CaCO ₃ |

2. Which salt is insoluble in water?

- | | |
|----------------------|---------------------|
| A) Potassium nitrate | C) Barium sulphate |
| B) Sodium chloride | D) Sodium carbonate |

3. What is formed when an acid reacts with a base?

- | | |
|----------------------|---------------------|
| A) Salt and hydrogen | C) Water and oxygen |
| B) Salt and water | D) Salt only |

4. Which reagent is used to test chloride ions?

- | | |
|-----------------------------|---------------------------|
| A) Sodium hydroxide | C) Aqueous silver nitrate |
| B) Dilute hydrochloric acid | D) Limewater |

5. What is a salt?

6. State two physical properties of inorganic salts

7. Why do soluble salts conduct electricity in aqueous solution?

8. What observation confirms the presence of carbonate ions?

B) Skills Questions

9. Describe how copper(II) sulphate crystals can be prepared from copper(II) oxide and sulphuric acid.

10. A student adds aqueous sodium hydroxide to a solution and observes a blue precipitate that dissolves in excess ammonia to form a dark blue solution. Identify the ion present.

11. Differentiate between saturated and unsaturated solutions.

12. Explain how calcium carbonate can be prepared by precipitation.

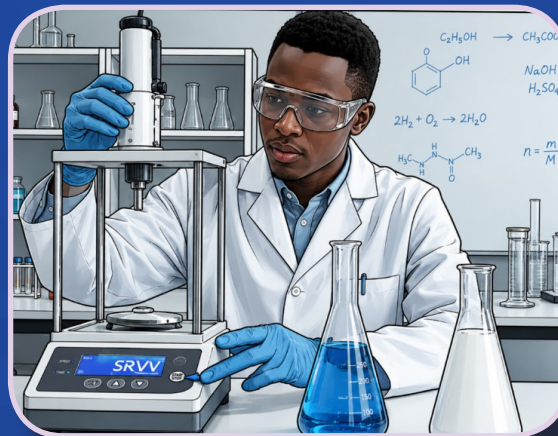
13. Describe the test and observation for chloride ions.

C) Attitude and values

14. Why is it important to follow safety rules when carrying out chemical reactions?

UNIT 5:

THE MOLE CONCEPT



Key unit competence

Apply the mole concept in quantitative chemical calculations involving mass, moles, and chemical formulas.

Learning Objectives

By the end of this unit you should be able to:

Knowledge and understanding

- Explain the mole concept.
- Explain the concepts of relative atomic mass, relative formula mass, relative molecular mass, molar mass, limiting reactant, empirical and molecular formulae.

Skills

- Calculate the mass composition of an element in a given compound.
- Apply the formula to calculate molar mass, relative atomic mass, relative formula mass, relative molecular mass, molar mass, limiting reactant, empirical and molecular formulae.
- Use Stoichiometric calculation in given chemical equations

Values and attitudes

- Develop a teamwork approach in research, group activities, and exercises.
- Demonstrate perseverance and confidence when solving numerical problems.
- Appreciate the importance of the mole concept in understanding chemical reactions and industrial processes.
- Work collaboratively and responsibly when performing calculations or practical tasks related to quantitative chemistry

Introduction

Have you ever wondered how chemists know exactly how much of a substance to use, even though atoms and molecules are far too small to count one by one? Just as we count eggs in dozens or measure rice in cups, chemists use a special way of counting particles in chemistry. This powerful idea is called the mole concept. It helps us link the tiny world of atoms and molecules to the visible amounts of substances we can weigh and measure in the laboratory. Understanding the mole concept will make chemical equations meaningful, experiments accurate, and problem-solving in chemistry much easier.



Introductory activity

Discovering the Mole in Everyday Life

Instructions

Think carefully about the questions below. Write a short report summarizing your thoughts and be ready to share your ideas with a classmate. Use examples from everyday life to support your answers.

Questions:

1. How many eggs are in a **dozen**?
2. How many eggs are in a **tray**?
3. Why is it impossible to count atoms one by one, unlike counting books, pencils, or other objects?
4. Why are some goods sold in groups rather than individually?
5. How is a dozen similar to a mole, and how are they different?
6. Why is the mole concept important in chemistry? Give reasons for its use in chemical calculations and reactions.

Reflection Tip:

Think of the mole as a “chemist’s dozen” that makes it easier to work with unimaginably tiny particles. How might this idea simplify chemical calculations in the lab?

5.1. Avogadro number and the mole concept



Learning activity 5.1

Linking number of particles to mass of particles

1. Why do people count some items in groups such as dozens, packets, or kilograms instead of counting one by one?
2. If you were asked to count all the grains of sand on a beach one by one, would it be practical? Why?
3. Atoms and molecules are extremely tiny and invisible to the naked eye. How do you think chemists measure or count them in the laboratory?
4. Why is it important for chemists to know the exact amount of substances used in a chemical reaction?
5. What might happen if too much or too little of a substance is used during an experiment or in manufacturing medicines?
6. The periodic table gives relative atomic masses of elements. What do you think these masses help chemists to determine?
7. How do you think the mass of a substance can help us estimate the number of particles it contains?
8. In everyday life, we use units such as a dozen for eggs or a ream for paper. Do you think chemistry also has a special counting unit for particles? Explain your idea.
9. Why do you think chemists cannot simply count atoms and molecules directly?
10. How do you think understanding the relationship between mass and number of particles can help in solving chemical problems?

In chemistry, substances are made of extremely tiny particles such as atoms, molecules, and ions, which cannot be counted one by one. To solve this problem, chemists use a special counting unit called the **mole**. The mole concept helps chemists measure and compare very large numbers of particles using measurable quantities such as mass.

The word mole was introduced around 1896 by the German physical chemist Wilhelm Ostwald, derived from the Latin word mole, meaning a pile or heap. Chemists think of a substance as a heap of atoms or molecules.

In everyday life, we use counting units like a dozen (12 objects) or a gross (144 objects) to handle large quantities. In chemistry, the unit used for counting atoms, ions, or molecules is the **mole**. SI symbol of mole is **mol**.

A mole is the amount of substance that contains a fixed number of particles called Avogadro's number denoted N_A , in honor of Italian scientist Amedeo Avogadro ($N_A = 6.022 \times 10^{23}$ particles). Therefore, Avogadro's number is the number of particles present in one mole of a substance. These particles may be atoms, molecules, ions, or electrons.

This means that:

- 1 mole of carbon atoms contains 6.022×10^{23} carbon atoms.
- 1 mole of water molecules contains 6.022×10^{23} water molecules.
- 1 mole of chloride ions contains 6.022×10^{23} ions

So, One mole of any particle (atom, molecule, or ion) contains 6.022×10^{23} particles.

From experiments, scientists found that 12 g of ^{12}C contains 6.022×10^{23} atoms. Therefore

Then, the mass of 1 mole of a substance equals its relative atomic or molecular mass in grams.

The atomic mass (A) of an element gives us the mass of one atom of that element in atomic mass units (u). To get the mass of 1 mole of atom of that element, we have to take the same numerical value but change the units from 'u' to 'g'. The mass of atoms in grams is also known as gram atomic mass.

Example:

Table 5.1: Relationship Between Atomic Mass, Gram Atomic Mass, and Number of Atoms

Element	Atomic mass (u)	Gram atomic mass (g)	Number of atoms in 1 g / 1 mol
Hydrogen (H)	1 u	1 g	6.022×10^{23} atoms
Oxygen (O)	16 u	16 g	6.022×10^{23} atoms
Carbon (C)	12 u	12 g	6.022×10^{23} atoms

In summary:

- 1 mole = 6.022×10^{23} particles
- Mass of 1 mole = relative atomic/molecular mass in grams
- Atomic mass units (u) indicate grams for mole calculations

Below is summary on how Avogadro number N_A can be related to mole and mass of any particle (atoms, molecules, ions)

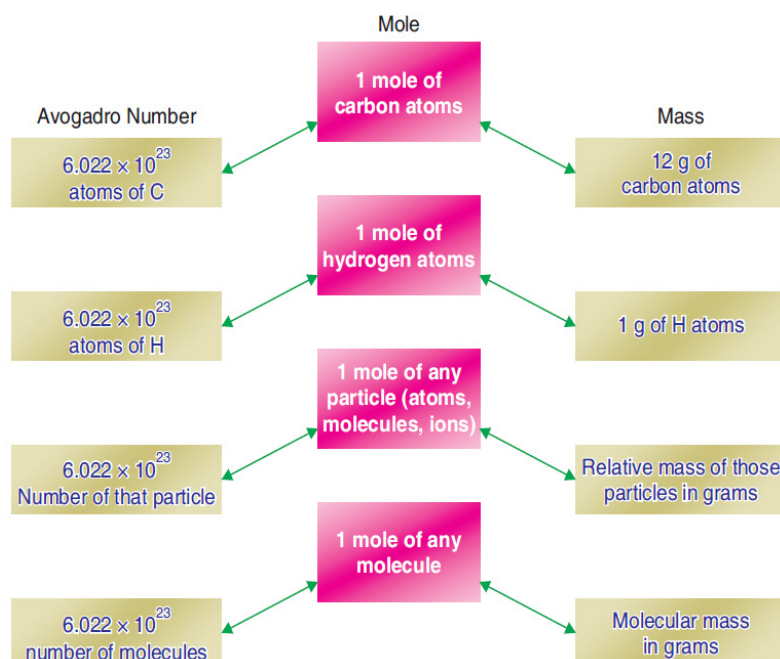


Figure 5.1: Relationship between Avogadro number, mole and mass.

This shows clearly how mass, number of particles, and the mole are all linked, forming the basis for chemical calculations.



Application activity 5.1

1. Avogadro number is the number of atoms in 1 gram-atom of an element. (True or False)
2. A group of 6.022×10^{23} particles (atoms, ions, or molecules) of a substance is called a _____ of that substance.
3. One mole of CO_2 contains _____ atoms of carbon.
4. The mass of one Avogadro number of nitrogen atoms is equal to:
a) 14 amu b) 14 g c) 28 g d) 14 kg
5. Mole is a link between the _____ of atoms (or molecules) and the _____ of atoms (or molecules).

5.2 Calculation of the number of moles



Learning activity 5.2

1. If 12g of carbon atom corresponds to 1 mole of the same atom, how many moles of carbon are 36g?
2. Calculate the number of moles of ions found in 3.011×10^{23} ions of a given element

The mole is the unit that allows us to convert between the number of particles (atoms, molecules, or ions) and the mass of a substance.

Ways of Calculating the Number of Moles

i) From the number of particles

If the total number of atoms or molecules is known, the number of moles is found by dividing by Avogadro's number, since 1 mole always contains 6.022×10^{23} particles.

$$\text{Number of moles} = \frac{\text{Number of particles}}{\text{Avogadro's number}}$$

ii) From the mass of an element

When the mass of an element is known, the number of moles is obtained by comparing the given mass with the molar mass (mass of 1 mole of the element).

$$\text{Number of moles} = \frac{\text{Mass of element (g)}}{\text{Molar mass of element (g/mol)}}$$

iii) From the mass of a molecule or compound

For molecules or compounds, the molar mass is the sum of atomic masses of all atoms in the molecule.

$$\text{Number of moles} = \frac{\text{Mass of substance (g)}}{\text{Molar mass of substance (g/mol)}}$$

Example 1

Calculate the number of moles of 12.044×10^{23} helium atoms.

Solution: Since 1 mole contains 6.022×10^{23} atoms, dividing the given number of atoms by Avogadro's number gives the number of moles.

$$\text{No. of moles} = n = \frac{\text{No. of particles}}{\text{Avogadro number}} = \frac{12.044 \times 10^{23} \text{ atoms}}{6.022 \times 10^{23} \text{ atoms/mol}} = 2 \text{ mols}$$

Example 2

Calculate the number of particles in 0.1 mole of carbon atoms.

Solution:

If 1 mole contains 6.022×10^{23} particles, then a fraction of a mole will contain the same fraction of that number.

$$\begin{aligned} \text{No. of particles} &= \text{No. of moles} \times \text{Avogadro number} \\ &= 0.1 \text{ mol} \times 6.022 \times 10^{23} \text{ atoms/mol} = 6.022 \times 10^{22} \text{ atoms} \end{aligned}$$

Example 3

Calculate the number of moles of 3.011×10^{23} hydrogen atoms.

Solution:

Here, the given number of atoms is exactly half of Avogadro's number, so the number of moles will also be half.

$$\text{No. of moles} = n = \frac{\text{No. of particles}}{\text{Avogadro number}} = \frac{3.011 \times 10^{23} \text{ atoms}}{6.022 \times 10^{23} \text{ atoms/mol}} = 0.5 \text{ mols}$$

$$\begin{aligned} \text{Number of moles} &= \frac{\text{number of particles}}{\text{Avogadro's number}} \\ &= \frac{3.011 \times 10^{23}}{6.023 \times 10^{23}} = 0.5 \text{ mole} \end{aligned}$$

Example 4

- a) Distinguish between N_2 and 2N .
- b) Calculate the mass of particles in the following:
- c) 0.5 mole of nitrogen atoms
- d) 0.5 mole of nitrogen molecules

(Given: Atomic mass of nitrogen atom = 14 g Molecular mass of nitrogen molecule = 28 g)

Solution

- a) N_2 stands for 1 mole of nitrogen molecules and 2N stands for 2 moles of nitrogen atoms.
- b) Mass of nitrogen atoms = Atomic mass of nitrogen atom $\times$ number of moles
Mass of nitrogen atoms = ?
Atomic mass of nitrogen = 14 g
Molecular mass of N_2 = 28 g
No. of mole = 0.5
The mass of 0.5 mole of nitrogen atom = $14 \times 0.5 = 7$ grams
- c) Mass of nitrogen molecules (N_2) = Molecular mass of molecules $\times$ no. of moles
 $= 28 \times 0.5 = 14$ grams

Example 5

Calculate the no. of moles in

- i) 0.478 g of magnesium
- ii) 12 g oxygen gas

Given: Atomic mass of Mg = 24

Atomic mass of O = 16.

Solution

i)

$$\text{No. of moles} = n = \frac{\text{Mass of Mg in grams}}{\text{Atomic mass}} = \frac{0.478 \text{ g}}{24 \text{ g/mol}} = 0.019 \text{ mols}$$

ii) In Oxygen gas, there are two atoms of Oxygen (O_2)

$$\text{Molecular mass of } \text{O}_2 = 16 \times 2 = 32 \text{ g}$$

$$\text{No of moles in 12g of } \text{O}_2 = \frac{12}{32} = 0.375 \text{ mol}$$

Example 6

Calculate the mass of 6.022×10^{23} nitrogen molecules.

Solution

The mass of nitrogen molecules (N_2) = $14 \times 2 = 28 \text{ g}$

1 mole of N_2 molecules = 6.022×10^{23} molecules of nitrogen.

1 mole of N_2 molecule weighs = 28 g

Therefore, 6.022×10^{23} molecules will weigh 28 grams.



Application activity 5.2

1. How many moles are there in 1.204×10^{24} molecules of water (H_2O)?
2. Calculate the number of **atoms** in **0.5 mole of magnesium (Mg)**.
3. How many moles are in 3.01×10^{23} molecules of oxygen (O_2)?
4. Calculate the number of **molecules** in **2 moles of carbon dioxide (CO_2)**.
5. How many **atoms of hydrogen** are in **1 mole of ammonia (NH_3)**?
6. Calculate the mass of **1 mole of oxygen atoms (O)**
7. Find the mass of 6.02×10^{23} molecules of CO_2 .
8. Find the mass of **2 moles of H_2O molecules**.
9. Calculate the mass of 3.01×10^{23} atoms of carbon (C).

5.3. Definition of relative atomic mass



Learning activity 5.3

Look at the periodic table and identify the atomic masses of some elements such as copper, tin, lead, iodine.

Just like mole, relative atomic mass is based on Carbon-12 isotope as a standard measure. The relative atomic mass (A_r) of an element is the average relative mass of an atom of an element as compared with an atom of ^{12}C taken as 12- atomic mass unit. Hence,

$$\text{Relative atomic mass of an element } (A_r) = \frac{\text{Average number of one atom}}{\frac{1}{12} \times (\text{mass of one carbon-12 isotope})}$$

The relative atomic mass (A_r) of an element is a pure number and it does not have a unit. The relative atomic masses of some elements are given in table

Table 5.3: Showing examples of relative atomic masses and their commonly used values

Element	Symbol	Relative Atomic Mass (A_r)	Commonly Used Value
Hydrogen	H	1.008	1
Carbon	C	12.011	12
Nitrogen	N	14.007	14
Oxygen	O	15.999	16
Sodium	Na	22.990	23
Magnesium	Mg	24.305	24
Aluminum	Al	26.982	27



Application activity 5.3

- Define the term relative atomic mass. Use a mathematical expression.
- Which element is used as reference in defining relative atomic mass?
- What is its relative atomic mass of the element identified in (b)?

5.4. Definition and calculation of relative molecular mass



Learning activity 5.4

- What is relative atomic mass?
- By reference to the definition of relative atomic mass give the definition of relative molecular mass and suggest its determination.

Like the atomic masses, the molecular masses of compounds are also very small, and these cannot be measured directly. The molecular masses of compounds are also expressed as relative molecular masses (M_r).

$$\frac{\text{Mass of one molecule of compound}}{\frac{1}{12} \text{ the mass of one molecule of carbon-12 isotope}}$$

a) Molecular mass of water:

The molecular formula of water is H_2O . Hence, Molecular mass of water = $(2 \times \text{Atomic mass of hydrogen}) + (1 \times \text{Atomic mass of oxygen}) = (2 \times 1) + (1 \times 16) = 18 \text{ u}$.

b) Molecular mass of nitric acid (HNO_3):

The molecular formula of nitric acid is HNO_3 . Hence, Molecular mass of nitric acid = $(1 \times \text{Atomic mass of hydrogen}) + (1 \times \text{Atomic mass of nitrogen}) + (3 \times \text{Atomic mass of oxygen}) = (1 \times 1) + (1 \times 14) + (3 \times 16) = 1 + 14 + 48 = 63 \text{ u}$.

Example

Find the molecular mass of the following compounds

- i) $\text{C}_6\text{H}_{12}\text{O}_6$ ii) $(\text{NH}_4)_3\text{PO}_4$

Given atomic masses ($\text{H}=1\text{u}$, $\text{C}=12\text{u}$, $\text{O}=16\text{u}$, $\text{N}=14\text{u}$, $\text{P}=31\text{u}$, $\text{S}=32\text{u}$)

Solutions

i) Step I

Multiply the number of atoms by atomic mass:

Carbon: $6 \text{ atoms} \times 12 = 72\text{u}$

Hydrogen: $12 \text{ atoms} \times 1 = 12\text{u}$

Oxygen: $6 \text{ atoms} \times 16 = 96\text{u}$

Step II

Sum the totals: $72 + 12 + 96 = 180\text{u}$

ii) Step I

Multiply the number of atoms by atomic mass:

Nitrogen: $3 \times 14 = 42\text{u}$

Hydrogen: $12 \times 1 = 12 \text{ u}$

Phosphorus: $1 \times 31 = 31\text{u}$

Oxygen: $4 \times 16.00 = 64\text{u}$

Sum up

$42 + 12 + 31 + 64 = 149\text{u}$



Application activity 5.4

Find the molecular mass unit of the following compounds

- i) $\text{Fe}_2(\text{SO}_4)_3$, ii) $\text{NH}_4\text{C}_2\text{H}_3\text{O}_2$, iii) $(\text{NH}_4)_2\text{CO}_3$, iv) $(\text{NH}_4)_3\text{PO}_4$

Atomic masses of: ($\text{Fe}=56\text{u}$, $\text{O}=16\text{u}$, $\text{S}=32$, $\text{C}=12\text{u}$, $\text{N}=14\text{u}$, $\text{P}=31\text{u}$,)

5.5 Definition and calculation of relative formula mass



Learning activity 5.5

1. "If one apple weighs 200 g, how much do 3 apples weigh?"
2. Relate to chemistry: atoms have tiny masses, so we calculate total mass using relative atomic masses (A_r) and formula masses (M_r).
3. Define M_r : "The sum of the relative atomic masses of all atoms in a formula unit of a compound."
4. Show formula to calculate M_r :

$M_r = \sum \text{Relative atomic masses of all atoms in the formula.}$

The sum of the atomic masses of all the atoms in a formula unit of a compound is called the formula unit mass of the compound. Always remember that the concept of the formula unit mass is used in case of ionic compounds. This is because in case of ionic compounds there are no separate individual molecules but they exist as aggregates, i.e., as cluster of ions. For example, solid sodium chloride is represented as $(\text{Na}^+\text{Cl}^-)_n$ and in the simplified form it is written as NaCl . The formula unit mass of ionic compounds is calculated in the same manner as in the case of the molecular mass described earlier. For example, the formula unit mass of NaCl = Atomic mass of Na + Atomic mass of Cl = $23 \text{ u} + 35.5 \text{ u} = 58.5 \text{ u}$

Examples

Calculate the formula unit masses of ZnO , Na_2O , and K_2CO_3 .

(Given atomic mass: $\text{Zn} = 65 \text{ u}$, $\text{Na} = 23 \text{ u}$, $\text{K} = 39 \text{ u}$, $\text{C} = 12 \text{ u}$ and $\text{O} = 16 \text{ u}$) Solution
Formula unit mass of ZnO = atomic mass of Zn + atomic mass of O = $65 \text{ u} + 16 \text{ u} = 81 \text{ u}$.
Formula unit mass of Na_2O = $(2 \times \text{atomic mass of Na}) + \text{atomic mass of O}$

Formula unit mass of Na_2O = $(2 \times \text{atomic mass of Na}) + \text{atomic mass of O}$ = $2 \times 23 \text{ u} + 16 \text{ u} = 46 \text{ u} + 16 \text{ u} = 62 \text{ u}$

Formula unit mass of K_2CO_3 = $(2 \times \text{atomic mass of K}) + \text{atomic mass of C} + (3 \times \text{atomic mass of O})$ = $2 \times 39 \text{ u} + 12 \text{ u} + 3 \times 16 \text{ u} = 78 \text{ u} + 12 \text{ u} + 48 \text{ u} = 138 \text{ u}$

Difference between Formula mass and molecular mass:

Molecular mass : Used for covalent compounds, where discrete molecules exist.

Formula mass : Used for ionic compounds, which exist as crystal lattices rather than individual molecules.

Both are calculated in the same way (by adding the atomic masses of all atoms present), but the terminology depends on the type of bonding in the substance.



Application activity 5.5

1. Calculate the relative molecular mass of water (H_2O)
2. Find the relative formula mass of sodium carbonate (Na_2CO_3).
3. Calculate the relative formula mass of ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$).
4. Calculate the relative formula mass of magnesium sulfate heptahydrate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$).

5.6 Calculation of molar mass



Learning activity 5.6

1. Using the periodic table give the mass of 1 mole of carbon, nitrogen and calcium.

We know that one methane molecule contains one carbon atom and four hydrogen atoms. 1 mole of methane molecules consists of 1 mole of carbon atoms and 4 moles of hydrogen atoms.

The mass of 1 mole of methane can be found by adding the masses of carbon and hydrogen present.

Mass of 1 mole carbon = $1 \times 12.01 \text{ g} = 12.01 \text{ g mol}^{-1}$

Mass of 4 moles hydrogen atoms = $4 \times 1.00 \text{ g} = 4.00 \text{ g}$

Mass of 1 mole methane = $12.01 + 4.00 = 16.01 \text{ g mol}^{-1}$

The quantity 16.01 g mol^{-1} is called the molar mass of methane.

1 mole methane has 6.022×10^{23} molecules so the mass of 6.022×10^{23} molecules is also 16.01 g. Calculate the mass of 12 moles of methane.

Mass of one mole of any substance is called its molar mass. Alternatively, the average mass of one mole of any substance is called its molar mass. Molar mass is represented by M and its unit is g mol^{-1} . It is given by,

$$\begin{aligned} \text{Molar Mass (Mr)} &= \frac{\text{Mass of the substance}}{\text{Amount of substance in moles}} = \frac{m}{n} \text{ Molar Mass Mr} \\ &= \frac{\text{Mass of the substance}}{\text{Amount of substance in moles}} = \frac{w}{n} \text{ where } w \text{ is the mass of substance} \end{aligned}$$

and n the number of moles

Molar mass of a substance = mass of one mole (6.022×10^{23}) of its chemical units

Molar mass of one hydrogen atom = mass of 6.022×10^{23} hydrogen atoms

Molar mass of one hydrogen molecule = Mass of 6.023×10^{23} molecules of hydrogen.

Calculate the molar mass of sulphur dioxide. Solution

Mass of 1 mol S = $1 \times 32.07 = 32.07 \text{ g mol}^{-1}$

Mass of 2 mol O = $2 \times 16.00 = 32.00 \text{ g mol}^{-1}$

Mass of 1 mol SO₂ Molar mass = 64.07 = 64.07 g mol⁻¹

The molar mass of SO₂ is 64.07 g mol⁻¹. It represents the mass of 1 mole of SO₂ molecules.



Application activity 5.6

Calculate the molar masses of the following substances:

- a) Ethyne, C₂H₂
- b) Hydrochloric acid, HCl
- c) Nitric acid, HNO₃
- d) Glucose, C₆H₁₂O₆

(Atomic masses: C = 12 u, H = 1 u, Cl = 35.5 u, N = 14 u, O = 16 u)

5.7. Relationship between number of moles, mass, and molar mass



Learning activity 5.7

1. Calculate the number of moles (n) in 24 g (mass) of Carbon -12 isotope and deduce the relationship between n and m.
2. Calculate the mass (m) of 5 moles of Sodium Na

Use the periodic table to do these calculations.

How many moles are there in a certain mass of a substance? The number of moles in a certain mass of a substance is calculated using the following formula:

$$\begin{aligned}\text{Number of moles of substance X} &= \frac{\text{Mass of substance in grams}}{\text{Molar mass of the substance in grams per mole}} \\ &= \frac{m}{Mr}\end{aligned}$$

Where m is the mass of the substance, and M is the molar mass of the substance.

How many molecules are there in a certain mass of a substance? The number of molecules present in a certain mass of a substance is calculated using the following formula:

Number of

$$\begin{aligned}\text{Molecules of a Substance} &= \frac{\text{Mass of a substance}}{\text{Molar mass of the substance}} \times \text{Avogadro's Number} \\ &= \frac{m}{M} \times 6.02 \times 10^{23} \text{ Particles/Mole} = \frac{m}{M} \times 6.02 \times 10^{23} \text{ Particles/Mole}\end{aligned}$$

$$\text{Molecules of a Substance} = \frac{\text{Mass of a substance}}{\text{Molar mass of the substance}} \times \text{Avogadro's Number}$$

$$\frac{m}{M} \times 6.02 \times 10^{23} \text{ Molecules} \quad \frac{m}{M} \times 6.02 \times 10^{23} \text{ Molecules}$$

Example

1. How many atoms of S_8 molecules are present in 50 g of Sulphur? The relative atomic mass of sulphur is 32.

Solution

32g of sulphur contain 6.023×10^{23} atoms of sulphur

50g of sulphur contain $6.023 \times 10^{23} \text{ atoms/mol} \times \frac{50}{32} \text{ mol}$

50g of sulphur contain $6.023 \times 10^{23} \text{ atoms/mol} \times \frac{50}{32} \text{ mol} = 9.411 \times 10^{23} \text{ atoms}$

8g of sulphur are present in 1 molecule of sulphur

9.411×10^{23} atoms of sulphur are present in $\frac{9.411 \times 10^{23}}{8}$ molecules

$= 1.176 \times 10^{23}$ molecules of sulphur

Example 2

The antibiotic penicillin has the molecular formula $\text{C}_{16}\text{H}_{18}\text{N}_2\text{SO}_4$. One injection of penicillin contains 500 mg of penicillin. When one intramuscular injection of penicillin is administered:

- a) How many moles of penicillin are administered?
- b) How many molecules of penicillin are administered?
- c) How many atoms of nitrogen are injected?

Solution

(a) Molar mass of penicillin ($\text{C}_{16}\text{H}_{18}\text{N}_2\text{SO}_4$)

$$= (16 \times 12 + 18 \times 1 + 2 \times 14 + 1 \times 32 + 4 \times 16) \text{ g/mol}$$

$$= 334 \text{ g/mol}$$

$$334 \text{ g of penicillin} = 1 \text{ mol}$$

$$0.500 \text{ g of penicillin} = \frac{0.500}{334} \text{ mol} = 1.497 \times 10^{-3} \text{ mol}$$

(b) 1 mole of penicillin = 6.023×10^{23} molecules of penicillin

1.497×10^{-3} moles of penicillin contain

$$6.023 \times 10^{23} \times 1.497 \times 10^{-3} \text{ molecules of penicillin}$$

$$= 9.016 \times 10^{20} \text{ molecules of penicillin}$$

(c) 1 molecule of penicillin contains 2 atoms of nitrogen
 9.016×10^{20} molecules of penicillin contain $2 \times 9.016 \times 10^{20}$
 $= 1.8032 \times 10^{21}$ molecules of nitrogen



Application Activity 5.7

1. Calculate the number of molecules of chloroform (CHCl_3) weighing 0.0239 g (H = 1, C = 12, Cl = 35.5).
2. Calculate the mass in grams of (a) 1 atom of nitrogen (b) one molecule of benzene (C_6H_6).
3. Find the number of atoms in the following: (a) 52 mol of Ne (b) 52 g of Ne.
4. Find: a) the number of molecules of Sulphur (S_8) in 16 g of solid Sulphur
b) the number of aluminium ions in 0.051 g of aluminium oxide.
5. Chlorophyll, the green coloring matter in plants involved in photosynthesis, contains 2.7% magnesium by mass. Calculate the number of magnesium atoms in 1.0 g of chlorophyll.
6. Calculate the mass of carbon dioxide which contains the same number of molecules as are contained in 40 g of oxygen.
7. Calculate the number of moles in the following masses a) 7.85 g Fe (at. mass 56) b) 65.5 μg of Carbon (at. mass 12) c) 4.68 mg of Si (at. mass 28) d) 1.46 metric tons of Al (at. mass 27) e) 7.9 g of Ca (at. mass 40). (Note: 1 metric ton = 103 kg).
8. Calculate the molar mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and the number of atoms of each kind in it.
9. How many molecules of water of hydration are present in 252 mg of oxalic acid, ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$)?

5.8. Calculation of mass percentage composition of an element in a compound



Learning activity 5.8

Aim: Determining the Percent Composition of magnesium in magnesium oxide

Safety precautions

The burning of magnesium generates an intense white light. Mixed in with the white light there is ultraviolet light as well.

Do not look at the magnesium as it is burning.



Learning activity 5.8

Wear goggles

Apparatus and Material Stand, ceramic tile hot plate, ring clamp, pipe clay triangle, crucible and lid, crucible tongs, Bunsen burner, 25 cm³ of Magnesium ribbon.

Procedure

1. Obtain a 25 cm piece of magnesium ribbon. The exact length is not important.
2. Find the mass of the magnesium ribbon and record it in the data table.
3. Set up the apparatus with the stand, ring clamp, and Bunsen burner sitting on the ceramic tile hot plate.
4. Find the mass of the empty crucible and record it in the data table.
5. Roll the ribbon of magnesium into a spiral or crush it into a compacted mass and place it in the crucible.
6. Cover the crucible and heat strongly for 5 minutes or until red hot.
7. With the crucible tongs, gently lift the crucible lid to let air into the crucible.
8. With the lid left slightly open, discontinue heating.
9. Remove the crucible lid and let the crucible cool before massing it again.
10. Record Mass the crucible and contents. The contents are your new compound of magnesium oxide. Record the mass on the data table.....

Data Table

Mass of magnesium ribbon = _____ g

Mass of crucible = _____ g

Mass of crucible and contents = _____ g

Calculations

- a) Find the mass of the contents of the crucible. This is your new compound of magnesium oxide.
- b) The percentage of magnesium in the compound can be found like this:

$$\text{Percentage of magnesium} = \frac{\text{Mass of Magnesium}}{\text{Mass of Magnesium oxide}} \times 100$$

- c) Find the percentage of oxygen in the compound.

Percentage composition shows the mass percentage of each element in a compound, calculated by dividing the total mass of element in the compound by the total molar mass of the compound then multiply by 100%; the sum of all percentages of the elements equal 100.

Steps to Calculate Percentage Composition

- Find the Molar Mass of Each Element:** Use the periodic table to find the atomic mass (in g/mol) for each element in the compound.
- Calculate Total Mass of Each Element:** Multiply the atomic mass of each element by the number of atoms (subscript) of that element in the chemical formula.
- Determine the Molar Mass of the Compound:** Add the total masses of all elements in the compound.
- Calculate Percent Composition for Each Element using the formula:**

$$\text{Percentage composition of Element} = \frac{\text{Mass of Element}}{\text{Molar mass}} \times 100$$

$$\text{Percentage composition of Element} = \frac{\text{Mass of Element}}{\text{Molar mass}} \times 100$$

Exercise 1

0.24 g sample of a compound of oxygen and boron was found by analysis to contain 0.096 g of boron and 0.144 g of oxygen. Calculate the percentage composition of the compound by mass.

$$\text{Percentage of boron} = \frac{\text{Mass of boron} \times 100\%}{\text{Mass of the compound}} = \frac{0.096\text{g} \times 100}{0.24\text{g}} = 40\%$$

$$\text{Percentage of oxygen} = \frac{\text{Mass of oxygen} \times 100\%}{\text{Mass of the compound}} = \frac{0.144\text{g} \times 100}{0.24\text{g}} = 60.0\%$$

Exercise 2

Find the percentage composition of glucose whose formula is $\text{C}_6\text{H}_{12}\text{O}_6$.

Solution

Molecular mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) = $6 \times 12\text{u} + 12 \times 1\text{u} + 6 \times 16\text{u} = 180\text{u}$

Mass percentage of carbon in glucose

$$\frac{72\text{u}}{180\text{u}} \times 100 = 40.0\%$$

Mass percentage of Hydrogen in glucose

$$= \frac{12\text{u}}{180\text{u}} \times 100 = 6.6\%$$

Mass percentage of oxygen in glucose

$$= \frac{96\text{u}}{180\text{u}} \times 100 = 53.33\%$$

Molecular mass of glucose ($C_6H_{12}O_6$) = $6 \times 12u + 12 \times 1u + 6 \times 16u = 180u$

Mass percentage of carbon in glucose = $\frac{72u}{180u} \times 100 = 40.0\%$

Mass percentage of Hydrogen in glucose = $\frac{12u}{180u} \times 100 = 6.6\%$

Mass percentage of oxygen in glucose = $\frac{96u}{180u} \times 100 = 53.33\%$

Exercise 3

Calculate the percentage of water of crystallization in washing soda whose formula is $Na_2CO_3 \cdot 10H_2O$.

Molecular mass of $Na_2CO_3 \cdot 10H_2O = 2 \times 23u + 12u + 3 \times 16u + 10(2 \times 1u + 1 \times 16u)$

= 286u 286g of washing soda contain 180g of water of crystallisation

100g of washing soda will contain $\frac{180u}{286u} \times 100g$ water of crystallisation = 62.94g

The amount of water of crystallisation in washing soda = 62.94% by mass

Exercise 4

Ferric sulphate is a non-crystalline compound of iron, $Fe_2(SO_4)_3$. It is used in water and sewage treatment to help the removal of suspended impurities. Calculate the mass percentage of iron, Sulphur and oxygen in the compound.

molecular mass of Iron(III) sulphate, $Fe_2(SO_4)_3$

= $2 \times 56u + 3(32 + 64) = 400u$

Mass percentage of Iron, Fe = $\frac{112u}{400u} \times 100 = 28\%$

Mass percentage of sulphur = $\frac{96u}{400u} \times 100 = 24\%$

Mass percentage of Oxygen = $\frac{192u}{400u} \times 100 = 48\%$



Application activity 5.8

1. Calculate the mass percentage of each element present in water.
2. Find the percentage composition of each element in glucose whose formula is $C_6H_{12}O_6$.
3. Calculate the percentage of water of crystallization in Ethanedioic acid whose formula is $C_2H_2O_4 \cdot 2H_2O$.
4. Find the percentage composition of each element in potassium permanganate whose formula is $KMnO_4$ (Atomic masses, H=1, C=12, O=16, K=39, Mn=55)
5. a) Find the percentage of iron in iron (III) oxide Fe_2O_3 (Fe= 56, O= 16)
b) Find the percentage of calcium in calcium carbonate, $CaCO_3$ (C=12, Ca= 40, O=16)

5.9. Empirical and molecular formulae



Learning activity 5.9

Determination of simplest ratio from mass data

Consider the following molecular formula of glucose $C_6H_{12}O_6$ and find its the simplest formula i.e the simplest ration of subscripts..

Question

Determine the empirical (simplest) formula of $C_6H_{12}O_6$.

Step 1: Write the subscripts: C=..., H:..., O:

Step 2: Find the greatest common divisor (GCD) of 6, 12, and 6:

GCD =

Step 3: Divide each subscript by GCD and write the simplest formula (Empirical formula)

a) Empirical formulae

The empirical formula of a compound is the simplest formula which represents its composition. It shows the elements present and the ratio of the amounts of elements present.

To find an empirical formula,

- Find the mass in grams of each element in the compound.
- Convert this mass to moles
- You need to work out the ratio of the amounts of the elements present by dividing each by the smallest value of moles.
- If the ratio obtained in above is not a whole number, then they should be multiplied by a suitable common factor to convert each of them to whole numbers.

Example:

A 0.4764 g sample of an oxide of iron was reduced by carbon monoxide. The mass of iron that remained was 0.3450 g. find the empirical formulae of the oxide.

Method

Elements present	Iron	Oxygen
Mass/g	0.3450	0.1314
Ar	56	16
Amount (mol)	$0.3450 / 56 = 6.16 \times 10^{-3}$	$0.1314 / 16 = 8.21 \times 10^{-3}$
Ratio of amounts	$\frac{6.16 \times 10^{-3}}{6.16 \times 10^{-3}} = 1$ 1 multiply by 3 we get 3	$\frac{8.21 \times 10^{-3}}{6.16 \times 10^{-3}} = 1.33$ 1.33 then multiply 3 we get 4

The empirical formula is Fe_3O_4 .

A compound has the following composition, Mg = 9.76%, S = 13.01%, O = 26.01, H_2O = 51.22, what is its empirical formula?

[Mg = 24, S = 32, O = 16, H = 1]

Solution

Element	%	Relative No. of moles	Simple ratio moles	Simplest Whole No. ratio
Magnesium	9.76	$9.76 \div 24 = 0.406$	$0.406 \div 0.406 = 1$	1
Sulphur	13.01	$13.01 \div 32 = 0.406$	$0.406 \div 0.406 = 1$	1
Oxygen	26.01	$26.01 \div 16 = 1.625$	$1.625 \div 0.406 = 4$	4
Water	51.22	$51.22 \div 18 = 2.846$	$2.846 \div 0.406 = 7$	7

Hence the empirical formula is $\text{Mg SO}_4 \cdot 7\text{H}_2\text{O}$.

b) Molecular formulae

The molecular formula is a simple multiple of the empirical formula. It shows the actual number of atoms of each of the elements in the formula of a compound.

Example 1: A compound has the empirical formula CH_3 . Given that its relative molecular mass is 30, find the molecular formula of the compound.

Solution

The relative molecular mass of $\text{CH}_3 = (12+3) = 15$

$n \times \text{empirical formula mass} = \text{formula mass}$

$n \times 15 = 30,$

$$n = 30 / 15 = 2 \text{ So,}$$

molecular formula = 2 × empirical formula, Thus molecular formula is C_2H_6

Example 2:

An organic compound consists of 84.90% carbon and 15.10% hydrogen. Its relative molecular mass is 70. Find the molecular formula of the compound.

Solution;

Step 1: Empirical formula calculation

Element	% Composition	Divide by Atomic Mass	Moles	Divide by Smallest	Ratio
C	84.90	$84.90 \div 12$	7.075	$7.075 \div 7.075$	1
H	15.10	$15.10 \div 1$	15.10	$15.10 \div 7.075$	2

Simplest Ratio

C : H

1 : 2

Empirical Formula

CH_2

Step 2: Molecular Formula Calculation

Molecular mass = 70 g/mol

Empirical formula = CH_2

Empirical formula mass (EFM):

$$\text{EFM} = 12 + (2 \times 1)$$

$$\text{EFM} = 14 \text{ g/mol}$$

We use the relationship:

Molecular mass = $n \times$ Empirical formula mass

$$70 = n \times 14$$

$$n = 70 \div 14$$

$$n = 5$$

Molecular formula = $(\text{CH}_2)_5$

$$= \text{C}_5\text{H}_{10}$$



Application activity 5.9

1. What is the simplest formula of the compound which has the following percentage composition: Carbon 80%, Hydrogen 20%? If the molecular mass is 30, calculate its molecular formula.
2. Butyric acid contains C, H, O elements. A 4.24 mg sample of butyric acid is completely burnt in oxygen. It gives 8.45 mg of carbon dioxide and 3.46 mg of water. What is the mass percentage of each element? Determine the empirical and molecular formula of butyric acid if molecular mass of butyric acid is determined to be 88 u.
3. An organic compound on analysis gave the following data : C = 57.82%, H = 3.6%, and the rest is oxygen. Its vapour density is 83. Find its empirical and molecular formula.
4. 2.746 g of a compound gave on analysis 1.94 g of silver, 0.268 g of sulphur and 0.538 g of oxygen. Find the empirical formula of the compound. (At. masses: Ag = 108, S = 32, O = 16)
5. A compound on analysis gave the following percentage composition: Na = 14.31%, S = 9.97%, H = 6.22%, O = 69.5%. Calculate the molecular formula of the compound on the assumption that all the hydrogen in the compound is present in combination with oxygen as water of crystallization. Molecular mass of the compound is 322. [Na = 23, S = 32, H = 1 and O = 16]

5.10 Stoichiometric calculations



Learning activity 5.10

1. What does Stoichiometry mean in chemistry?
2. Explain steps in stoichiometric calculations
3. Balance the following chemical equations:
 - a) $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 - b) $\text{Mg}(\text{OH})_2 + \text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O}$

Stoichiometry is the branch of chemistry that deals with the quantitative relationships between reactants and products in a chemical reaction.

It uses a balanced chemical equation to calculate how much of a substance is needed or produced.

In simple terms, **stoichiometry is the chemistry of counting and measuring substances involved in chemical reactions.**

Using stoichiometry, you can answer questions such as:

- How many grams of a reactant are required?

- How many moles of a product will be formed?
- How many molecules or atoms are involved in a reaction?
- Which reactant limits the amount of product formed?

Key Steps in Stoichiometric Calculations

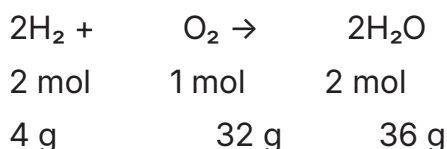
1. Write and balance the chemical equation.
2. Convert given quantities to moles.
3. Use mole ratios from the balanced equation.
4. Convert moles to the required unit (mass, volume, particles, etc.).

Examples

1. How many grams of oxygen (O_2) are required to completely react with 0.200 g of hydrogen (H_2) to yield water (H_2O)? Also, calculate the amount of water formed. (Molar masses: $H_2 = 2 \text{ g mol}^{-1}$; $O_2 = 32 \text{ g mol}^{-1}$).

Solution

The balanced equation for the reaction is:



Now, 4 g of H_2 requires oxygen = 32 g

$$0.200 \text{ g of } H_2 \text{ requires oxygen} = \frac{32}{4} \times 0.2 \text{ g} = 1.6g \frac{32}{4} \times 0.2 \text{ g} = 1.6g$$

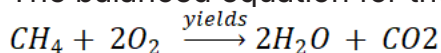
Again, 4 g of H_2 produce $H_2O = 36 \text{ g}$

$$0.200 \text{ g of } H_2 \text{ produces } H_2O = \frac{36}{4} \times 0.2 \text{ g} = 1.8g \frac{36}{4} \times 0.2 \text{ g} = 1.8g$$

2. Calculate the amount of water (g) produced by the combustion of 10 g of methane.

Solution

The balanced equation for the reaction is:

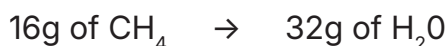


From the above equation,

1 mol of $CH_4(g)$ gives 2 mol of $H_2O(g)$.

Using mass: 16g of methane gives 32 g of water.

$$2 \text{ mol of water } (H_2O) = 2 \times (2 + 16) = 2 \times 18 = 36 \text{ g}$$



$$10g \text{ of } CH_4 \rightarrow \frac{32g}{16g} \times 10g = 20g \text{ of water}$$

3. How many moles of methane (CH_4) are required to produce 252 g of $CO_2(g)$ after combustion?

Solution

Step 1: Write the balanced chemical equation for the combustion of methane.



Step 2: Determine the molar mass of CO_2 .

$$\text{Molar mass of } \text{CO}_2 = 12 + (2 \times 16) = 44 \text{ g/mol}$$

Step 3: Calculate the number of moles of CO_2 produced.

$$\text{Number of moles} = \text{mass} / \text{molar mass}$$

$$\text{Number of moles of } \text{CO}_2 = \frac{252\text{g}}{44\text{g/mol}} = 5.73\text{mol}$$

Step 4: Use the mole ratio from the balanced equation.

From the equation, 1 mol CH_4 produces 1 mol CO_2 (Equal number of moles from the balanced equation).

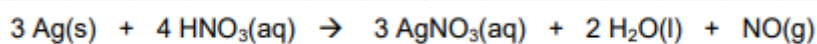
Therefore, moles of CH_4 required = 5.73 mol.

Final Answer: 5.73 moles of methane (CH_4) are required.



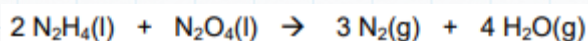
Application activity 5.10

1. Silver and nitric acid react according to the following balanced equation:



- How many moles of silver are needed to react with 40 moles of nitric acid?
- From the amount of nitric acid given in a) how many moles of silver nitrate will be produced?
- From the amount of nitric acid given in a), how many moles of water will be produced?
- From the amount of nitric acid given in a), how many moles of nitrogen monoxide will be made?

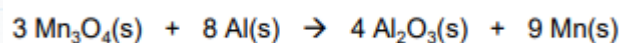
2. Given the balanced equation:



- How many moles of dinitrogen tetrahydride are required to produce 57 moles of nitrogen?
- How many moles of dinitrogen tetroxide are required to produce 57 moles of nitrogen?
- How many moles of water are produced when 57 moles of nitrogen are made?

Calculate the mass of aluminum oxide (Al_2O_3) produced when 3.75 moles of aluminum burn in oxygen.

4. At a very high temperature, manganese is isolated from its ore, manganomanganic oxide, via the following balanced equation:



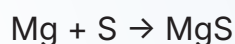
- How many manganese atoms are liberated if 54.8 moles of Mn_3O_4 react with excess aluminum?
 - How many moles of aluminum oxide are made if 3580 g of manganomanganic oxide are consumed?
 - How many moles of manganomanganic oxide will react with 5.33×10^{25} atoms of aluminum?
 - If 4.37 moles of aluminum are consumed, how many molecules of aluminum oxide are produced?
5. Write the balanced chemical equation of each reaction:
- Calcium carbide (CaC_2) reacts with water to form calcium hydroxide ($\text{Ca}(\text{OH})_2$) and acetylene gas (C_2H_2).
 - If 300g of CaC_2 reacted, what is the number of moles of C_2H_2 produced?
 - What is the mass and the moles of $\text{Ca}(\text{OH})_2$ produce?

5.11. Limiting reactants



Learning activity 5.11

1. When 2.00 g of magnesium and 2.00 g of sulphur are mixed to react, by the reaction



Which reactant will be used up before the other and stops the reaction from continuing.

First of all each of the masses are expressed in moles

$$2.00\text{g of Mg} = \frac{2.00}{24.3} = 0.0824 \text{ moles of Mg}$$

$$2.00\text{g of S} = \frac{2.00}{32.1} = 0.0624 \text{ moles of S}$$

From the equation, $\text{Mg} + \text{S} \rightarrow \text{MgS}$, it follows that one mole of Mg reacts with one mole of S. We are given more moles of Mg than of S. Therefore, Mg is in excess and some of it will remain unreacted when the reaction is over.

Let express all of the masses in moles:

$$2.00\text{ g of Mg} = \frac{2.00}{24.3} = 0.0824 \text{ moles of Mg}$$

$$2.00\text{ g of S} = \frac{2.00}{32.1} = 0.0624 \text{ moles of S}$$

From the equation, $\text{Mg} + \text{S} \rightarrow \text{MgS}$, it follows that one mole of Mg reacts with one mole of S. We are given more moles of Mg than of S. Therefore, Mg is in excess and some of it will remain unreacted when the reaction is over. S will be used up before Mg and stops the reaction from continuing.

S is the limiting reagent and will control the amount of product.

From the equation we note that one mole of S gives one mole of MgS, so 0.0624 mole of S will react with 0.0624 mole of Mg to form 0.0624 mole of MgS. Molar mass of MgS = 56.4 g

$$\text{Mass of MgS formed} = 0.0624 \times 56.4 \text{ g} = 3.52 \text{ g of MgS}$$

$$\text{Mole of Mg left unreacted} = 0.0824 - 0.0624 \text{ moles of Mg}$$

$$= 0.0200 \text{ moles of Mg}$$

$$\text{Mass of Mg left unreacted} = \text{moles of Mg} \times \text{molar mass of Mg} = 0.0200 \times 24.3 \text{ g of Mg}$$

$$= 0.486 \text{ g of Mg}$$

In many chemical reactions, reactants are not always present in the exact proportions required. As a result, one reactant is used up before the others and stops the reaction from continuing. This reactant is called the **limiting reactant**, and understanding it helps us predict how much product can be formed in a chemical reaction.

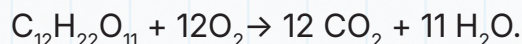
A **limiting reactant** (or limiting reagent) is the reactant that is completely consumed first in a chemical reaction. Because it runs out first, it determines the maximum amount of product that can be formed.



Application activity 5.11

1. Identify the limiting reactant in the reaction of hydrogen and oxygen to form water if 61.0 g of O_2 and 8.40 g of H_2 are combined. Determine the amount (in grams) of excess reactant that remains after the reaction is complete. Determine also the mass of water formed.
2. Calculate the limiting reactant. Zinc metal reacts with hydrochloric acid by the following reaction: $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
3. If 0.30 mol Zn is added to hydrochloric acid containing 0.52 mol HCl, how many moles of H_2 are produced?
4. If 23 grams of iron (II) chloride reacts with 41 grams of sodium phosphate, what is the limiting reagent? How much sodium chloride can be formed?

5. For the combustion of sucrose:



6. There are 10.0 g of sucrose and 10.0 g of oxygen reacting. Which is the limiting reagent? Find the mass of CO_2 formed.

7. Suppose 316.0 g of aluminum sulfide reacts with 493.0 g of water. What mass of the excess reactant remains? The balanced equation is: $\text{Al}_2\text{S}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{Al}(\text{OH})_3 + 3\text{H}_2\text{S}$

5.12. Calculation of molar gas volume under standard conditions



Learning activity 5.12

1. Determine the molar mass of $\text{O}_2(\text{g})$, $\text{CaCl}_2(\text{s})$
2. What is the meaning of STP?
3. At STP what is the temperature and the pressure?
4. What is Molar gas volume at STP?

Molar gas volume is the volume occupied by **one mole of any gas** under specified conditions of temperature and pressure.

The molar gas volume at STP is the volume occupied by one mole of any gas at 0°C and 1 atm pressure, equal to 22.4 L.

At Standard Temperature and Pressure (STP):

- Temperature = **0°C (273 K)**
- Pressure = **1 atm (101.3 kPa)**

One mole of any gas occupies 22.4 L at STP.

1 mol gas at STP = 22.4 L

Examples

- 1 mol of O_2 occupies **22.4 L** at STP.
- 1 mol of H_2 occupies **22.4 L** at STP.
- 1 mol of CO_2 occupies **22.4 L** at STP.

Although these gases have different masses, they occupy the **same volume** at STP because equal numbers of gas particles occupy equal volumes under the same conditions.

Formula

Volume of gas at STP = Number of moles $\times$ 22.4 L

Find the volume of 2 mol of oxygen gas at STP:

$$V = n \times 22.4$$

$$V = 2 \times 22.4 = 44.8$$

Standard formulas for calculations

Volume Type	Formula	Conditions / Notes
Volume (L or dm ³)	$V = n \times 22.4$	At STP (0°C, 1 atm)
Volume (L or dm ³)	$V = n \times 24$	At RTP (25°C, 1 atm)
Moles	$n = V / V_m$	$V_m = 22.4$ L at STP, $V_m = 24$ L at RTP

Notes

1. Always specify the conditions: STP (0°C) or RTP (25°C).
2. Use liters (L or dm³) only to keep it simple no mL nor cm³

Example 1:

Practice converting between units with a simple grid at RTP

Gas Name	Amount (mol)	Calculation	Volume (dm ³)
Hydrogen	3.0	3×24	72.0
Carbon Dioxide	0.25	0.25×24	6.0
Ammonia	0.02	0.02×24	0.48

Example 2

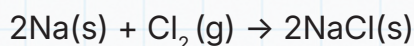
Calculate the number of moles in each of the following: (i) 11 g of CO₂ (ii) 3.01×10^{22} molecules of CO₂ (iii) 1.12 litres of CO₂ at S.T.P.

Part	Given	Formula	Calculation	Moles
i)	11 g CO ₂	$n = \text{mass} / \text{molar mass}$	$n = 11 / 44$	0.25 mol
ii)	3.01×10^{22} molecules CO ₂	$n = \text{number of molecules} / \text{Avogadro's number}$	$n = (3.01 \times 10^{22}) / (6.022 \times 10^{23}) = 0.050$ mol	0.05 mol
iii)	1.12 L CO ₂ at STP	$n = \text{volume} / 22.4$	$n = 1.12 / 22.4$	0.05 mol



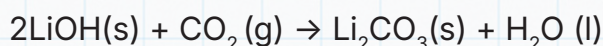
Application activity 5.12

1. a) How many moles are in 40.0 grams of water?
b) How many grams are in 3.7 moles of Na_2O ?
c) How many atoms are in 14 moles of cadmium?
d) How many moles are in 4.3×10^{22} molecules of H_3PO_4 ?
2. Sodium reacts with chlorine to produce sodium chloride:



Calculate the mass of sodium chloride produced from 960 cm^3 of chlorine and an excess of sodium. (M_r of NaCl = 58.5, molar volume = 24,000 cm^3)

3. Lithium hydroxide absorbs carbon dioxide from the air:



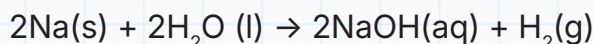
Calculate the mass of lithium hydroxide needed to absorb 12 dm^3 of carbon dioxide. (M_r of LiOH = 24, molar volume = 24 dm^3)

4. Calcium carbonate decomposes to form calcium oxide and carbon dioxide when it is heated:



Calculate the volume of carbon dioxide produced from completely decomposing 1.0 g of calcium carbonate. (M_r of CaCO_3 = 100, molar volume = 24,000 cm^3)

5. 4.6 g of sodium reacts completely with excess water:



Calculate the volume of hydrogen produced. (A_r of Na = 23, molar volume = 24 dm^3)



5.13 End of unit assessment

A) Knowledge and understanding

1. a) State the value for Avogadro's number.
b) Calculate the mass of 6.022×10^{23} nitrogen molecules.

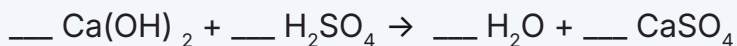
B) Skills

2. a) How many moles of copper react with 3.50 moles of silver nitrate?
$$\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu(NO}_3)_2 + 2\text{Ag}$$

b) If 89.5 grams of silver were produced, how many grams of copper reacted?



2. Given the following equation:



How many grams of calcium sulfate will be formed if 130 grams of calcium hydroxide react with an excess of sulfuric acid?

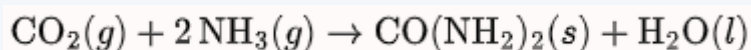
3. I_2 is produced by the reaction of 0.4235 mol of CuCl_2 according to the following equation: $2\text{CuCl}_2 + 4\text{KI} \rightarrow 2\text{CuI} + 4\text{KCl} + \text{I}_2$

a) How many molecules of I_2 are produced?

b) What mass of I_2 is produced?

5. In an accident, a solution containing 2.5 kg of nitric acid was spilled. Two kilograms of Na_2CO_3 were quickly spread on the area and CO_2 was released by the reaction. Was sufficient Na_2CO_3 used to neutralize all of the acid? (In this reaction, water and sodium nitrate are the other two products)

6. Urea, $\text{CO(NH}_2)_2$, is manufactured on a large scale for use in producing urea-formaldehyde plastics and as a fertilizer.



Calculate the maximum mass of urea that can be produced using the carbon dioxide generated from the combustion of 1.00×10^3 kg of carbon, assuming the CO_2 is fully used in the urea synthesis reaction.

7. Determine.

a) The number of moles and the mass (in grams) of chlorine, Cl_2 , required to react with 10.0 g of sodium metal, Na, to produce sodium chloride, NaCl.

b) The number of moles and the mass (in milligrams) of diatomic oxygen formed by the decomposition of 1.252 g of mercury(II) oxide.

c) The number of moles and the mass (in g) of sodium nitrate, NaNO_3 , required to decompose and produce 128 g of diatomic oxygen, where NaNO_2 is the other product.

d) The number of moles and the mass (in kg) of carbon dioxide formed by the combustion of 20.0 kg of carbon in an excess of diatomic oxygen.

e) The number of moles and the mass (in kg) of copper(II) carbonate needed to decompose in order to produce 1.500 kg of copper(II) oxide, where CO_2 is the other product.

f) The number of moles and mass (in grams) of C_2H_4 required to react with water to produce 9.55 g $\text{C}_2\text{H}_6\text{O}$.

8. 10.0 kg of $\text{N}_2(g)$ and 10.0 kg of $\text{H}_2(g)$ are mixed to produce $\text{NH}_3(g)$. Calculate the $\text{NH}_3(g)$ formed.



9.6. 28 g of sodium sulfate hydrate is heated. The mass after heating is 4.61 g. Calculate the value of x in $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$.

10. The empirical formula of **X** is $\text{C}_4\text{H}_{10}\text{S}$. The relative formula mass (M_r) of **X** is 180.

Find the molecular formula of **X**. ($A_r(\text{C}) = 12$, $A_r(\text{H}) = 1$, $A_r(\text{S}) = 32$)

11. A major textile dye manufacturer developed a new yellow dye. The dye has a percent composition of 75.95% C, 17.72% N, and 6.33% H by mass with a molar mass of about 240 g/mol. Determine the molecular formula of the dye.

C) Attitude and values

12. Why should a student be careful and precise when measuring substances for mole calculations in the laboratory?

13. How does the correct use of the mole concept help in avoiding wastage of chemicals during experiments?

UNIT 6:

PREPARATION OF OXYGEN AND ITS PROPERTIES.



Key unit competence

Analyze the preparation, properties, and roles of oxygen and its allotropes in supporting life and maintaining environmental balance

Learning objectives

At the end of this unit, you should be able to:

Knowledge and understanding

- physical properties of oxygen.
- Describe different laboratory methods for preparing oxygen gas.
- Describe the reaction of oxygen with metals and non-metals.
- Explain the importance of O_2 and O_3 in daily life.
- Describe different methods of gas collection and reason why they are appropriate.
- Explain the consequences
- Recall the consequences of ozone layer depletion.

Skills

- Prepare oxygen gas using appropriate laboratory procedures.
- Test the presence of oxygen
- Apply methods to collect gases based on their properties.
- Write equations for the reactions of oxygen with other elements

Values and attitudes

- Develop a sense of responsibility when carrying out experiments.
- Appreciate the importance of oxygen gas and the ozone layer in daily life.
- Develop a sense of environmental protection.

Introduction

Oxygen makes up about 46% of the mass of Earth's crust, about 89% of the mass of ocean water, and about 21% of Earth's atmosphere by volume. That alone makes it a very important element. Then we add the fact that we need it to breathe, and we see the importance of oxygen. On Earth the production of oxygen gas (also called elemental oxygen) comes from photosynthesis carried out by plants and algae. In this unit, we will prepare it by using different methods and explore its properties, allotropes and uses.



Introductory activity

Investigating the Role of Oxygen in Burning

1. Set-up

Place a small candle in the center of a heat-proof dish and carefully light it. Pour a small amount of water into the dish around the candle.

2. Procedure

Cover the burning candle with a transparent glass jar. Ask students to observe what happens to the flame. After a short time, the flame flickers and then goes out.

3. Observation

Students note that the flame becomes weaker and eventually extinguishes while the c

4. Guide students to discuss why the candle went out.

5. Students explain the observations.

6.1. Methods of preparation of oxygen gas



Learning activity 6.1.a

Experiment to prepare oxygen from hydrogen peroxide

Apparatus and reagents

Weighing balance, trough, water, stopper, delivery tube, flat-bottomed flask, dropping funnel, matchbox, splints, blue and red litmus papers, gas jar, hydrogen peroxide, and manganese dioxide.

Procedure

1. Weigh 5g of manganese dioxide using the weighing balance and put in the flat-bottomed flask.
2. Set up the apparatus as shown in Fig.6.1 below.
3. Add hydrogen peroxide from the dropping funnel to the manganese dioxide in the flat-bottomed flask. Observe and note what happens.
4. Test the gas collected using a glowing splint and the litmus papers.

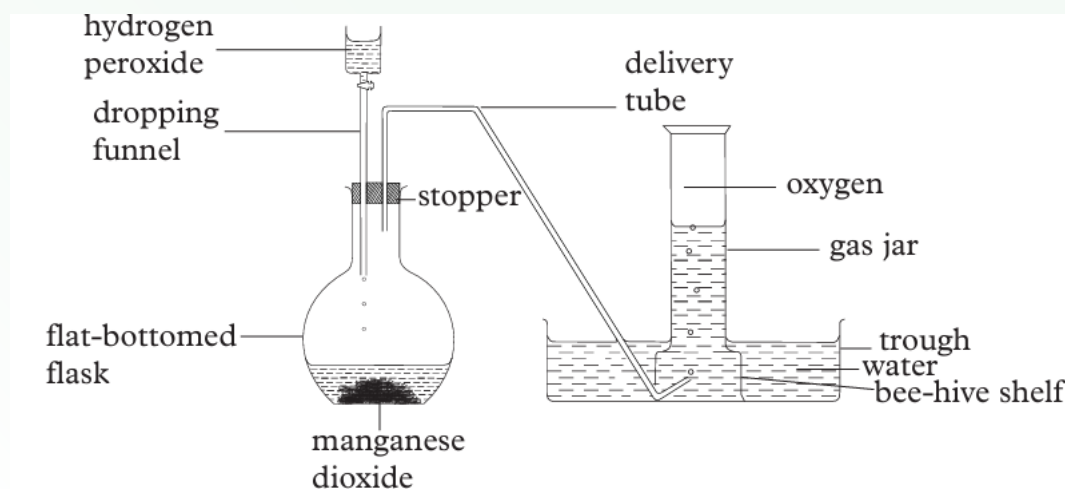


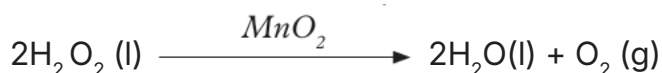
Fig 6.1.a: Set-up to prepare and collect oxygen gas in the laboratory from H_2O_2

1. What properties of the gas did you consider when collecting the gas?
2. What is the colour of the gas collected?
3. What happened to the (a) glowing splint? (b) red litmus paper? (c) Blue litmus paper?
4. What can you conclude about the nature of oxygen gas?

When we are collecting a gas, we consider its physical properties such as density and solubility in water. Oxygen is a colourless, odourless gas. It is also a neutral gas.

Hydrogen peroxide decomposes to produce oxygen. The decomposition of the peroxide is sped up by manganese dioxide, which acts as a catalyst.

Hydrogen peroxide $\xrightarrow{\text{manganese dioxide}}$ water + oxygen



The oxygen gas produced has no effect on moist red or blue litmus paper. It is a neutral gas. The gas relights a glowing splint. This is the confirmatory test for oxygen gas.



Learning activity 6.1.b

Experiment to prepare oxygen from metal peroxide and water

Procedure

1. Weigh about 5 g of sodium peroxide using the weighing balance and place it in the flat-bottomed flask.
2. Set up the apparatus as shown in Fig. 6.1 below, with the flask connected to a delivery tube leading to the gas jar over water.
3. Carefully add water to the sodium peroxide in the flask. Observe the reaction and the gas produced.
4. Collect the gas in the gas jar over water.
5. Test the gas using a glowing splint and by exposing it to blue and red litmus papers. Record your observations.

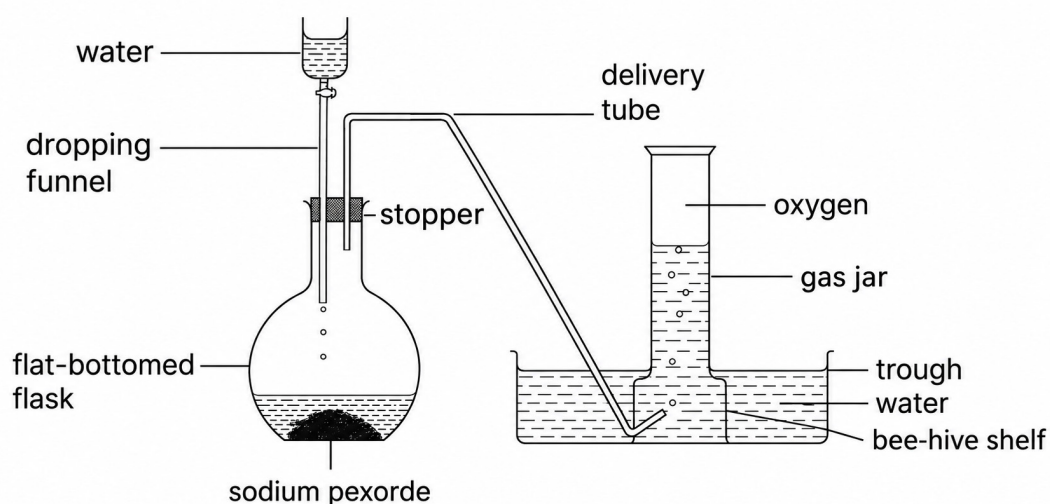
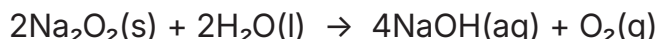


Fig.6.1.b: Set-up to prepare and collect oxygen from sodium peroxide

Sodium peroxide reacts vigorously with water to produce oxygen gas and a solution of sodium hydroxide.

Reaction



A method to prepare oxygen in the laboratory involves heating solid nitrates of group I metals, such as sodium nitrate and potassium nitrate, as well as heating potassium chlorate. This process can be conducted using a setup similar to the one described below:



Learning activity 6.1 c

Experiment to prepare oxygen from potassium chlorate

Apparatus and reagents

Boiling tube, delivery tube, test tube, trough, distilled water, splints, wet red and blue litmus papers, potassium chlorate, manganese dioxide, Bunsen burner.

Procedure

1. Place a mixture of 4 g of potassium chlorate and 1g of manganese dioxide in a boiling tube.
2. Set up your apparatus as shown in Fig 6.2 below.

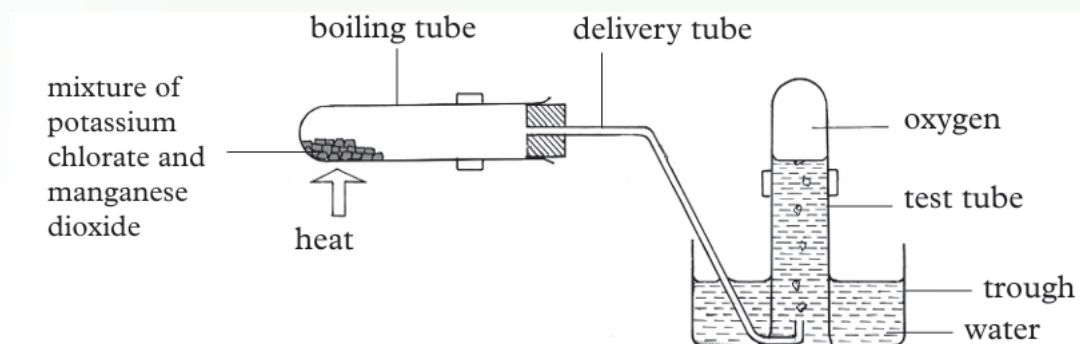


Fig 6.1.c: Preparation of oxygen using potassium chlorate

3. Heat the mixture gently then strongly.
4. Test the gas collected using a glowing splint and wet red and blue litmus papers. Write your observations.

On heating a mixture of potassium chlorate and manganese dioxide, the mixture decomposes into a white solid and a colourless gas. The gas relights a glowing splint and has no effect on moist blue or red litmus papers.

The white solid formed after heating is potassium chloride and the gas is oxygen. Manganese dioxide acts as a catalyst for the reaction.





The oxygen produced in all these cases relights a glowing splint. This is the confirmatory test for oxygen. The gas is collected over water since it is slightly soluble in water.

Note: Allotropes of oxygen are Dioxygen and Ozone

Table 6.1: Allotropes of oxygen

Allotrope	Formula	Physical State	Key Feature	Main Use
Dioxygen	O ₂	Gas	Supports life & combustion	Breathing, industrial processes
Ozone	O ₃	Gas	Strong oxidizer, UV absorber	Disinfection, UV protection



Application activity 6.1

1. Write the formula for oxygen gas-----
2. Oxygen gas is prepared in the laboratory by various methods. However, decomposition of hydrogen peroxide is the most preferred method. Explain
3. Write a balanced chemical equation with state symbols, for the decomposition of hydrogen peroxide
4. Manganese (iv) oxide is added during decomposition of hydrogen peroxide. What is the purpose of manganese (iv) oxide

6.2 Physical and chemical properties of oxygen gas

6.2.1 Physical properties of oxygen gas



Learning activity 6.2.1

Experiment to investigate the physical properties of oxygen

Apparatus and reagents

Two test tubes full of oxygen, litmus papers, a trough, and distilled water.

Procedure

1. Investigate the smell of oxygen. Caution! Always smell a gas by wafting it towards your nose.
2. Drop moist blue and red litmus papers in one test tube full of oxygen.
3. Insert the second test tube full of oxygen in a trough with distilled water and note the level of water inside the gas jar. Leave the setup for 20 minutes and note any change in the water level.

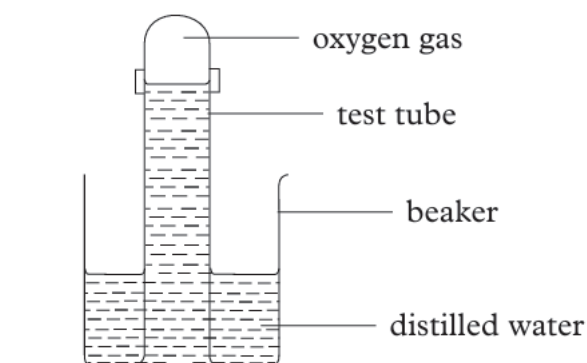


Fig 6.2.1: Dissolving oxygen in water

1. What is the smell of oxygen?
2. What is the effect of oxygen on the moist litmus papers?
3. What changes do you observe in the test tube inverted in water? Explain your observation.
4. From the results of the experiment, state physical properties of oxygen.

Oxygen is a colourless odourless gas. It has no effect on moist litmus papers. When a test tube full of oxygen is inverted in water, the level of water rises slightly in the tube.

Some of the physical properties of oxygen include:

1. It is slightly denser than air.
2. Oxygen is slightly soluble in water.
3. It is a neutral gas.

Table 6.2: Summary of Physical Properties of Oxygen

State	Gas at room temperature.
Color/Odor/Taste:	None.
Density	1.429 g/L (slightly denser than air).
Solubility	Slightly soluble in water (vital for aquatic life).
Boiling Point	-183°C
Melting Point:	-218.4°C.
Conductivity	Poor conductor of heat and electricity

6.2.2 Chemical properties of oxygen gas



Learning activity 6.2.2

Experiment to demonstrate that part of air is used in burning.

Apparatus and reagents

Candle, trough, water, cork, matchbox, gas jar

Procedure

1. Put some water in a trough as shown below.
2. Put a short candle on a cork and float the cork with the candle on the water as shown in figure 6.4

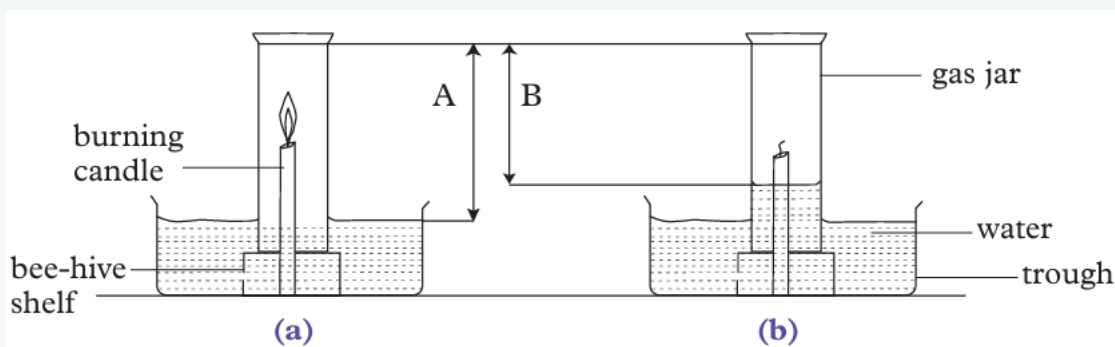


Fig 6.2.2: Experimental set-up

3. Cover the candle with a gas jar and mark on the gas jar the level of the water. Measure the height of the air column (A) and record it.
4. Now remove the gas jar and light the candle. Gently cover it with the gas jar again and observe what happens.
5. Mark the final level of water inside the gas jar when there is no further change. Measure and record the height of the air column (B) once more.

Question

1. State and explain what happens to the candle after some time.
2. State and explain what happened to the level of water at the end of the experiment.
3. What was the length of the air column in the gas jar before and after burning?
4. Using the results of this experiment, calculate the percentage of air used up in burning.
5. Is it necessary to leave the apparatus to cool before making the final reading? Explain.
6. State any sources of errors in this experiment.

After burning for some time in a fixed volume of air, the candle goes out and the water level in the gas jar rises. The candle extinguishes because the active part of air (oxygen) needed for burning is used up, leaving the inactive part of air behind. As the **active air** is consumed, a partial vacuum is formed inside the gas jar. The higher atmospheric pressure outside then pushes water up into the jar, causing the water level to rise.

Oxygen, which makes up about 21% of air, is the active part that supports burning. When it is used up, the remaining air mainly nitrogen does not support combustion. Dilute sodium hydroxide can replace water because it absorbs carbon dioxide present and produced during burning. The apparatus must be allowed to cool before measuring the final air volume since heating causes gases to expand.



Learning activity 6.2.3

Experiment to demonstrate reaction of oxygen with metals

Apparatus and reagents

Gas jars full of oxygen, deflagrating spoon, tongs, litmus papers, Bunsen burner, magnesium, iron, copper, calcium, distilled water.

Procedure

1. Place a piece of calcium in a deflagrating spoon. Heat to burn then lower the burning metal into a gas jar of oxygen. What do you observe?
2. Lower a piece of burning magnesium ribbon by the use of tongs into a jar of oxygen. How does it burn? What is the colour of the product formed?
3. Add some water to the product formed, shake and test the solution formed with litmus paper. Repeat this for all other metals in this experiment. What do you observe?

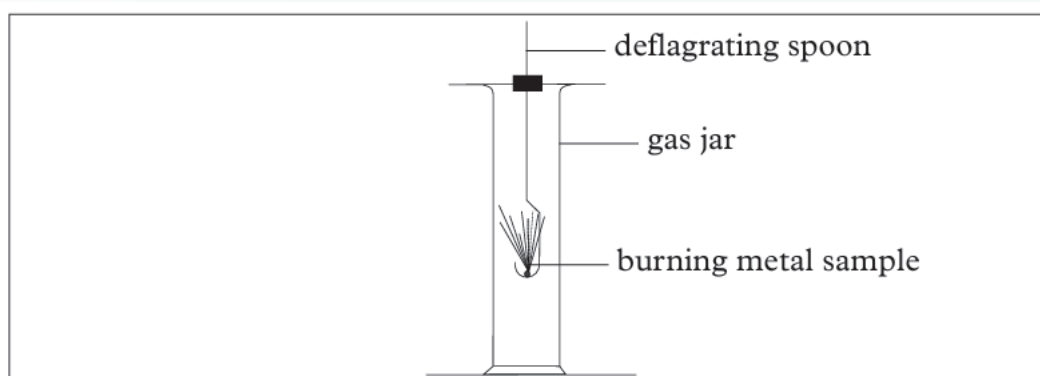


Fig 6.2.3: Reaction of metals with oxygen

4. Attach a small piece of iron wire at the end of a deflagrating spoon, heat until the wire glows red. Lower the glowing wire into a gas jar of oxygen. Record any observations. Repeat the experiment using a piece of copper instead of iron wire. Record your observations.



Question

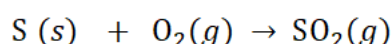
1. Why do you think red litmus paper turns to blue while the blue one retains its colour in procedure 3?
2. Write symbolic equations for the reactions.
3. What is the general name given to the products formed when metals burn in oxygen?

When elements burn in oxygen, they form oxides, which are compounds made of oxygen and one other element. This process is called oxidation. Most metals form basic oxides when they burn in oxygen, and some of these oxides dissolve in water to produce alkaline solutions.

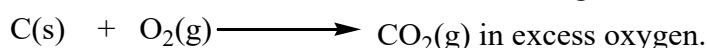
No.	Word Equation	Balanced Chemical Equation	Effect in Water / Litmus
(i)	Sodium + Oxygen → Sodium oxide	$2\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow \text{Na}_2\text{O(s)}$	Na_2O dissolves in water to form NaOH , which turns red litmus paper blue (basic)
		$\text{Na}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightarrow \text{NaOH(aq)}$	
(ii)	Calcium + Oxygen → Calcium oxide	$2\text{Ca(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{CaO(s)}$	CaO dissolves slightly in water to form Ca(OH)_2 , which turns red litmus paper blue (Lime water is a filtrate of Ca(OH)_2 .)
		$\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)}$	
(iii)	Iron + Oxygen → Iron(III) oxide	$4\text{Fe(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Fe}_2\text{O}_3\text{(s)}$	Insoluble in water; no effect on litmus paper
(iv)	Copper + Oxygen → Copper(II) oxide	$2\text{Cu(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{CuO(s)}$	Insoluble in water; no effect on litmus paper

When non-metals are oxidised, they form acidic oxides. Acidic oxides cause air pollution when released into the atmosphere.

Sulphur + Oxygen → sulphur dioxide gas $\rightarrow$ (burns with a blue flame)



Carbon + Oxygen $\longrightarrow$ Carbon dioxide gas



In limited Oxygen carbon monoxide gas is formed



Application activity 6.2

1. State three physical properties of oxygen.
2. Complete the following table to show how the metals react when burnt in oxygen.

Metal	How it burns in oxygen	Appearance of product	Name of the oxide	Solubility of the oxide in water	Effect of the oxide solution on litmus
Calcium			Calcium oxide		
Copper					No effect on litmus
Sodium				Soluble	
Magnesium		White solid			
Iron	Glows red-hot with sparks				

3. a) The table below is an incomplete record of observations made when non metals are burnt in oxygen. Complete the table by filling in the blank spaces.

Non-metal	How it burns in oxygen	Name of the oxide formed	Appearance of the oxide	Effects of the oxide solution on litmus paper
Phosphorus	Burns with a bright flame			
Carbon			Colorless gas	
Sulphur			White fumes (cloudy gas)	

- b) Write symbolic chemical equations for the reactions that take place when the three non-metals are burnt in oxygen.
4. Why do you think we should economise on the use of reagents when performing experiments?

6.3. Methods of collecting gases



Learning activity 6.3

Provide students with a list of gases (e.g. NH_3 , HCl , H_2) and their properties (solubility and density). Ask them to match each gas to the correct laboratory setup.

Use a reaction between magnesium and hydrochloric acid to produce hydrogen. Students must collect the gas using a **gas syringe** or **water displacement** to record the volume produced every 10 seconds.

Ask students to design a setup to collect a specific volume of "mystery gas" that is both soluble in water and denser than air (solution: downward delivery).

Density Demonstration: Pour CO_2 (denser than air) from one jar into another to extinguish a candle, illustrating why downward delivery works for this gas.

Collection of gases in the laboratory depends on the physical properties of the gas. Such properties include:

- Solubility in water
- Density of the gas

Gases can be collected using the following methods:

- Over water
- Downward delivery
- Upward delivery

1. Collection of gases over water

This method is appropriate for gases that are insoluble or slightly soluble in water and those that do not react with water. The gas jar must be filled with water first and the beehive shelf placed under water in the trough as shown in fig 6.6.

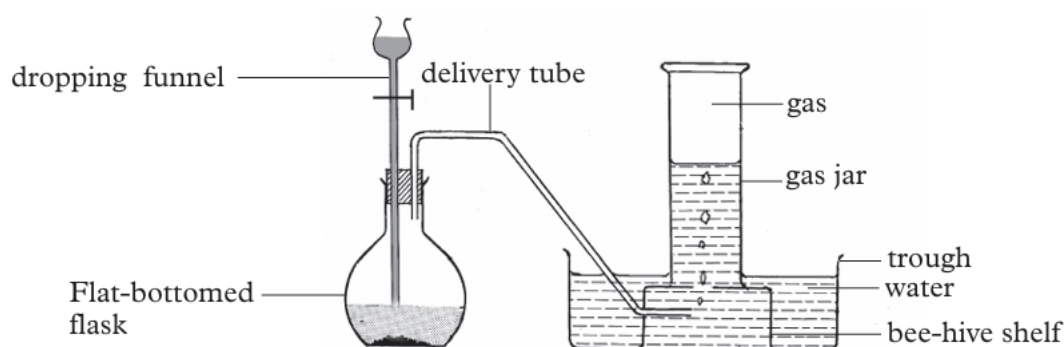


Fig 6.3.a: Collection of a gas over water

This method can be used to collect gases such as oxygen, nitrogen, dinitrogen oxide, nitrogen monoxide, carbon monoxide and hydrogen sulphide.

The advantage of this method of gas collection is that you can always see when the gas jar is full. If the gas is required dry, it is passed through concentrated sulphuric acid or anhydrous calcium chloride.

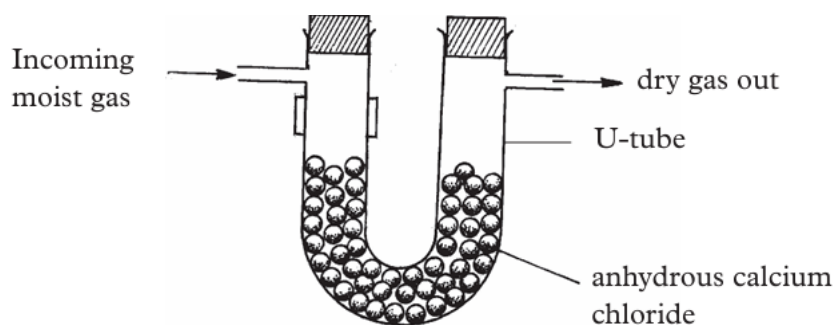


Fig 6.3.b Drying of a gas using anhydrous calcium chloride

Gases, such as ammonia, sulphur dioxide and hydrogen chloride are highly soluble in water. Therefore, they should never be collected over water. The following examples illustrate preparation and collection of gases over water.

a) Preparation and collection of nitrogen gas.

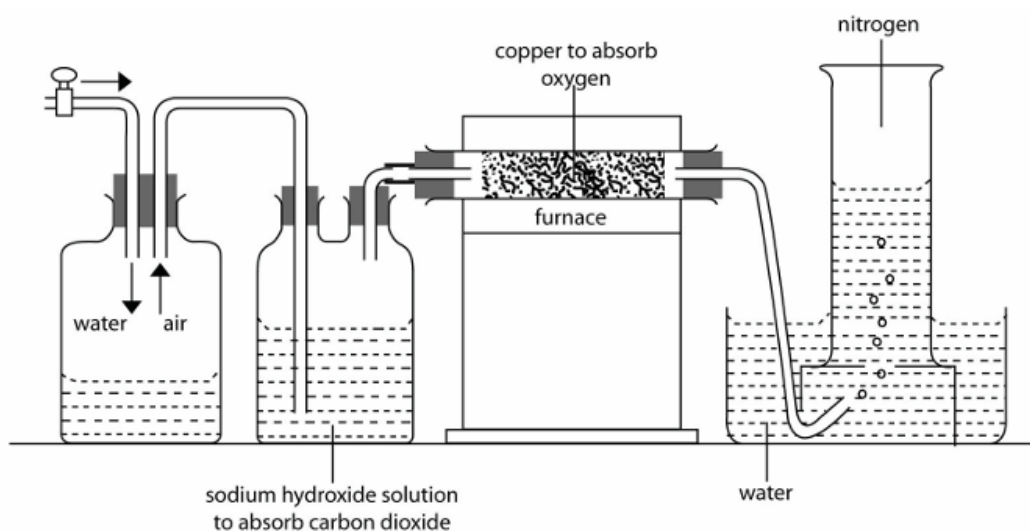


Fig 6.3.c: Set-up to prepare and collect dry nitrogen gas in the laboratory

b) Preparation and collection of oxygen.

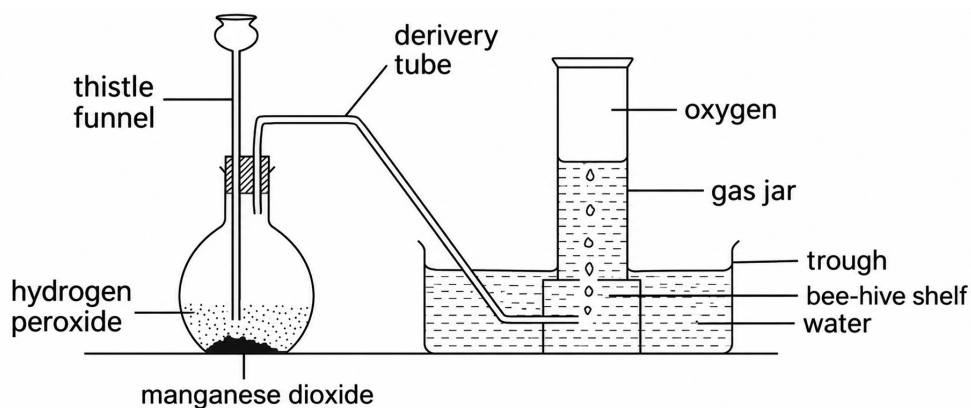


Fig 6.3.d: Set-up to prepare and collect dry oxygen gas in the laboratory

2. Collection of gases by downward delivery

This method is used for collection of gases that are denser than air. It is also called upward displacement of air as illustrated in fig 6.11. The set-up to be used is as shown below.

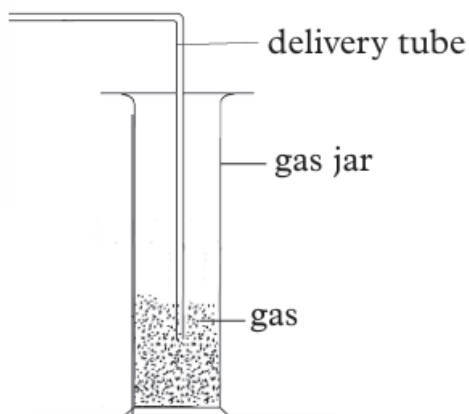


Fig 6.3.e: Collection of a gas by downward delivery

The main disadvantage of this method is that gases diffuse so fast that the gas collected will almost certainly contain some air. Also, unless the gas is coloured, it may not be easy to know when the gas jar is full. Gases that can be collected by this method include carbon dioxide, sulphur dioxide, chlorine, nitrogen dioxide and hydrogen sulphide. The following examples illustrate collection of gases by downward delivery.

a) Preparation and collection of chlorine.

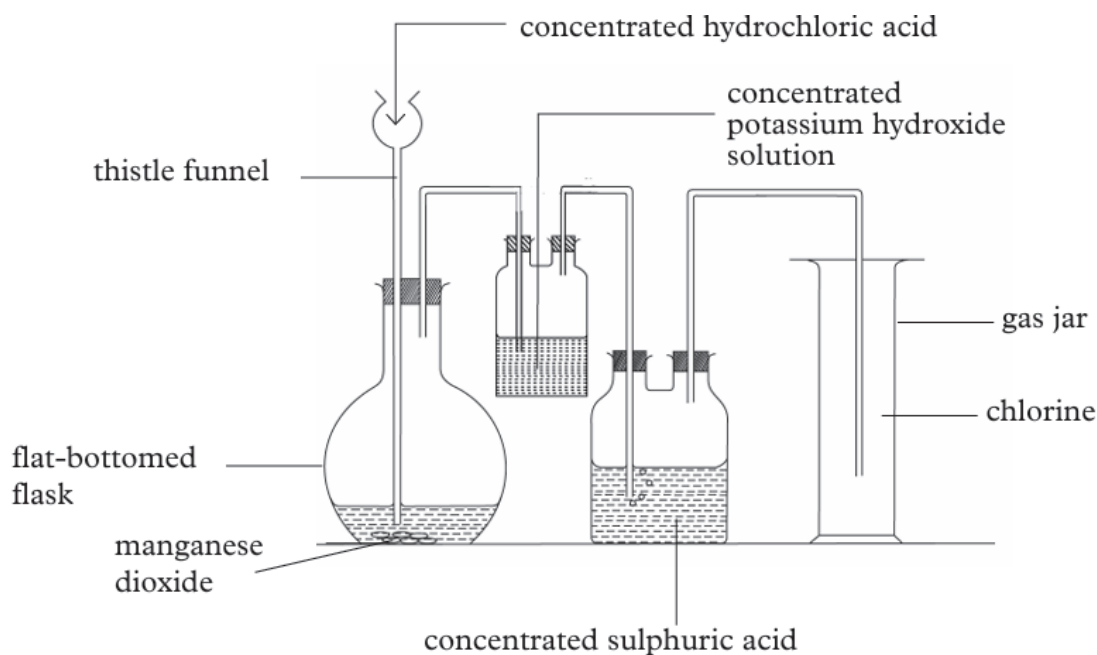


Fig 6.3.f: Set-up to prepare and collect dry chlorine gas

b) Preparation and collection of dry carbon dioxide.

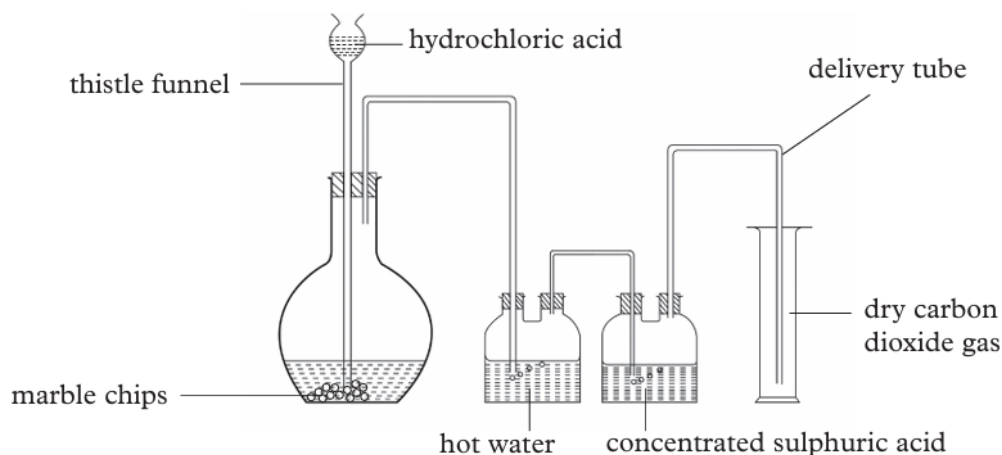


Fig 6.3.g: Set-up to prepare and collect dry carbon dioxide gas

3. Collection of gases by upward delivery

This method is used for collecting gases that are less dense than air. This method is also called downward displacement of air as illustrated in fig 6.14

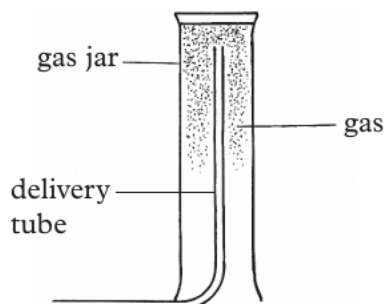


Fig 6.3.h: Set-up for collecting gases by upward delivery method

The disadvantage of this method is that you may not easily know when the gas jar is full. Also because of diffusion of gases, it may be contaminated with air. Gases that can be collected by this method are hydrogen and ammonia. The following example illustrates collection of gases by upward delivery.

a) Preparation and collecting of ammonia

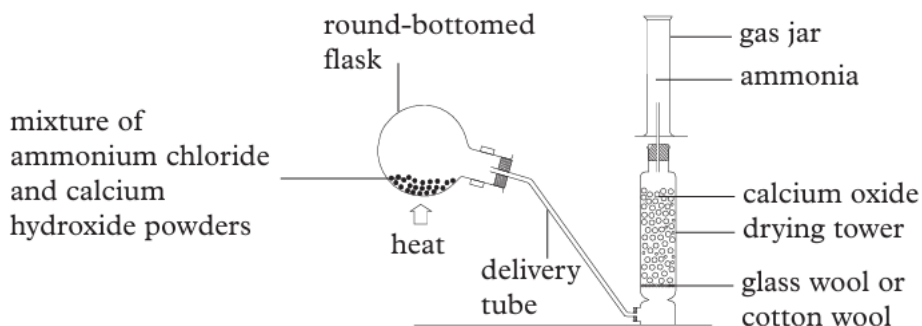


Fig 6.3.i: Set-up to prepare and collect dry ammonia gas

Note: It is important to ensure that we choose the appropriate method of collecting a gas during preparation. This will ensure that we do not misuse expensive laboratory chemicals.



APPLICATION ACTIVITY 6.3

1. State the physical properties considered when collecting a gas.
2. Identify the method that can be used to collect the following gases:
 - a) Oxygen
 - b) Carbon dioxide
 - c) Ammonia
3. Draw well-labelled diagrams to illustrate the following methods of gas collection.
 - a) Upward delivery
 - b) Downward delivery

6.4. Uses of oxygen gas



Learning activity 6.4

1. List and explain the uses of oxygen gas.

1. Oxygen is necessary for respiration. When we eat food, it is oxidised to produce energy for various body activities. Such activities include movement, growth, reproduction and circulation of blood.

Food + oxygen $\rightarrow$ energy + carbon dioxide + water vapour

Thus, plants and animals oxidise food to get energy.

Note: Always plant trees each time you cut one and also plant where none existed before. Trees increase concentration of oxygen in the atmosphere.

2. Medical Uses:

- Oxygen is supplied to patients who have difficulty breathing.
- It is used in oxygen therapy for heart and lung patients.

3. Industrial Uses:

- Used in steel and metal production (e.g., in blast furnaces).
- Supports combustion in welding and cutting metals.

4. Water Treatment: Oxygen helps in purifying water by supporting aerobic bacteria that decompose waste.

5. Fire / Combustion Support: Oxygen supports burning of fuels, e.g., in furnaces or engines.



Application activity 6.4

1. Discuss uses of oxygen in: (a) Hospitals (b) Welding (c) Respiration
2. Why do you think it is necessary to plant a tree each time you cut one and also plant more?
3. "Without oxygen there will be no life on earth" Discuss.



6.5 End unit assessment

A) knowledge and understanding

Multiple Choice Questions

1. Oxygen gas can be prepared in the laboratory by decomposing hydrogen peroxide using
 - A) Copper
 - B) Manganese (IV) oxide
 - C) Zinc
 - D) Iron
2. Oxygen is usually collected over water because it is
 - A) Very soluble in water
 - B) Insoluble in water
 - C) Slightly soluble in water
 - D) Heavier than water
3. Which of the following is a use of oxygen?
 - A) Making fertilizers
 - B) Respiration
 - C) Preventing rusting
 - D) Making plastics

4. True or False

5. Oxygen supports combustion.

True / False

Open-ended Questions

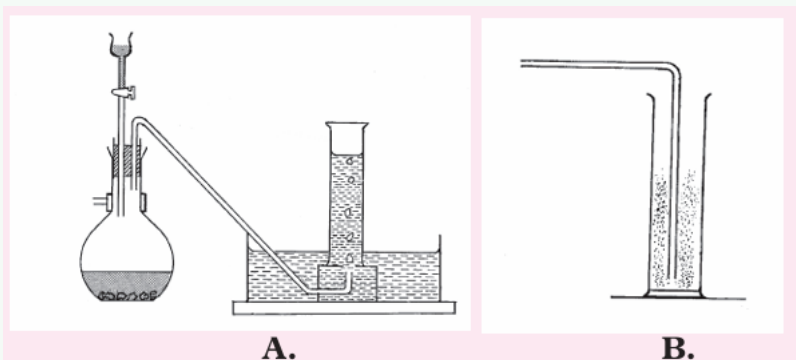
6. State two physical properties of oxygen gas.
7. Describe one method used to prepare oxygen gas in the laboratory and write the balanced chemical equation.

B) Skills

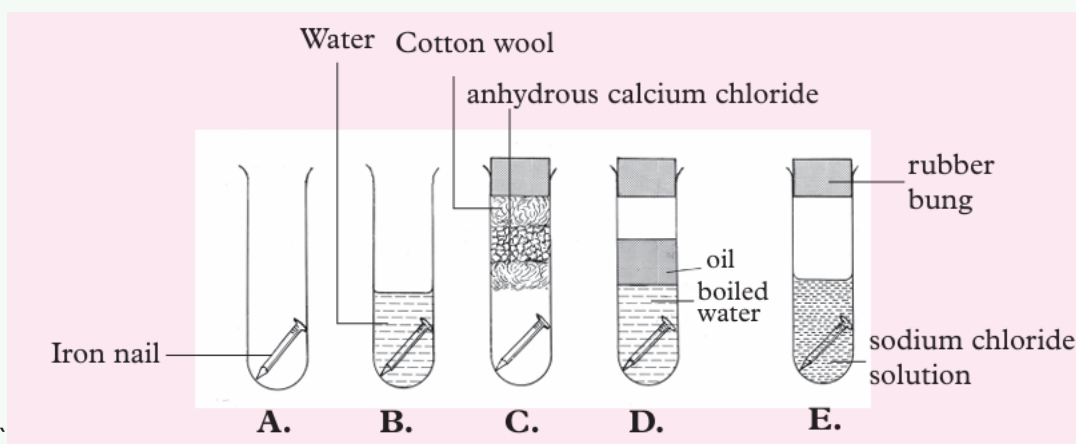
8. a) Write balanced symbolic equations to show the formation of the following compounds from the constituent elements.
 - i) MgO
 - ii) Fe_2O_3



9. The following diagrams represent some methods which can be used to collect various gases in the laboratory.



- Identify the two methods of gas collection.
 - Name the gases that can be collected as shown in A and B.
 - State at least one property of the gases collected by each of the two methods.
9. The diagram below shows an experiment to determine conditions under which iron rusts.



- State the reason for using boiled water in tube D.
- State the purpose of the layer of oil in tube D.
- What is the effect of anhydrous calcium chloride in tube C?
- State what happens to the nails in tube A.
- From this experiment, state two conditions necessary for rusting to take place.



C) Attitudes and values

10. a) Name the product of burning sulphur in oxygen.
b) Account for the effects of the product formed in 7(a) above to the atmosphere.
11. Explain why mountain climbers and deep-sea divers need to have cylinders of oxygen.
12. Explain why protecting forests and planting trees is important for maintaining sufficient oxygen in the environment.

UNIT 7:

PREPARATION AND CLASSIFICATION OF OXIDES



Key unit competence

- Use an appropriate method for preparing oxides
- Classify oxides based on their properties

You should be able to:

Knowledge and understanding

- Explain how different oxides are formed.
- State the uses of different oxides.
- Describe the reaction of oxides with water, acids, and bases.
- Describe the process of producing slaked lime.

Skills

- Prepare different oxides from elements and compounds
- Classify oxides into alkaline, acidic, amphoteric, and neutral.
- Prepare the slaked lime in the laboratory.

Values and attitudes

- Develop a team approach and a sense of responsibility in group activities.
- Respect for the procedures while performing experiments.
- Care about harmful oxides like sulphur dioxide and nitrogen dioxide, during experiments
- Appreciate the uses of slaked lime in balancing the pH of soil.

Introduction

Oxides are binary chemical compounds formed by the combination of oxygen with another element. They are found extensively in nature, forming the majority of the Earth's crust.

The classification of oxides is done into neutral, amphoteric, and basic or acidic based on their acid-base characteristics.

Oxides are crucial in modern technology, used for everything from catalysts (like TiO_2 in cars) and semiconductors (ZnO in displays) to construction (CaO in cement), ceramics, coatings, energy storage, and even biomedicine.



Introductory activity

Take a sample of magnesium, iron filings, sulphur and carbon. Burn each sample one by one. Observe the product formed and gas produced. Was any residue left in case of non-metals? Can you name the gas produced when carbon burns in air? Write the chemical equation for each reaction.

7.1. Preparation of oxides

a) Preparation of oxides by direct combination or synthesis method



Learning activity 7.1 a)

"Burning Elements to create Oxides."

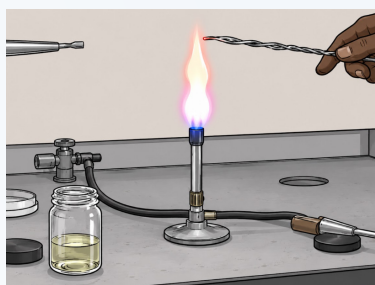
Objective: To observe how metals and non-metals combine directly with oxygen and to classify the resulting oxides based on their properties.

1. Materials Required

- Magnesium ribbon, iron filings, sulfur powder, and carbon (charcoal), Bunsen burner, deflagrating spoon (combustion spoon), gas jar with lid, tongs, and litmus paper (red and blue).

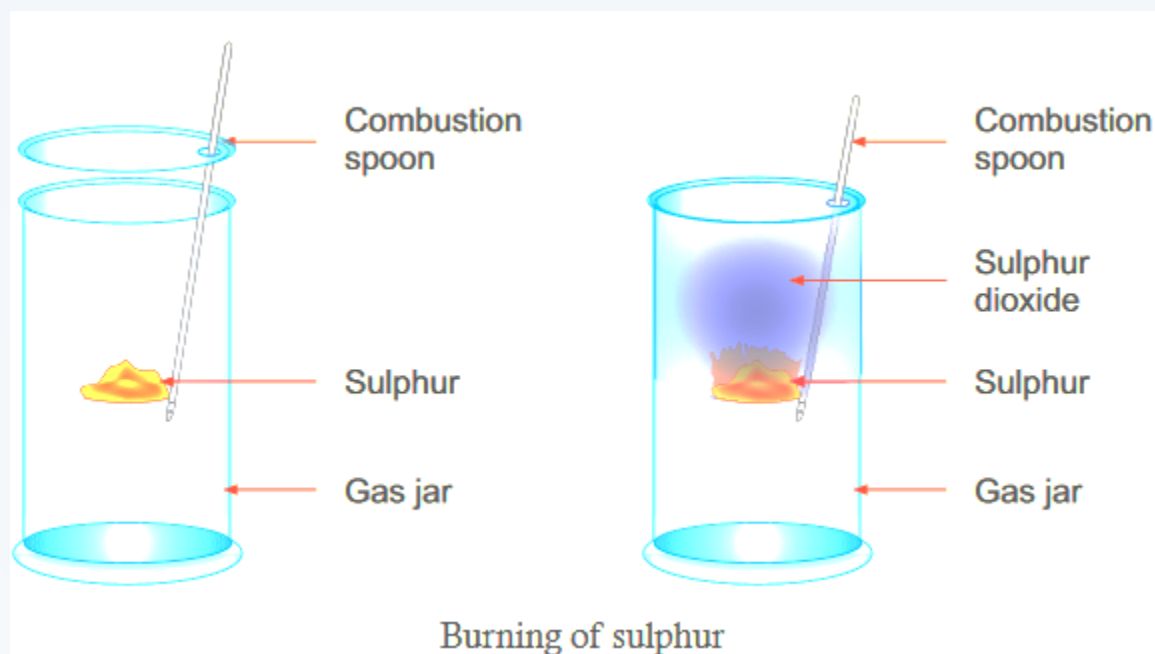
2. Procedure

- Hold a piece of magnesium ribbon with tongs and heat it in a Bunsen burner flame. It will burn with a dazzling white flame to form a white powder (magnesium oxide).





Non-Metal Reaction (Sulfur): Place a small amount of sulfur in a deflagrating spoon and heat it until it starts burning with a blue flame. Lower the spoon into a gas jar of oxygen/air and cover it to collect the gas (sulfur dioxide).



Alternative Reaction (Iron): Sprinkle iron filings into a flame; they will burn vigorously to form iron (III) oxide.



3. Testing the Products

For Metals: Dissolve the magnesium oxide residue in a small amount of water. Test the solution with red litmus paper; it will turn blue, indicating the oxide is **basic**.

For Non-Metals: Add a small amount of water to the gas jar containing sulfur dioxide and shake. Test with blue litmus paper; it will turn red, indicating the oxide is **acidic**.



Element type	Example reaction	Balanced equation
Metal	Magnesium + oxygen	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
Metal	Iron + oxygen	$4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$
Non-metal	Carbon + oxygen	$\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
Non-metal	Sulphur + oxygen	$\text{S} + \text{O}_2 \rightarrow \text{SO}_2$

b) Preparation of oxides by thermal decomposition of carbonates, nitrates, and hydroxides of metals



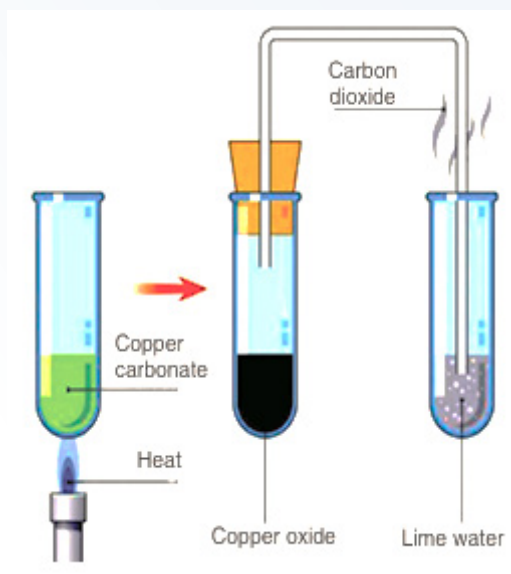
Learning activity 7.1 b)

Activity : Thermal Decomposition of Carbonates

This activity prepares metal oxides while releasing carbon dioxide.

Procedure:

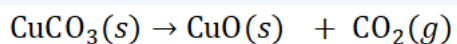
1. Place a spatula of **copper (II) carbonate** (green) into a boiling tube.
2. Fit a delivery tube to the boiling tube and dip the other end into a test tube of **limewater**.
3. Heat the carbonate gently at first, then strongly.



Observations: The green powder turns into **black copper (II) oxide**. The limewater turns milky/cloudy, confirming the release of CO_2



Copper carbonate $\rightarrow$ Copper oxide + Carbon dioxide

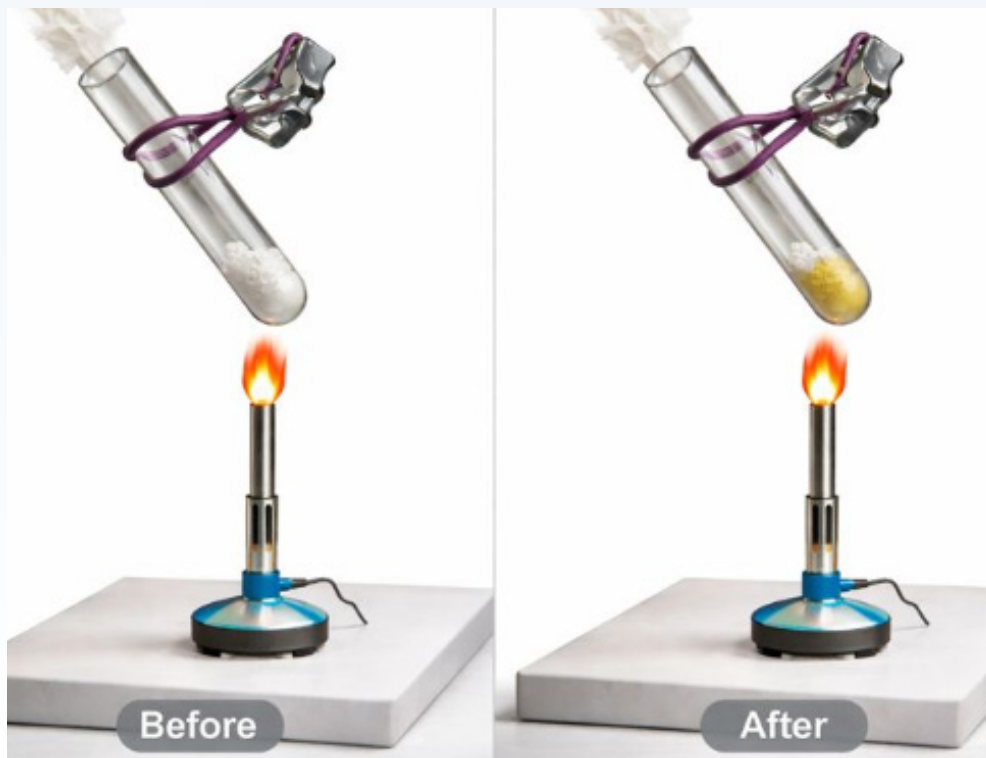


Activity ii: Thermal Decomposition of Nitrates

This activity is used to prepare oxides, typically characterized by the evolution of brown nitrogen dioxide fumes.

Procedure:

1. Place a small amount of **lead (II) nitrate** crystals (white) into a dry test tube.
2. Heat strongly over a Bunsen burner.



3. Test the evolved gases with a glowing splint (for O_2) and damp blue litmus paper (for NO_2).

Observations: Brown fumes (NO_2) are produced, and a glowing splint relights (O_2). A **yellow residue of lead (II) oxide** remains.

Equation: Lead nitrate $\rightarrow$ LeadOxide + Nitrogen dioxide gas + Oxygen gas



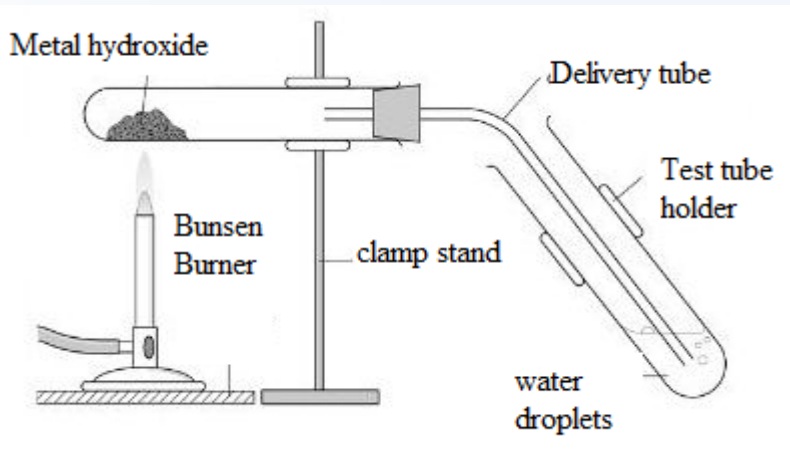


Activity iii: Thermal Decomposition of Hydroxides

Metal hydroxides decompose into metal oxides and water vapor.

Procedure:

1. Heat a sample of **magnesium hydroxide** or **calcium hydroxide** in a hard glass test tube.



2. Observe the changes in the solid and any condensation at the mouth of the tube.

Observations: The hydroxide decomposes into a white powder (**magnesium oxide** or **calcium oxide**). Droplets of water often form near the cooler mouth of the test tube.

Thermal decomposition reactions

1. Metal Carbonates

When heated, most metal carbonates decompose into a metal oxide and carbon dioxide gas.

General equation:

Metal carbonate $\rightarrow$ Metal oxide + CO_2

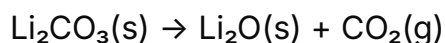
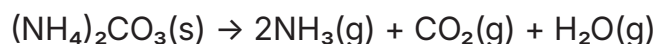
Example:



Special Cases

Group 1 carbonates

Carbonates of Group 1 metals (except lithium carbonate) do not decompose easily on heating.

Example exception:**Ammonium carbonate****2. Metal nitrates**

On heating, most metal nitrates decompose to form a metal oxide, nitrogen dioxide (brown gas), and oxygen.

General equation:

Most metal nitrates $\rightarrow$ Metal oxide + NO_2 + O_2

Group 1 nitrates (except LiNO_3) form nitrites + O_2 .

Example:**3. Metal hydroxides**

Most metal hydroxides decompose on heating to form a metal oxide and water.

General equation:

Metal hydroxide $\rightarrow$ Metal oxide + H_2O

Example:**Application activity 7.1**

1. What is meant by *thermal decomposition*?
2. What happens when metal carbonates are heated?
3. Write the balanced chemical equation for the decomposition of CaCO_3 .
4. Why are sodium and potassium carbonates stable to heat?
5. Write the word equation for heating potassium nitrate.
6. Write the word equation for heating copper (II) hydroxide.
7. What products are formed when calcium hydroxide is heated?

7.2 Classification of oxides as acidic, basic, neutral and amphoteric



Learning activity 7.2

Materials and chemicals

Materials: Beakers, test tubes, test tubes rack,

Chemicals: Magnesium Oxide (MgO), Calcium Oxide (CaO), Carbon Dioxide (CO₂) can be generated from dry ice or reacting vinegar/baking soda), Zinc Oxide (ZnO), Distilled water, dilute hydrochloric acid (HCl), Red and blue litmus paper or universal indicator, sodium hydroxide (NaOH).and Aluminium Oxide (Al₂O₃).

Procedure:

Step 1: Used prepared oxide solutions

1. Label 5 test tubes for each oxide: MgO, CaO, ZnO, Al₂O₃, CO₂.
2. Add **about 0.5–1 g** of each oxide to its test tube.
3. Add **5–10 mL of distilled water** to each test tube.
4. Stir the mixture thoroughly with a stirring rod.

Question: Do all oxides dissolve in water? Which ones form clear solutions, and which ones remain cloudy?

Step 2: Test with Litmus paper

Take a piece of red litmus paper and dip it into each solution.

Take a piece of blue litmus paper and dip it into each solution.

Observations to record:

Oxide	Red Litmus	Blue Litmus	Interpretation
MgO			
CaO			
ZnO			
Al ₂ O ₃			
CO ₂			

Question: Does red litmus turn blue? Does blue litmus turn red? What does this tell you about the solution (acidic, basic, or neutral)?

Step 3: Test with Universal Indicator (Optional)

1. Add a few drops of universal indicator to each solution.
2. Observe the color change and compare it with the **pH color chart**.

Observation/Question:

- What color appears for each oxide?
- Does it match acidic (red/orange), basic (blue/purple), or neutral (green)?



Step 4: Reaction with acid and base

1. Take small portions of **ZnO** and **Al₂O₃** in separate test tubes.
2. Add a few drops of **dilute HCl** and stir. Observe any reaction.
3. In another set, add a few drops of **NaOH solution** to ZnO and Al₂O₃. Observe.

Questions:

- Which oxide reacts with acid? Which reacts with base?
- What does this tell you about amphoteric oxides?
- Why are ZnO and Al₂O₃ called amphoteric?
- How can we use these observations to predict the behavior of oxides in reactions?

Basic Oxides: Formed by metals (e.g., Na₂O, MgO). They react with water to form bases or with acids to form salt and water.

Acidic Oxides: Formed by non-metals (e.g., CO₂, SO₂). They react with water to form acids or with bases to form salt and water.

Amphoteric Oxides: Formed by specific metals (e.g., ZnO, Al₂O₃). They react with both acids and bases to form salt and water.

Neutral Oxides: Show neither acidic nor basic properties (e.g., CO, NO). They do not react with acids or bases.

- Basic oxides (MgO, CaO): Turn red litmus blue, slightly or strongly soluble in water, react with acids.
- Acidic oxide (CO₂): Turns blue litmus red, reacts with bases, forms an acidic solution in water.
- Amphoteric oxides (ZnO, Al₂O₃): Do not change litmus paper in water, but react with both acids and bases.

Oxide	Reaction with Water	Litmus Test	Reaction with Acid/ Base	Classification
MgO	Slightly soluble, forms Mg(OH) ₂	Red → Blue	Reacts with acid	Basic oxide
CaO	Reacts vigorously to form Ca(OH) ₂	Red → Blue	Reacts with acid	Basic oxide

Oxide	Reaction with Water	Litmus Test	Reaction with Acid/ Base	Classification
ZnO	Insoluble in water	No change	Reacts with acid & base	Amphoteric oxide
Al ₂ O ₃	Insoluble in water	No change	Reacts with acid & base	Amphoteric oxide
CO ₂	Dissolves to form H ₂ CO ₃	Blue → Red	Reacts with base	Acidic oxide

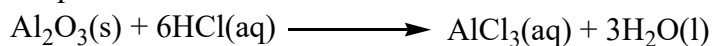
A table showing a summary of the classification and reactions of oxides

Classification	Nature of element	Reaction with water	Reaction with acids/bases	Examples
Basic oxides	Metals, e.g., group 1 and 2	Forms alkaline solutions/bases	Reacts with acids to form salt and water	Na ₂ O, CaO, MgO, CuO
Acidic oxides	Non-metals	Reacts with water to form acidic solutions./ acidic	Reacts with bases to form a salt and water.	SO ₂ , CO ₂ , SO ₃ , P ₄ O ₁₀
Amphoteric Oxide	Specific metals / metalloids	Usually insoluble in water	React with both acids and bases to form a salt and water	Al ₂ O ₃ , PbO, ZnO,
Neutral oxides	Certain non-metals	Does not react with water	Does not react with either acids or bases	CO, NO, N ₂ O, H ₂ O

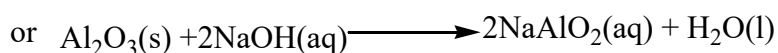
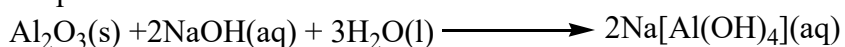
Chemical equations:



Amphoteric Oxide+ Acid:



Amphoteric Oxide + Base:





Application activity 7.2

1. a) What is an oxide?
b) Name the four main classes of oxides.
2. a) What is an acidic oxide?
b) Give two examples of acidic oxides.
c) Write an equation to show the reaction of an acidic oxide with water.
3. a) Define a basic oxide.
b) Give two examples of basic oxides.
c) Write an equation showing the reaction of a basic oxide with an acid.
4. a) What is an amphoteric oxide?
b) Name two amphoteric oxides.
c) Write one equation showing an amphoteric oxide reacting with an acid
5. a) What is a neutral oxide?
b) Give two examples of neutral oxides.
6. a) Calcium oxide is a basic oxide and reacts with acids to form salt and water. True/false
b) Carbon dioxide is an acidic oxide and turns blue litmus paper red when dissolved in water. True/ false
c) Zinc oxide reacts only with acids and not with bases./ true/false
7. Match the oxide in Column A with its reaction behavior in Column B.

Column A	Column B
Magnesium oxide (MgO)	A. Reacts with both acids and bases
Silicon dioxide (SiO ₂)	B. Reacts with acids only
Zinc oxide (ZnO)	C. Reacts with bases only

7.3. Uses of oxides and production of slaked lime (Ishwagara)

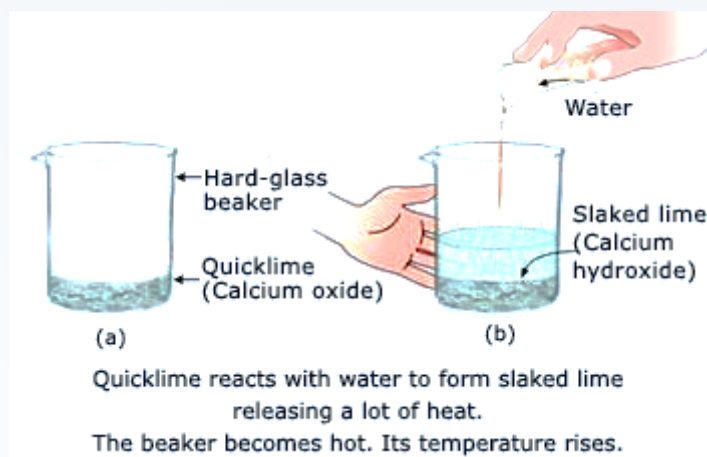


Learning activity 7.3

Materials: Quicklime (CaO), water, hard beaker, glass stirring rod, and a thermometer.

Procedure:

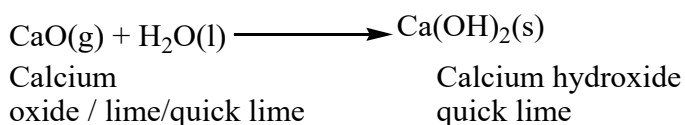
- Place a small amount of quicklime in the beaker.
- Slowly add water to the beaker.



- Stir the mixture. What do you notice? You will notice the quicklime crumbles into a fine white powder and the beaker becomes very hot, indicating an **exothermic reaction**

Notice: the quicklime crumbles into a fine white powder and the beaker becomes very hot, indicating an **exothermic reaction**.

Calcium oxide (lime or quicklime) reacts vigorously with water to form calcium hydroxide (slaked lime) releasing a lot of heat (exothermic reaction)



Uses of slaked lime (calcium hydroxide):

- Construction:** Used as a binder in mortar, plaster, and for **whitewashing** walls.
- Agriculture:** Applied to **neutralize acidic soil** and improve fertility.
- Water Treatment:** Used to soften water and remove impurities in wastewater treatment plants.
- Industry:** Used in sugar refining to eliminate contaminants and in the production of bleaching powder.

Uses of other oxides

1.Construction & Materials	• Calcium Oxide (Lime): Used in steelmaking, cement, and water purification. • Silicon Dioxide: Key for glass, cement, optical fibers, and heat-resistant silicones. • Aluminum Oxide: Forms durable ceramics for aerospace, automotive, and electrical insulation.
Electronics & Energy	• Indium Oxide, Zinc Oxide: Used in semiconductors for displays, solar cells, and sensors. • Titanium Dioxide: Used in photo catalysis (pollution breakdown) and as a white pigment.
Catalysis & Environment	• Metal Oxides: Act as catalysts in industrial processes like sulfuric acid production ($2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ Catalyst: $\text{V}_2\text{O}_5(\text{s})$). • Titanium Dioxide, Zinc Oxide: Help break down pollutants and sense contaminants.
Pigments & Coatings	• Iron Oxides: Provide red, yellow, and brown colors in paints, coatings, and ceramics. • Cobalt/Copper Oxides: Used to create blue and green colors in glazes and stains for pottery.
Health & Food	• Magnesium Oxide: Used as an antacid and dietary supplement. • Silicon dioxide: Used as an anticaking agent in some foods



Application activity 7.3

1. Match the oxide with its primary real-world use.

Oxide	Real-world use
Carbon dioxide	Forms a natural protective coating on aluminium foil to prevent corrosion
Aluminium oxide	Used by plants for photosynthesis and in fire extinguishers to displace oxygen
Calcium oxide	The primary component of sand used extensively in glass manufacturing
Iron(iii)oxide	Acts as a catalyst in industrial chemical synthesis for the oxidation of hydrocarbons and the decomposition of hydrogen peroxide in the preparation of oxygen gas.
Manganese(iv) oxide	Used a rust –preventing primer (red oxide) for protecting metal work
Silicon dioxide	Also known as quick lime, it's widely used in glass and ceramic production and to treat acidic soils in farming.

2. Calcium oxide is an important industrial oxide.
- Name the raw material used to manufacture calcium oxide.
 - Write the chemical equation for the preparation of calcium oxide in a lime kiln.
 - State three uses of calcium oxide.
 - Explain why calcium oxide is suitable for neutralizing acidic soils.
3. Silicon dioxide occurs naturally in large amounts.
- Name two natural sources of silicon dioxide.
 - State the physical properties of silicon dioxide that make it useful industrially.
 - Describe the use of silicon dioxide in the manufacture of glass.
 - State one use of silicon dioxide in construction
4. Aluminium oxide has both chemical and physical importance.
- State two properties of aluminium oxide.
 - Explain why aluminium oxide is used as a refractory material.
 - State two uses of aluminium oxide in industry.
 - Explain the use of aluminium oxide in the extraction of aluminium.
5. Carbon dioxide is a common non-metal oxide with many uses.
- State two physical properties of carbon dioxide.
 - Explain how carbon dioxide is used in fire extinguishers.
 - Describe the use of carbon dioxide in the manufacture of soft drinks.
 - State one environmental use of carbon dioxide in plant growth.



7.4 End of unit assessment

A) Knowledge and understanding

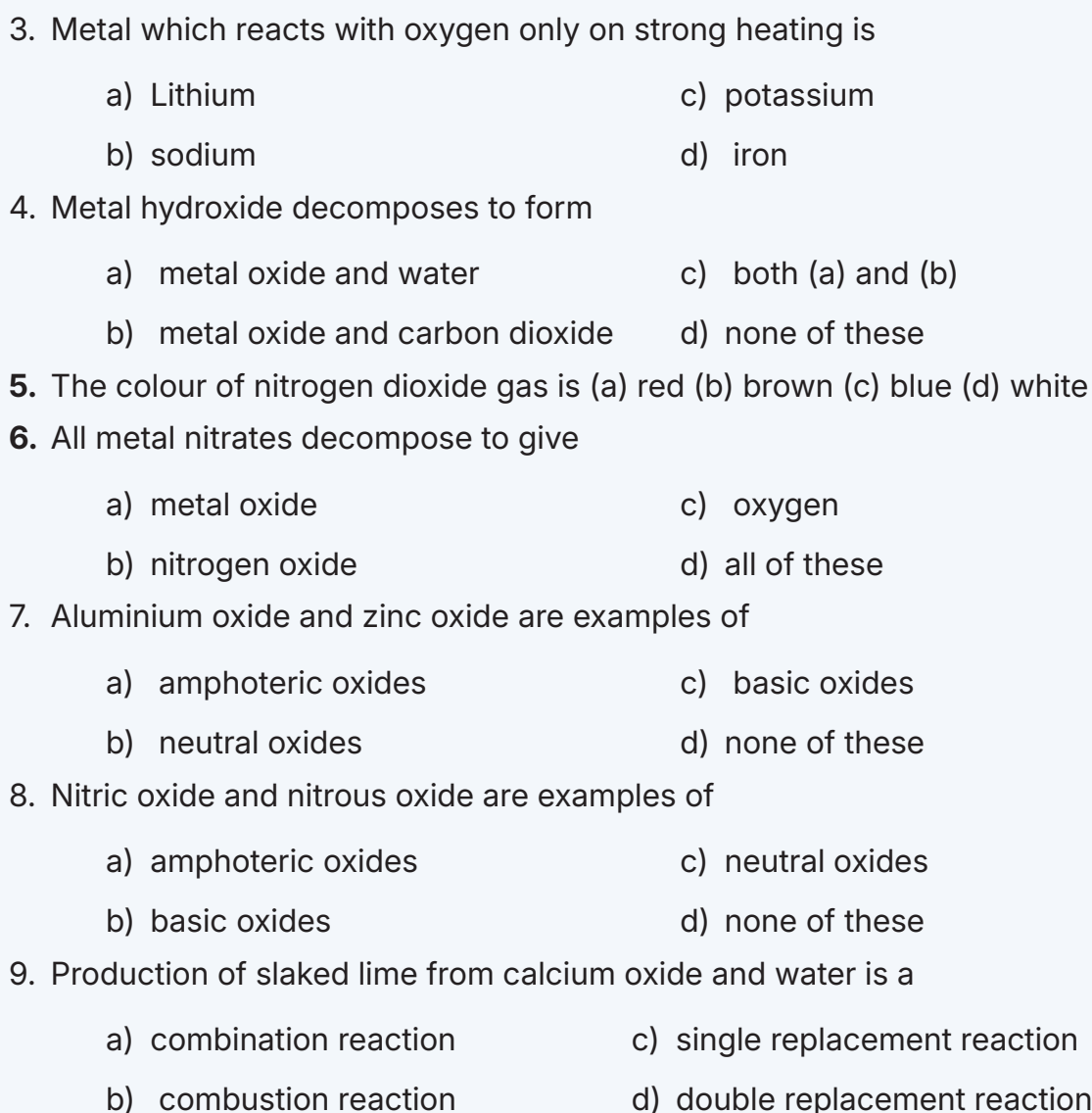
i) Multiple Choice Questions

1. Metals react with oxygen to produce

- | | |
|----------------------|------------------------|
| a) basic oxides | c) both (a) and (b) |
| b) amphoteric oxides | d) neither (a) nor (b) |

2. Non-metals react with oxygen to give

- | | |
|-------------------|---------------------|
| a) acidic oxides | c) both (a) and (b) |
| b) neutral oxides | d) none of these |



10. List two examples of each

- i) Acidic oxide
- ii) Basic oxide
- iii) Amphoteric oxide
- iv) Neutral oxide
11. Can you name some compounds that decompose to produce metal oxide and carbon dioxide?
12. State one property of **acidic oxides**.

13. Compare between acidic oxide and basic oxide.

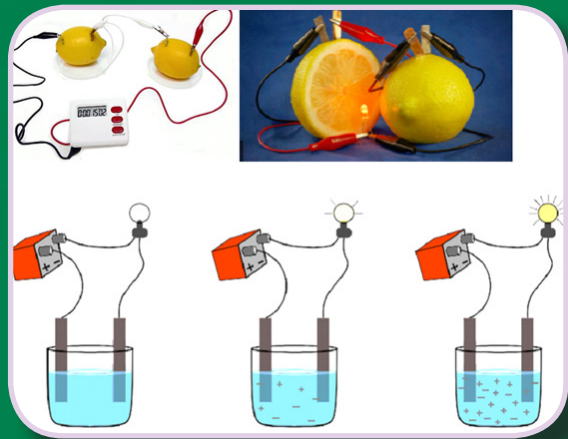
14. Distinguish between amphoteric and neutral oxides.



15. Explain the preparation of oxide by thermal decomposition of carbonate.
 16. State the properties of acidic oxides and basic oxides.
 17. What happens when
 - i) calcium oxide reacts with water
 - ii) magnesium combines with oxygen
 - iii) sulphur burns in oxygen
 - iv) lithium carbonate decomposes
 - v) carbon dioxide dissolves in water
 18. Write a balanced chemical equation for each reaction of Q6.
 19. Make a table showing a list of acidic and basic oxides.
 20. Explain why aluminium oxide is classified as an amphoteric oxide
- C) Attitudes and values**
21. Why should you respect the procedures while performing experiments?

UNIT 8:

ELECTROLYTES AND NON-ELECTROLYTES



Key unit competence

Distinguish between non-electrolytes, weak electrolytes, and strong electrolytes.

Learning Objectives

You should be able to:

Knowledge and understanding

- Demonstrate understanding of electrolytes and non-electrolytes.
- Differentiate weak and strong electrolytes from non-electrolytes by using example.
- State applications of electrolytes in daily life

Skills

- Use an experiment to differentiate electrolytes from non-electrolytes.
- Classify solutions as strong electrolytes, weak electrolytes, or non-electrolytes through experiment.

Values and attitudes

- Develop an awareness of safety issues when carrying out experiments.
- Appreciate the importance of electrolytes in daily life, like sulphuric acid in a car battery.

Introduction

Some substances, when dissolved in water, undergo either a physical or a chemical change that results in constituent cations and anions in the solution. This process of dissociation of a substance into its ions is known as ionization, and the substances that undergo ionization are called electrolytes. Substances that do not ionize when dissolved in water are called non-electrolytes.



Introductory activity

Given the following substances:

Table salt, Sugar, Distilled water, and Lemon juice or dilute acid

Question: *Which of these do you think can conduct electricity when dissolved in water? Why?*

Substance	Prediction (Conducts / Does not conduct)
Salt solution	
Sugar solution	
Distilled water	
Lemon juice	

8.1 Electrolytes and non-electrolytes



Learning activity 8.1

The Conductivity Circuit

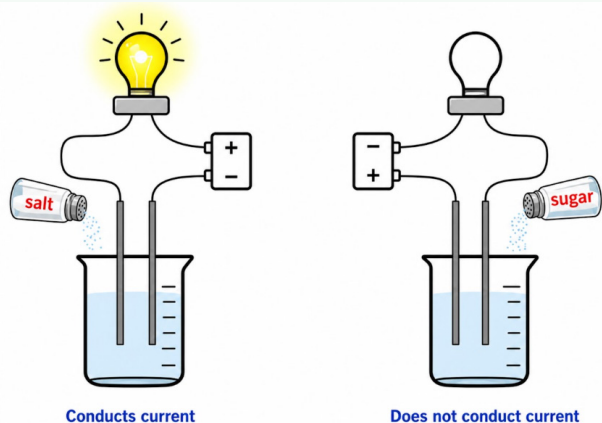
1. Materials Needed

Apparatus: 9V battery, LED or small light bulb with holder, two carbon electrodes (or copper wires), and a beaker.

Test Substances: Distilled water, table salt (NaCl), sugar (sucrose), vinegar (acetic acid), and rubbing alcohol (ethanol).

2. Procedure

Build the Circuit: Connect the battery and the light bulb in series. Leave a «gap» in the circuit where the two electrodes are placed.



Test Pure Water: Place the electrodes in a beaker of distilled water. Observe if the bulb glows (it should not).

Test Solutions: Dissolve a small amount of each test substance in separate beakers of distilled water and insert the electrodes.

Observe and Clean: Record the brightness of the light bulb for each solution.

Crucial: Rinse the electrodes with distilled water between every test to avoid cross-contamination.

Table of results

Substance	Observation	Why it happens	Classification
Table salt (NaCl) Solution	Bright glow	The substance dissociates into ions (charged particles) that carry current	Strong electrolyte
Vinegar	Dim glow	The substance dissociates partially to form ions	Weak electrolyte
Sugar solution	No glow	The substance dissolves as neutral molecules; no ions are present to carry current	Non-electrolyte
Distilled water	No glow	The substance dissolves as neutral molecules; no ions are present to carry current	Non-electrolyte



Conclusion: Substances that allow electricity to pass through them are called electrolytes, while substances that do not allow electricity to pass are called non-electrolytes. **Guiding questions**

1. Which solutions made the light bulb glow? What do these substances have in common? (They form ions in water, are ionic compounds/strong acids/bases).
2. Which solutions did not light up the bulb? (Covalent, molecular compounds like sugar).
3. Why did saltwater glow brighter than vinegar? (Strong vs. weak electrolytes, more ions vs. fewer ions).

Electrolytes are substances that dissolve in water to form solutions that conduct electricity.

The aqueous solutions of **non-electrolytes**, on the other hand, **do not conduct electricity**

Electrolytes are substances that dissolve in water to produce ions and conduct electricity.

- **Strong electrolytes** (e.g., table salt) dissociate completely: bright bulb glow.
- **Weak electrolytes** (e.g., vinegar) dissociate partially: dim bulb glow.

Non-electrolytes (e.g., sugar) dissolve in water but do not ionize to form ions: no bulb glow.

Key Point: More ions: stronger conductivity. Substances that produce light in a circuit are electrolytes; those that do not are non-electrolytes



Application activity 8.1

- a) Define the term **electrolyte**.
- b) Define the term **non-electrolyte**.
4. a) Explain why an electrolyte conducts electricity in aqueous solution or molten state.
b) Explain why a non-electrolyte does not conduct electricity.
5. Distinguish between **electrolytes** and **non-electrolytes** using the following aspects:
 - a) Presence of ions
 - b) Electrical conductivity
 - c) Examples

4. a) State two examples of electrolytes.
- b) State two examples of non-electrolytes.
- c) Classify the following as electrolyte or non-electrolyte:
 - i) Sodium chloride solution
 - ii) Sugar solution
 - iii) Dilute hydrochloric acid.
5. a) what is meant by **strong electrolyte** and **weak electrolyte**?
- b) Give one example of a strong electrolyte and one example of a weak electrolyte.
- c) Explain how a strong electrolyte differs from a weak electrolyte.

8.2. Conductivity of electricity by electrolytes



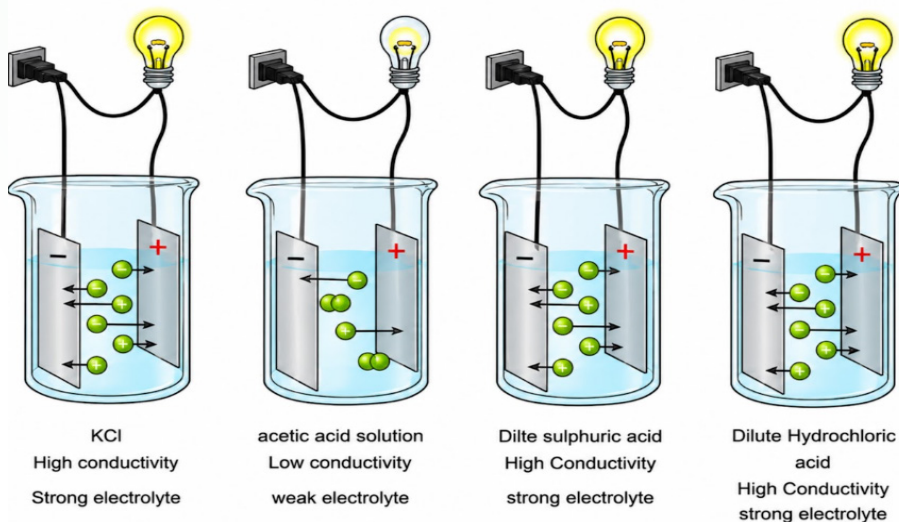
Learning activity 8.2

Setup: Create a simple circuit with a battery, wires, a resistor, a light bulb, and two electrodes.

Baseline: Test deionized (DI) water first; the bulb should not light.

Test Strong Electrolyte: Dissolve a strong electrolyte (e.g., NaCl) in deionized water. Place electrodes in the solution; the bulb will glow brightly. Repeat with mineral acids such as dilute sulphuric acid, dilute hydrochloric acid or nitric acid. The bulb will glow brightly as in sodium chloride.

Test Weak Electrolyte: Dissolve a weak electrolyte (e.g., acetic acid/vinegar) in deionized water. The bulb will glow dimly.



Strong Electrolytes: Dissociate *completely* into ions (e.g., $\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$), creating many charge carriers for high conductivity.

Weak electrolytes ionize/dissociate partially, leaving many undissociated molecules and fewer ions.

Strong vs. Weak

If the bulb glows dimly (as with vinegar or lemon juice), the substance is a weak electrolyte, meaning it only partially dissociates into ions. A bright glow indicates a strong electrolyte that dissociates completely.

Strong: High ions, high conductivity, bright light, complete dissociation.

Weak: Low ions, low conductivity, dim light, partial dissociation



Application activity 8.2

- Define the term **strong electrolyte**.
 - Define the term **weak electrolyte**.
 - State the main difference between strong and weak electrolytes in terms of ionization in solution.
- Explain why strong electrolytes conduct electricity well in aqueous solution.
 - Explain why weak electrolytes conduct electricity poorly.
 - State one experimental observation that shows the difference in conductivity between the two.
- State the type of substances that usually form strong electrolytes.
 - State the type of substances that usually form weak electrolytes.
 - Give two examples of strong electrolytes and two examples of weak electrolytes.
- What is meant by the degree of dissociation?
 - Compare the degree of dissociation of strong electrolytes with that of weak electrolytes.
 - State how the degree of dissociation affects the number of ions in solution.
- Name one strong acid and one weak acid.
 - Classify hydrochloric acid, ethanoic acid, sodium chloride solution, and ammonia solution as strong or weak electrolytes.
 - Explain why ethanoic acid is a weak electrolyte while hydrochloric acid is a strong electrolyte

8.3 Applications of electrolytes in daily life



Learning activity 8.3

1. Explain why sulphuric acid is useful in car batteries but harmful to skin.

The following are applications of electrolytes:

- Sodium chloride (NaCl) is more commonly known as “salt” or “table salt,” and is the major ingredient in the edible salt that is sprinkled on food to make it taste better. Sodium chloride is present in the ocean and is the main thing that makes ocean water taste so salty.
- Nitric acid (HNO_3) is a corrosive acid. It is used in the manufacture of fertiliser. It is also used as one of the ingredients in certain types of liquid rocket fuel.
- Hydrochloric acid (HCl) is a strong acid widely used in the chemical industry.
- Calcium chloride (CaCl_2) is also a salt and is different from table salt. Calcium chloride is a compound made of calcium and chloride, and is one of the types of salt used to control ice on sidewalks and roadways. The compound is often produced from limestone.
- Potassium nitrate (KNO_3) is used in a wide variety of substances. It can be used as a food additive, but is also an ingredient in various types of rocket fuels and in fireworks. Potassium nitrate was an ingredient in gunpowder for many years.
- Sodium hydroxide (NaOH) is an important ingredient in many detergents, soaps, and drain cleaners. It is highly dangerous because of its ability to decompose lipids and proteins in skin, causing burns when not handled properly.
- Sulphuric acid (H_2SO_4) is used in car batteries to get the electricity to flow. Like sodium hydroxide, it can cause very severe chemical burns when it comes into contact with skin, so it is very important to handle it with extreme caution.
- Ammonium chloride (NH_4Cl) is a main component of dry cell. A dry cell is commonly known as a Leclanché cell. The other electrolyte used in a dry cell is manganese dioxide (MnO_2).
- Sodium acetate (CH_3COONa) is an electrolyte that includes sodium as a main ingredient. It is often used to seal concrete so that it is protected against bad weather. It is also used as an ingredient in certain types of foods, such as salt and vinegar potato chips, because of its salty and tangy flavour when mixed with other seasonings.
- Magnesium hydroxide ($\text{Mg}(\text{OH})_2$) is also known as milk of magnesia because of its milk-like appearance. It is a main component of many types of laxatives and antacids, and in underarm deodorants and antiperspirants. It can also be applied to the scalp for dandruff control.



Application activity 8.3

1. Fill in the Blanks
 - a) Nitric acid is a _____ acid.
 - b) Sodium hydroxide is used in the making _____ and _____. _____ is used in car batteries for electricity.
 - c) _____ is a main component of dry cell.
 - d) Magnesium hydroxide is also known as _____.
2. Why is ammonium chloride used in dry cells?
3. Why is sodium chloride important in the human body?
4. Why does a salt solution conduct electricity?



8.4. End unit assessment

A) Knowledge and understanding

i) Multiple Choice Questions

1. Sugar solution does not contain (a) atom (b) molecule (c) ion
2. Sodium chloride and copper sulphate are examples of
 - a) electrolytes
 - b) compounds
 - c) both (a) and (b)
 - d) elements
 - e) None of these
3. Sucrose and urea are good examples of
 - a) strong electrolyte
 - b) weak electrolyte
 - c) non-electrolyte
 - d) all of these
4. During electrolysis of water, hydrogen gas is produced _____ times as much as oxygen.
 - a) two times
 - b) three times
 - c) four times
 - d) five times
 - e) Ammonium hydroxide
 - f) All of these
5. Choose the correct statement(s).
 - a) Hydrochloric acid is widely used in chemical industry
 - b) Magnesium hydroxide is also known as milk of magnesia
 - c) Nitric acid is used in the manufacture of fertiliser
 - d) All of the above



6. Electrolyte used in car batteries to get the electricity is ____ .

- a) sulphuric acid
- b) Hydrochloric acid
- c) nitric acid
- d) All of these

II) Short Answer Questions

7. Give two examples of (a) Electrolytes (b) Non-electrolytes

8. Explain non-electrolytes.

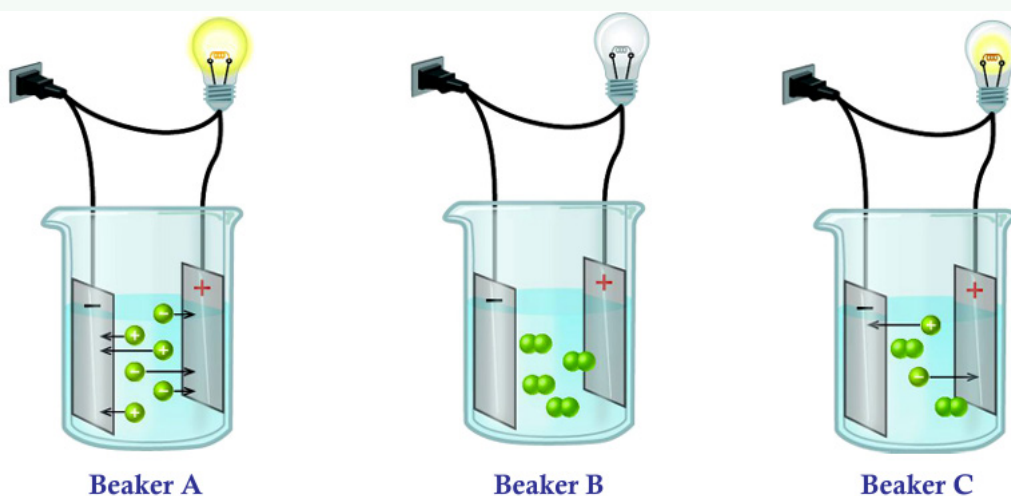
9. Name the electrolyte used in/as

- a) car batteries
- b) table salt
- c) fertilizer
- d) detergents

10. Distinguish between strong and weak electrolytes. Give examples.

11. Carry out an experiment to differentiate between electrolytes and non-electrolytes.

12. Look at the figures and answer the following questions.



- a) Which of the beakers contains a non-electrolyte? (a) Beaker A (b) Beaker B (c) Beaker C
- b) Which beaker contains a solution of a weak electrolyte? (a) Beaker A (b) Beaker B (c) Beaker C
- c) Which beaker contains a solution of a strong electrolyte? (a) Beaker A (b) Beaker B (c) Beaker C
- d) Which beaker contains a solution of an organic acid? (a) Beaker A, (b) Beaker B, (c) Beaker C, (d) None of these



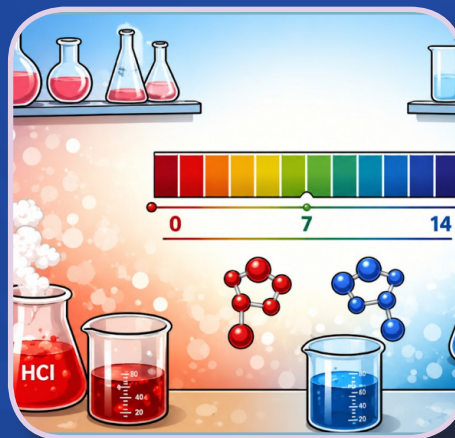
- e) The beaker B may contain
- i) Pure water (b) Glucose solution
 - ii) All of these
 - iii) None of these

C) Attitudes and values

13. How is a strong acid stored in a laboratory?

UNIT 9:

CHEMICAL PROPERTIES OF ACIDS AND BASES



Key unit competence:

Apply knowledge of the preparation and properties of acids and bases to explain their reactions and uses in daily life.

At the end of this unit, you should be able to:

Knowledge and understanding

- Describe how acids and bases can be prepared by the displacement or addition of water to soluble metal oxides
- Describe the chemical properties of acids and bases.

Skills

- Compare the reactions of acids and bases with metals, metal oxides, carbonates, and hydrogen carbonates
- Handle acids and bases.
- Predict the products of the reactions of acids and bases with other substances.

Values and attitudes

- Develop a culture of working in a team.
- Develop orderliness, especially when performing experiments.
- Appreciate the potential dangers associated with strong acids and bases.
- Appreciate the uses of acids in biological processes, especially in the digestion of human beings and other species.
- Appreciate the uses of a base in balancing the pH of soil.

Introduction

The word 'acid' is commonly used in everyday life. It is regarded as a highly corrosive substance with a sour taste. On the other hand, bases are bitter in taste and have a soapy feel. Acids and bases react differently with different substances. This unit aims to help to understand the chemical properties of acids and bases and how these substances behave during chemical reactions. This unit develops skills to identify acids and bases, explain their reactions, and relate these reactions to everyday and industrial applications while observing safety precautions.



Introductory activity

Look at the pictures below. Do the things in the pictures look familiar? Name them.



A



B



C

Fig 9.1: Substances/solutions

1. What are the things made of? Name other substances found at home made of acids and bases. Come up with a list of uses of acids and bases based on your findings above.

9.1. Preparation of acids and bases

9.1.1. Preparation by displacing a volatile acid by a less volatile one



Learning activity 9.1.1

Displacing a volatile acid by a less volatile one

Apparatus and reagents

Sodium chloride, concentrated sulphuric acid, round bottomed flask, thistle funnel,

tripod stand, Bunsen burner, wire gauze, wash bottle, delivery tubes, gas jar, rubber, bung.

Procedure

1. Place sodium chloride in a round bottomed flask.
2. Set up the apparatus as follows.

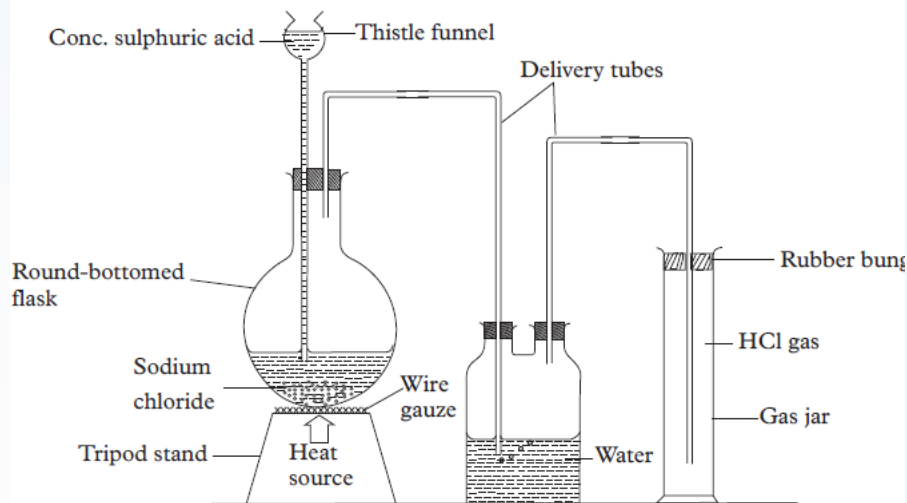


Fig 9.1.1: Laboratory preparation of hydrogen chloride gas

3. Slowly add concentrated sulphuric acid to the sodium chloride.
4. Heat the mixture gently.

Guidance questions

1. What do you observe when a mixture of sodium chloride and concentrated sulphuric acid is heated gently?
2. Write an equation for the reaction that occurs in the flask.
3. Why is an inverted funnel necessary in this experiment?

When the mixture of sodium chloride and concentrated sulphuric acid is heated, Effervescence occurs and a colourless gas is produced. The blue litmus paper turns red, showing that the gas is acidic. This gas evolved is hydrogen chloride (HCl).

Chemical equation of the reaction: $\text{NaCl(s)} + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{NaHSO}_4(\text{aq}) + \text{HCl(g)}$

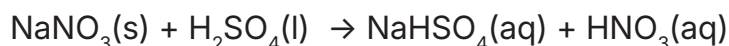
In this reaction, the less volatile concentrated sulphuric acid displaces the more volatile hydrochloric acid from sodium chloride. **Volatile acids** evaporate easily.

General Rule:

A less volatile acid displaces a more volatile acid from its salt.

Hydrogen chloride gas is highly soluble in water hence it is dissolved in distilled water using an inverted funnel arrangement to avoid "sucking back". A solution of hydrogen chloride gas in water is called hydrochloric acid.

The less volatile sulphuric acid also displaces the more volatile nitric acid from nitrates (sodium nitrate or potassium nitrate) according to the following equation.



In this preparation, all glass apparatus is used since nitric acid attacks cork and rubber fittings. The nitric acid vapour is then condensed to give nitric acid.

9.1.2. Preparation of a base by the addition of water to soluble metal oxides

Most metal oxides are insoluble in water, except oxides of Group 1 metals (Na_2O , K_2O) and some Group 2 metals (CaO , BaO)



Learning activity 9.1.2

Apparatus and reagents

Sodium oxide, distilled water, stirrer, beaker, conical flask or beaker.

Procedure

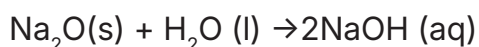
1. Place some sodium oxide in a conical flask.
2. Add distilled water to the sodium oxide and stir the mixture. What do you observe?
3. Test the resulting solution using blue and red litmus paper. What do you observe?



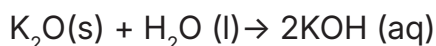
Study question

1. What is the purpose of stirring?
2. Do you think all bases can be prepared using the oxides of their metals? Explain.
3. What do you expect from the effect of litmus paper?

Sodium oxide is a white solid formed when sodium metal burns in air. On addition of water, the sodium oxide dissolves, forming a colourless solution. The solution turns red litmus paper blue, this shows that it is an alkali (NaOH).



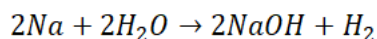
When potassium oxide is dissolved in water, it also forms an alkali called potassium Hydroxide (KOH).



Stirring increases contact between sodium oxide and water, allowing complete reaction and the uniform formation of a sodium hydroxide solution.

Not all bases can be prepared using the oxides of their metals. Only soluble metal oxides form bases with water.

Note that Group 1 metals can be used to prepare bases, as they react directly with water to form alkaline solutions, which are basic in nature. For example, sodium reacts with water to produce sodium hydroxide (NaOH) and hydrogen gas:



Application activity 9.1

1. a) Describe how hydrochloric acid can be prepared in the laboratory.
b) Nitric acid can be prepared in the laboratory by heating a mixture of potassium nitrate and concentrated sulphuric acid.
c) What type of reaction occurs?
d) Which of the two acids mentioned is more volatile? Explain.
2. Starting with sodium oxide, describe how a solution of sodium hydroxide can be prepared in the laboratory.
3. Complete the following chemical equations:
 - a) $\text{BaO} + \text{H}_2\text{O} \rightarrow$
 - b) $\text{Rb}_2\text{O} + \text{H}_2\text{O} \rightarrow$
4. Which ones in the following oxides cannot be used to prepare the corresponding bases Cs_2O , CaO , PbO , ZnO ? Explain.

9.2 Reactions of acids and bases with metals, metal oxides, carbonates and hydrogen carbonates.

9.2.1 Reactions of acids with metals, metal oxides, carbonates, and hydrogen carbonates.



Learning activity 9.2.1

Apparatus and reagents

Dilute hydrochloric acid, magnesium metal, calcium carbonate, sodium hydrogen carbonate, magnesium oxide, sodium hydroxide, phenolphthalein indicator, test tubes, splint, Bunsen burner, and beakers.



Procedure

1. Place 5 cm³ of dilute hydrochloric acid in a test tube. To the acid, add a piece of magnesium ribbon. Record your observations. Introduce a burning splint in the test tube. What do you observe?
2. Place 5 cm³ of dilute hydrochloric acid into a test tube. Add half spatula endful of calcium carbonate. What do you observe? Insert a burning splint into the test tube where the reaction is occurring. Record your observations.
3. Repeat procedure 2 using sodium hydrogen carbonate in place of calcium carbonate.
4. To 5 cm³ of dilute hydrochloric acid in a test tube, add magnesium oxide and heat it. What do you observe?
5. Place 5 cm³ of sodium hydroxide in a test tube and add 2 drops of phenolphthalein indicator. What do you observe? To this solution, add dilute hydrochloric acid slowly until the colour of the solution just turns colourless.

Guiding questions

1. What is observed when dilute HCl reacts with magnesium ribbon?
2. What is the purpose of the burning splint in step 1 of the procedure above?
3. What is observed when dilute HCl reacts with calcium carbonate and sodium hydrogen carbonate?
4. Why was there heating in step 4 of the procedure?
5. How do you expect the changes in phenolphthalein?
6. What would happen if methyl orange was used in place of phenolphthalein?
7. Suggest the equations for the reaction in step 1-5?
8. Describe all the possible observations in the procedure above.

When a piece of magnesium ribbon is dropped in dilute hydrochloric acid, effervescence occurs and a colourless gas that burns with a 'pop' sound is evolved. The gas is hydrogen.

Equation of the reaction: $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(g)} + \text{H}_2\text{(g)}$

A similar reaction occurs when dilute sulphuric acid is reacted with magnesium metal.

Equation of the reaction: $\text{Mg(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)}$

Generally, metals above hydrogen in the reactivity series react with dilute acids to liberate hydrogen.

NB: If nitric acid is used, the hydrogen liberated is oxidised to water since nitric acid is a powerful oxidising agent.

Generally, when a metal reacts with a dilute acid, the products are a salt and hydrogen gas.

Metal + dilute acid → salt + hydrogen gas.

When calcium carbonate and sodium hydrogen carbonate are added to dilute hydrochloric acid in separate test tubes, effervescence occurs, and a colourless gas is liberated. The gas extinguishes a burning splint. The gas is carbon dioxide.



Generally, dilute acids react with carbonates to form salt, carbon dioxide, and water.

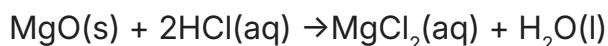
Carbonate + dilute acid → salt + carbon dioxide + water

However, the reaction between dilute sulphuric acid and calcium carbonate starts and then soon stops. This is due to the formation of insoluble calcium sulphate, which coats the carbonate, preventing any further contact between the reactants, the reaction stops.

What happens when dilute hydrochloric acid is reacted with lead (II) carbonate?



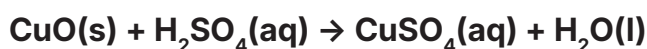
When a mixture of magnesium oxide and hydrochloric acid is heated, the magnesium oxide dissolves, reacting with the acid to form magnesium chloride and water. If a universal indicator is placed in the dilute hydrochloric acid, it turns red. If magnesium oxide is continuously added a little at a time with heating, the indicator eventually turns green, showing the solution has become neutral. This is a neutralisation reaction.



Generally, when an acid reacts with a metal oxide, salt and water are formed.

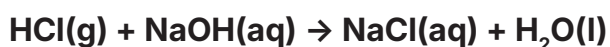
Dilute acid + metal oxide → salt + water.

Dilute sulphuric acid, when warmed with copper (II) oxide, forms the black copper (II) oxide reacts with the acid to form blue copper (II) sulphate and water.



Note: The acid used determines the type of salt formed. Hydrochloric acid gives chlorides, sulphuric acid gives sulphates whereas nitric acid gives nitrates.

When dilute hydrochloric acid was slowly added to sodium hydroxide solution to which phenolphthalein indicator was added, the colour of the solution turns from pink to colourless at some point. Dilute hydrochloric acid reacts with sodium hydroxide to form a salt, sodium chloride, and water on



This is a neutralisation reaction. Generally acids react with alkalis to form salt and water only. This is called a **neutralisation reaction**.

Dilute acid + alkali → salt + water

Dilute sulphuric acid neutralises potassium hydroxide solution to form potassium sulphate and water. $2\text{KOH}(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{K}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$

9.2.2 Reactions of bases with metals, metal oxides, carbonates, and hydrogen carbonates.



Learning activity 9.2.2

Apparatus and reagents

Sodium hydroxide, burette, test pipette, dilute hydrochloric acid, pipette, conical flask, methyl orange indicator.

Procedure

1. Pipette 25cm^3 of 1M sodium hydroxide solution into a conical flask and add three drops of methyl orange indicator. What do you observe?
2. Run 1M hydrochloric acid to the sodium hydroxide solution until the end Point/ colour change. How do you tell when the end point is reached?

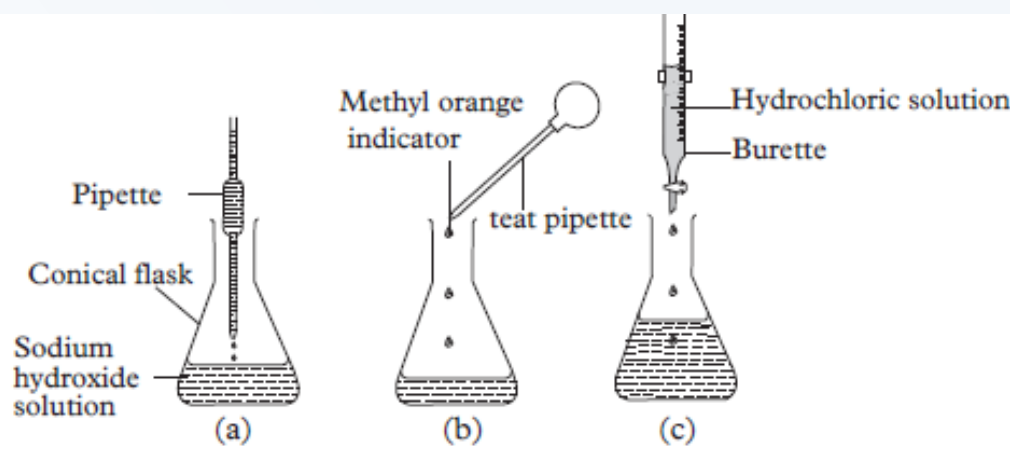
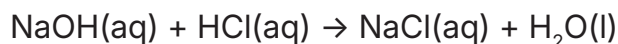


Fig 9.2.2: Experimental setup for the reaction between sodium hydroxide and dilute hydrochloric acid

Guiding questions

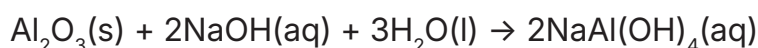
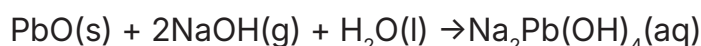
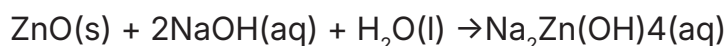
1. What name is given to the reaction that occurs?
2. Write equation of the reaction that occurs.
3. Why was methyl orange added?

When methyl orange indicator is added to a sodium hydroxide solution, the solution turns yellow. On the addition of hydrochloric acid, neutralisation takes place, and sodium chloride and water are formed.



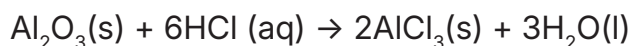
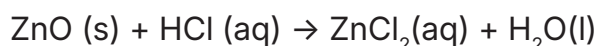
The endpoint is reached when one drop of the hydrochloric acid changes the colour of the solution from yellow to orange.

Some amphoteric oxides, such as zinc oxide, lead(II) oxide, and aluminium oxide, react with bases to form complex salts.



These oxides, which react with both alkalis and acids, are known as **amphoteric oxides**.

Reactions with acids:



Application activity 9.2

- Copy and complete the reaction below.
 - Magnesium carbonate + hydrochloric acid $\rightarrow$ ____ + ____ + ____
 - Sodium carbonate + nitric acid $\rightarrow$ ____ + ____ + ____
 - Calcium carbonate + sulphuric acid $\rightarrow$ ____ + ____ + ____
 - Potassium + hydrochloric acid $\rightarrow$ - ____ + ____ + ____
- When acid is added to an alkali, what products are formed?
- Complete the chemical equation $\text{K}_2\text{O} + \text{HCl}$
- What is the product of the reaction between nitric acid and ammonia?
- Give four chemical properties of acids.
- What is neutralisation?
 - Write an equation to show the reaction between calcium oxide and dilute hydrochloric acid.
- State one chemical property of bases.
 - Potassium hydroxide is neutralised by dilute sulphuric acid. Write an equation for the reaction that takes place.
- What is an amphoteric oxide?
 - Name two amphoteric oxides.

9.3. Uses and dangers of acids and bases.

9.3.1 Uses of acids and bases

Acids and bases play important roles in daily life, industry, and biological processes, but they can also pose environmental dangers if misused.



Learning activity 9.3.1

1. State some applications of acids and bases in our daily life.
2. Explain the role of hydrochloric acid in the human stomach.

a) Uses of Acids

Acids find a variety of uses in our daily lives

- Tannic acid is used in leather tanning, where it makes animal skins insoluble and durable.
- Hydrochloric acid is present in the human stomach, creating suitable conditions for digestion. It activates pepsin, an enzyme that breaks down proteins into smaller peptide chains.
- Sulphuric acid is used in lead–acid batteries for motor vehicles and solar power storage.
- Ascorbic acid (vitamin C) found in citrus fruits is essential for body health.
- Ethanoic acid is the main component of vinegar, used in food seasoning.
- Acids are widely used in the manufacture of fertilizers such as ammonium nitrate and ammonium sulphate.

b) Uses of Bases

- Ammonia is used in making fertilizers and household cleaning agents.
- Aluminium hydroxide is used in producing aluminium compounds and as an antacid.
- Calcium hydroxide is used in cement production, soil acidity neutralisation, and sewage treatment.
- Sodium hydroxide is used in the manufacture of soaps, detergents, and cleaning agents.
- Magnesium hydroxide is used in antacid tablets and antiperspirant deodorants.
- Calcium oxide is used to reduce soil acidity.

9.3.2 Dangers of acids and bases in daily life



Learning activity 9.3.2

1. Match the household substance with its possible danger.

Substance	Danger
1. Battery acid	A) Eye irritation
2. Bleach	B) Skin burns
3. Soap solution	C) Corrosion
4. Toilet cleaner	D) Poisonous fumes

2. Multiple Choice Question

Why are strong bases dangerous to the skin?

- A) They are cold
- B) They can cause burns
- C) They smell bad
- D) They evaporate quickly

Acids and bases can cause several **undesirable effects** on the environment and human health when present in excess.

a) Acid rain

- Industrial activities release **acidic gases** such as sulphur dioxide, carbon dioxide, and nitrogen oxides into the atmosphere.
- These gases dissolve in rainwater to form **acid rain**.
- Effects of acid rain include:
 - Damage to buildings and monuments by peeling paint and dissolving marble (limestone).
 - Corrosion of metals and statues.
 - Death of fish and aquatic organisms when acidic runoff enters rivers and lakes.
 - Loss of soil nutrients, weakening trees and plants.
 - Root damage in trees due to the release of aluminium ions, leading to poor water uptake and wilting.

b) Tooth decay

- Bacteria in the mouth convert trapped food into acids.
- These acids dissolve tooth enamel, causing tooth decay and toothache.
- Tooth decay begins when the pH of the mouth falls below 5.8.
- Toothpaste is basic and helps neutralise acids in the mouth.

c) Soil pH

- Soil acidity or alkalinity affects plant growth.
- High concentration of hydrogen ions (H^+) makes soil acidic, while low concentration makes it basic.
- Soil acidity is caused by excessive use of fertilisers and acid rain.
- Acidic soils reduce crop productivity.
- Lime (calcium oxide and calcium hydroxide) is added to neutralise excess acidity in soil.

d) Stomach indigestion

- Hydrochloric acid in the stomach aids digestion.
- Excess acid secretion causes indigestion, stomach burns, and ulcers.
- Stress and overeating increase acid production.
- Antacid tablets contain mild bases such as magnesium hydroxide, which neutralise excess acid and relieve pain.

While acids and bases are essential in natural and industrial processes, excessive amounts can harm the environment and human health, making **control and neutralisation** very important.



Application activity 9.3

1. a) State the uses of some acids in our daily life.
b) State the uses of some bases in our daily life.
2. Give three dangers caused by acids and bases
3. Give three safety precautions when using cleaning chemicals.
4. What should you do if acid splashes on your skin?



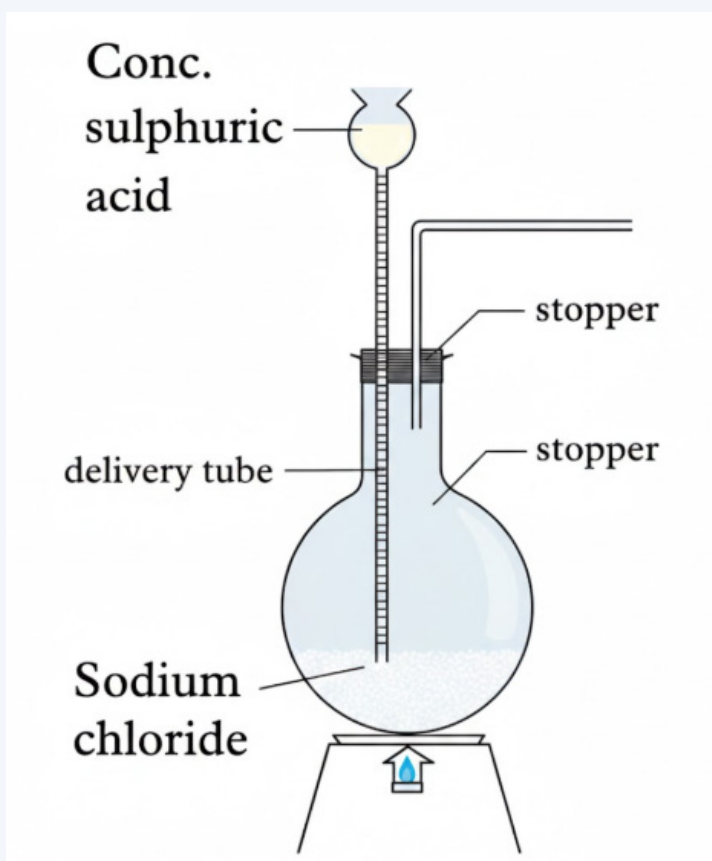
9.4 End of unit assessment

A) Knowledge and understanding

1. What is observed when a mixture of sodium chloride and concentrated sulphuric acid is heated?
2. Which gas is produced when sodium chloride reacts with concentrated sulphuric acid, and how can its acidic nature be tested?
3. Write the chemical equation for the preparation of hydrogen chloride gas from sodium chloride and concentrated sulphuric acid.
4. In the preparation of hydrogen chloride, which acid is less volatile and which one is more volatile?



5. State the general rule for the displacement of one acid by another during acid preparation.
6. Why is hydrogen chloride gas dissolved in water, and what is the name of the solution formed?
7. Why is an inverted funnel arrangement used when dissolving hydrogen chloride gas in water?
8. Which metal oxides can react with water to form bases, and why do most metal oxides not form bases with water?
9. Explain the role of **stirring** when sodium oxide reacts with water to form sodium hydroxide.
10. The following diagram gives an incomplete setup of the apparatus that can be used to prepare a sample of hydrochloric acid in the laboratory.



Study it and answer the questions that follow.

- i) Complete the diagram to show how a sample of hydrochloric acid may be collected.
- ii) Write an equation for the reaction that occurs.
- iii) Name another substance that can be used in place of sodium chloride.



- c) Dilute hydrochloric acid was added to zinc metal.
- State and explain the observations made.
 - Describe a test that can be used to confirm the gaseous product.
 - Write an equation for the reaction that occurs.
11. Sodium oxide was added to 5cm^3 of water in a boiling tube.
- What observations were made?
 - Name the product formed in this reaction.
 - Write an equation for the reaction that occurs.
 - A sample of the solution formed was placed in a test tube. One drop of methyl orange indicator was added, followed by addition of hydrochloric acid drop by drop until the colour change just occurred.
 - What type of reaction took place?
 - State two observations made on addition of methyl orange indicator up to the colour change.
 - Write an equation for the reaction that occurs.
12. Sodium metal was placed in water in a trough.
- State and explain any three observations made.
 - Write an equation for the reaction that occurs.
 - If lead (II) oxide was added to a sample of the solution formed, write an equation for the reaction that occurs.
13. a) What is acid rain?
b) Describe the formation of acid rain.
c) State two environmental effects of acid rain.

B) Multiple-Choice Questions (MCQs) with Answers

14. Which acid is used to displace hydrochloric acid from sodium chloride?
- | | |
|----------------------|-------------------|
| A) Nitric acid | C) Sulphuric acid |
| B) Hydrochloric acid | D) Carbonic acid |
15. The gas produced when NaCl reacts with concentrated H_2SO_4 is:
- | | |
|-------------|----------------------|
| A) Chlorine | C) Hydrogen chloride |
| B) Hydrogen | D) Sulphur dioxide |



16. Which substance turns blue litmus paper red?

- | | |
|---------------------|--------------------------|
| A) Sodium hydroxide | C) Hydrogen chloride gas |
| B) Potassium oxide | D) Calcium oxide |

17. Which oxide reacts with water to form an alkali?

- | | |
|--------------------------------------|--------------------------------------|
| A) A. CuO | C) C. Na ₂ O |
| B) B. Fe ₂ O ₃ | D) D. Al ₂ O ₃ |

18. The alkali formed when potassium oxide reacts with water is:

- | | |
|------------------------|------------------------|
| A) Sodium hydroxide | C) Calcium hydroxide |
| B) Potassium hydroxide | D) Magnesium hydroxide |

C) Skills

19. Why is sulphuric acid suitable for preparing hydrochloric acid in the laboratory?

20. Why are glass apparatuses used in the preparation of nitric acid?

21. Explain why copper(II) oxide cannot be used to prepare a base by adding water.

22. How would you confirm that a sodium hydroxide solution has been formed?

23. What safety precautions should be taken when dissolving hydrogen chloride gas in water?

D) Attitudes and Values

24. Why should laboratory preparation of acids be done carefully?

25. What responsible behaviour should learners show when handling acids and bases?

26. Why is it important to use correct chemicals rather than substitutes in experiments?

27. How does learning about acids and bases help you appreciate safety in daily life?

28. What value is developed by following correct laboratory procedures?

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